Preface

This book is intended primarily to cater for the needs of mathematics 1 students at Vaal University of Technology (VUT). Areas covered include:

- Unit 1—Basic algebra, functions and trigonometry
- Unit 2— Differentiation
- Unit 3— Integration
- Unit 4— Matrices and vectors
- Unit 5— Complex numbers
- Unit 6— Statistics

Special care has been taken in the presentation of steps in problem solving for easy understanding. The mathematical steps are so simple that an average student may follow it independently. Worked examples and revision exercises are incorporated in each unit. The exercises could be done in class or as homework. Even though the learning material is presented in such a way that it can easily be followed, students are encouraged to consult their lecturers and tutorial assistants should they encounter any problems from their studies. It is therefore expected that when this book is used alongside classroom tuition it will train the student to systematically attempt any question, however complicated, with the utmost confidence.

Although every care has been taken to check mistakes and misprints, some errors may be expected. Please draw the editors' attention to any error that you may come across. All such errors when brought to the attention of the editors will be corrected and incorporated in the next edition.

Several lecturers in the department of mathematics have contributed to this edition.

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Table of contents

1. Basic algebra, functions and trigonometry 1
1.1 Introduction 1
1.2 Revision: exponents, logarithms and manipulation of formula 1
1.3 Functions 3
1.4 Trigonometry 7
1.5 Binomial Expansion 10
1.6 Radian measure 15
1.7 Revision exercises 21

2. Differentiation
2.1 Introduction 23
2.2 Limit of a function 23
2.3 Differentiation from first principles 27
2.4 Basic differentiation rules 29
2.4.1 Derivatives of polynomials 29
2.4.2 Derivative of a sum or difference 29
2.4.3 The product rule 30
2.4.4 The quotient rule 31
2.4.5 The chain rule 32

2.5. Derivatives of trigonometric functions 33

2.6 Derivatives of exponential functions 35

2.7. Derivatives of logarithmic functions 35

2.8. Higher order derivatives 36

2.9. Derivatives of implicit functions 37

2.10. Logarithmic differentiation 39

2.11. Applications of differentiation 40

2.11.1 Curve sketching 40

2.11.2 Velocity and acceleration 42

2.11.3 Newton-Raphson method 45

2.10 Revision exercise 46

3. Integration
3.1 Introduction 49
3.2 Indefinite integrals 49

3.2.1 Integration by substitution 50

3.2.2 Integration involving trigonometric functions 52

3.2.3 Integration involving exponential and logarithmic functions 53

3.3 Definite integrals 54

3.4 Applications of integration 55

3.4.1 Area and the definite integral 55

3.4.2 Area between curves 56

3.4.3 Numerical integration (Simpson's rule) 58

Revision exercise 60

4. Matrices and Vectors
4.1 Introduction 62
4.2 Matrices 62
4.2.1 Elementary operations of 2x2 matrices 63
4.2.2 Determinant of a 2x2 matrix 64
4.3 Vectors 64
4.3.1 Notation 64
4.3.2 Representation of vectors 65
4.3.3 Magnitude/ Length of a vector 65
4.3.4 Resolution of a vector 65
4.3.5 Unit vectors 67
4.3.6 Direction vector 67
4.3.7 Negative or inverse of a vector 68
4.3.8 Multiplication of a vector by a scalar 68
4.3.9 Parallel vectors 68
4.3.10 Addition and subtraction of vectors 69
4.3.11 Scalar or dot product 69
4.3.12 Angle between two vectors 70
4.3.13 Work done 71
4.3.14 Vector or cross product 71
4.3.15 Moment of a vector 72

4.4 Revision exercise 73

5. Complex numbers 74
5.1 Introduction 74
5.2 Representation of a complex number 74
5.3 Addition and subtraction of complex numbers 75
5.4 Multiplication of complex numbers 75
5.5 Division of complex numbers 76
5.6 Equality of complex numbers 76

5.7 Argand diagram and polar form of complex numbers 77
5.8 De Moivre's theorem 79
5.9 Calculating roots 80
5.10 Revision exercise 81

6. Statistics 83
6.1 Introduction 83
6.2 Data collection 83
6.3 Tabulation 83
6.4 Data representation 84
6.5 Measures of central tendency 86
6.6 Measures of dispersion 93
6.7 Correlation and regression 96
6.8 Revision exercise 105

Answers to exercises and Revision exercises 111

Test and examination questions 117

UNIT 1

BASIC ALGEBRA, FUNCTIONS AND TRIGONOMETRY

1.1. INTRODUCTION

In Unit 1 the following topics are revised: exponents, logarithms, manipulation of formulae, functions (domain, range and inverse), trigonometric functions and inverse trigonometric functions. These are followed by binomial expansion, radian measure and its applications.

By the end of this Unit the learner should be able to:

- Apply the laws of exponent and logarithm
- Manipulate formulae to solve equations
- Expand $(a+b)^n$ where n is a rational number, using binomial theorem
- Convert from degree to radian measure of angles.
- To determine the arc-length, area of a sector and area of a segment of a circle.

1.2. REVISION: EXPONENTS, LOGARITHMS, FRACTIONS AND MANIPULATION OF FORMULA.

In this section exponents, logarithms, fractions and manipulation of formula are briefly introduced as part of the revision process.

The following are the exponential laws:

1.2.1. Exponents

If a and b are bases, m and n exponents, then the following exponential laws apply:

a) $a^m \times a^n = a^{m+n}$ b) $\frac{a^m}{a^n} = a^{m-n}$ a$\neq 0$

c) $(a \times b)^m = a^m b^m$ d) $a^0 = 1, a \neq 0$ e) $a^{-n} = \frac{1}{a^n}, \ a \neq 0$

1.2.2. Logarithms

A logarithm can be deduced from an exponential equation. The components of an exponential equation are indicated below:

exponent/index

$$a^b = c$$

number

base

i.e. $a^b = c \Leftrightarrow \log_a c = b$. For example, $log_3 \frac{1}{27} = -3$ because $3^{-3} = \frac{1}{27}$,

$log_5 625 = 4$ because $5^4 = 625$, $log 0{,}01 = -2$ because $10^{-2} = 0{,}01$.

Note that in the last example base 10 is not indicated. Whenever the base is not indicated, the assumption is that it is 10. A logarithm of a number with 10 as the base is called a **common logarithm**. The table below shows the laws of logarithm.

Table 1: Laws of Logarithm

Logarithm with base $a, a > 0$	Common logarithm, base 10	Natural logarithm, base e
$log_a MN = log_a M + log_a N$	$logMN = logM + logN$	$lnMN = lnM + lnN$
$log_a \frac{M}{N} = log_a M - log_a N$	$log \frac{M}{N} = logM - logN$	$ln \frac{M}{N} = lnM - lnN$
$log_a M^n = nlog_a M$	$logM^n = nlogM$	$lnM^n = nlnM$
	Change of base: $log_a M = \frac{log_c M}{log_c a}$	

Exercise 1.2.

1. Express the following equations in logarithmic form:

   a) $25^{-\frac{1}{2}} = \frac{1}{5}$ b) $\left(\frac{1}{16}\right)^{-\frac{1}{2}} = 4$ c) $3^2 = 9$ d) $81^{\frac{3}{4}} = 27$ e) $13^0 = 1$

2. Express the following equations in exponential form:

   a) $\log_5 \frac{1}{25} = -2$ b) $\ln\left(\frac{1}{e^{2x}}\right) = -2x$ c) $\log_{243} 3 = 0.2$ d) $\ln \frac{1}{e} = -1$

   e) $\log_{\frac{1}{6}} 36 = -2$

3. Determine x in each case:

   a) $\log_{32}(\frac{1}{16}) = x$ b) $\ln x = 4$ c) $e^{\ln x} = 5$ d) $\log_8(x-2) = -\frac{2}{3}$

   e) $a^{\log_a 43} = x$ f) $\log_x 2 = \log_2 x$

4. Solve for the indicated variable:

   a) $\log \frac{N}{N_0} = -kt, \quad N_0$ b) $C = \frac{k}{\log_e\left(\frac{R_1}{R_2}\right)}, \quad R_1$ c) $W = k \log_3 \frac{V_2}{V_1}, \quad V_2$

   d) $y = x^2 + 2x + 1, \quad x$ e) $N = \frac{af_1 + bf_2}{cf_1 + df_2}, \quad f_1$

5. Express y in terms of x

   a) $\log_b y = \log_b 2 + \log_b x$ b) $\log_3 y = -2\log_3(x+1) + \log_3 7$

   c) $\frac{\log y}{\log x} = \log y - \log x$ d) $(\log y)^2 = 2\log x - 1$

   e) $\pi \log_4 x + \log_4 y = 1$

1.3. FUNCTIONS

1.3.1. Definition

A **function** is a rule which gives only one output for every given input. The rule is often expressed in the form of an equation $y = f(x)$.

In terms of mapping of elements, a **function** is either a one-to-one or a many-to-one relation. A function can also be described in terms of a formula or algorithm that tells how to compute the output for a given input. An example is the function that relates each real number x to its square x^2.

This is written as $f(x) = x^2$

The output of a function f corresponding to an input x is denoted by $f(x)$ (read "f of x").

$f(\pm 3) = 9$. This is a many-to-one relation, and thus a function. Not all formulae define a function. For instance, if $f(x) = \pm\sqrt{x}$, two outputs, $f(x) = 2$ or -2 correspond to the same input $x = 4$. This makes the relation a one-to-many relation, and thus the relation is not a function.

1.3.2. Domain and range of a function

The domain of the function is the set of elements of the inputs for which the function is defined. The range is the set of outputs corresponding to the inputs.

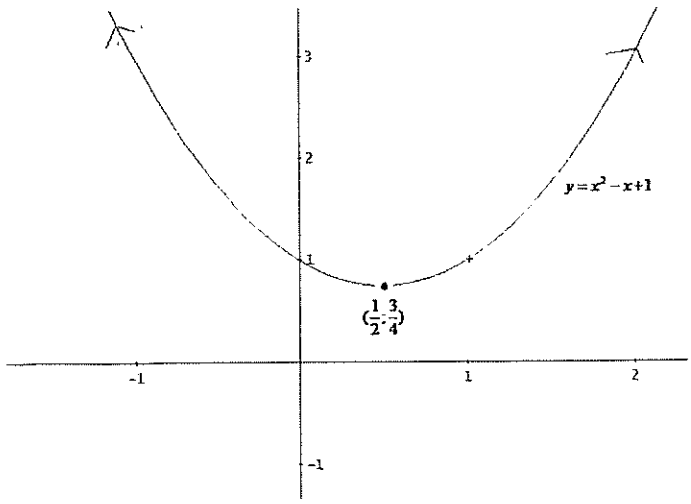
Notation: $Domf$ means the domain of the function f

$Rangef$ means the range of the function f

Examples.

1. $f(x) = x^2 - x + 1 \Rightarrow f(x) = \left(x - \frac{1}{2}\right)^2 + \frac{3}{4}$

This is a parabola with axis of symmetry $x = \frac{1}{2}$ and turning point $\left(\frac{1}{2}; \frac{3}{4}\right)$ as the following diagram shows:



The domain is the set of all real numbers, while the range $= \left\{y: y \geq \frac{3}{4}, y \in \Re\right\}$

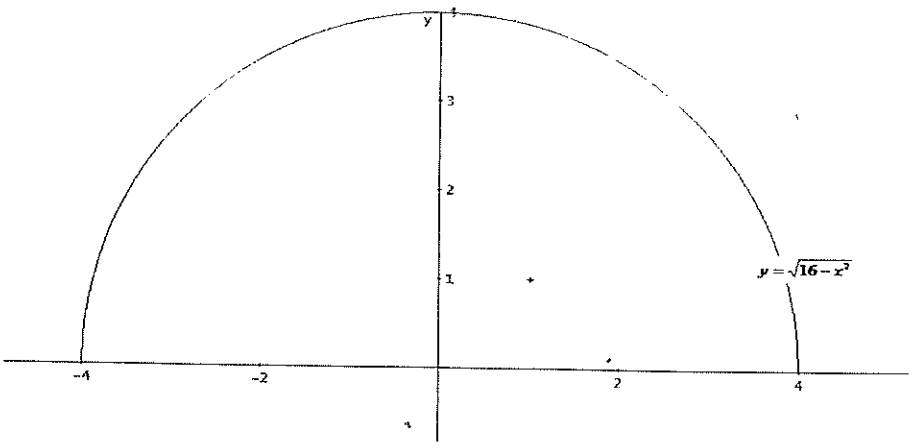
2. $f(x) = x^2 - x + 1 \qquad 1 \leq x \leq 4$

   The graph is actually part of $f(x) = x^2 - x + 1$

   $Dom\ f = \{x: 1 \leq x \leq 4\} \qquad Range\ f = \{y: 1 \leq y \leq 13\}$

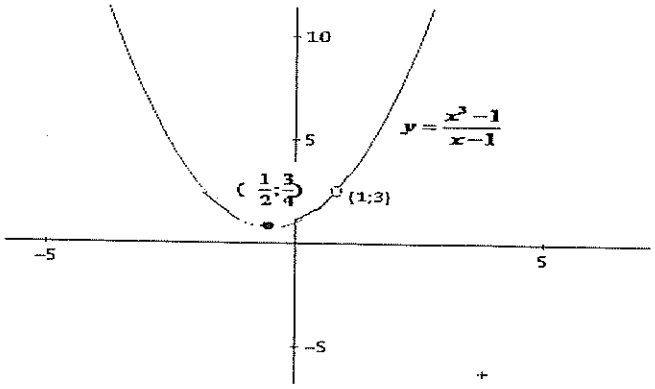
3. $f(x) = \sqrt{16 - x^2}$

   This is the equation of the semi-circle centre at origin with radius 4, as shown below.



$Dom\ f = \{x: -4 \leq x \leq 4\} \qquad Range\ f = \{y: 0 \leq y \leq 4\}$

4. $f(x) = \frac{x^3 - 1}{x - 1} = \frac{(x-1)(x^2+x+1)}{x-1} = x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4}$



$Dom\ f = \{x \in \Re, x \neq 1\} \qquad Range\ f = \left\{y \in \Re, y \neq 3, y \geq \frac{3}{4}\right\}$

Exercise 1.3.2.

1. Determine the domain and range of the functions defined by the following equations:

   a) $y = -\sqrt{-2x + 3}$ b) $y = -x^4 + 4$ c) $y = \sqrt{x - 5}$ d) $y = x^2 + 2$ e) $y = \frac{1}{t+2}$

f) $y = \ln(4 - t)$ g) $y = \sin t$

1.3.3. Inverse Functions

a) Definition

Let f be a one-to-one function with domain A and range B. Then the inverse function f^{-1} has the domain B and range A and is defined by

$f^{-1}(y) = x \Leftrightarrow f(x) = y$ for any y in B.

Examples

1. Determine the inverse of the function $y = 3x - 2$

Solution

Method 1: Interchange the variables and make y the subject

$\boldsymbol{y = 3x - 2}$

$\boldsymbol{x = 3y - 2 \rightarrow 3y = x + 2 \therefore y = \frac{x+2}{3}}$ **i.e** $\boldsymbol{f^{-1}(x) = \frac{x+2}{3}}$

Method 2: First make the other variable the subject of the formula, then interchange

$y = 3x - 2$

$3x = y + 2 \therefore x = \frac{y+2}{3}$. Interchanging the variables, we have

$f^{-1}(x) = \frac{x+2}{3}$

2. Determine the inverses of the following functions:

   a) $y = 2^x$ b) $y = x^2 - 3x + 9, x \geq \frac{3}{2}$ c) $f(x) = \frac{x-9}{x+3}$

Solutions

a) $y = 2^x$ $x = 2^y \rightarrow \log_2 x = y \log_2 2$ $[\log_2 2 = 1]$

$\therefore y = \log_2 x$ hence $f^{-1}(x) = \log_2 x$

b) $y = x^2 - 3x + 9, x \geq \frac{3}{2}$

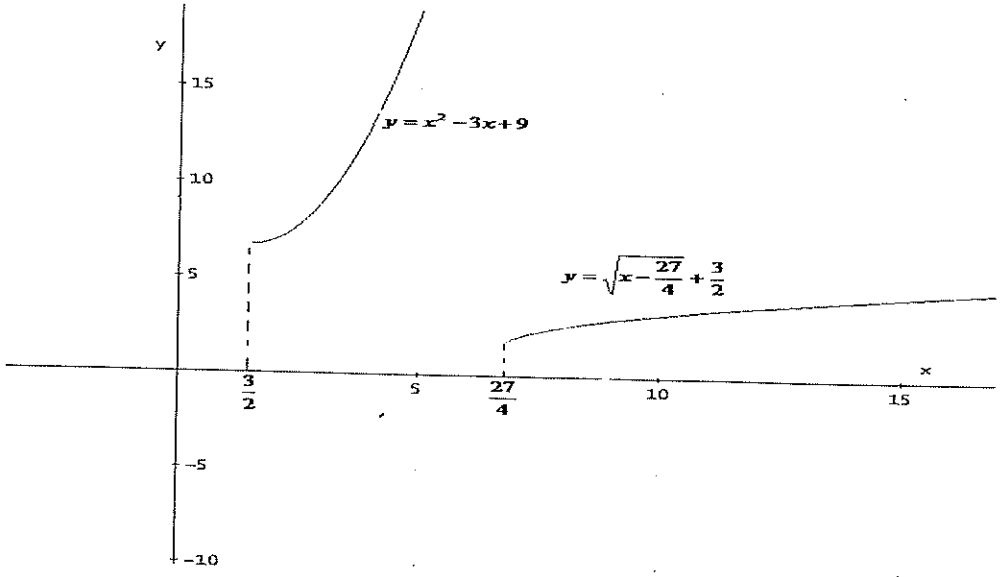
$x = y^2 - 3y + 9$

$x - 9 = \left(y - \frac{3}{2}\right)^2 - \frac{9}{4}$

$$x - \frac{27}{4} = \left(y - \frac{3}{2}\right)^2$$

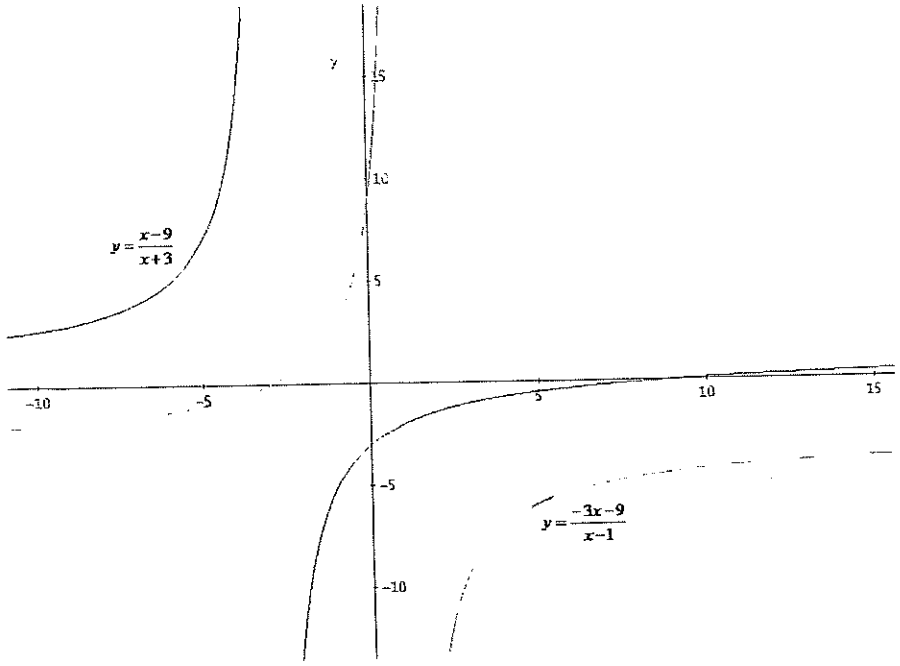
$$y = \sqrt{x - \frac{27}{4}} + \frac{3}{2} \text{ hence } f^{-1}(x) = \sqrt{x - \frac{27}{4}} + \frac{3}{2}, x \geq \frac{27}{4}$$

This is how the graph and its inverse look like:



c) $y = \dfrac{x - 9}{x + 3} \Rightarrow x = \dfrac{y - 9}{y + 3} \Rightarrow (y + 3)x = y - 9 \Rightarrow y(x - 1) = -3x - 9 \Rightarrow f^{-1}(x) = \dfrac{-3x - 9}{x - 1}$

The graphs of the function and its inverse are found below:



Exercise 1.3.3

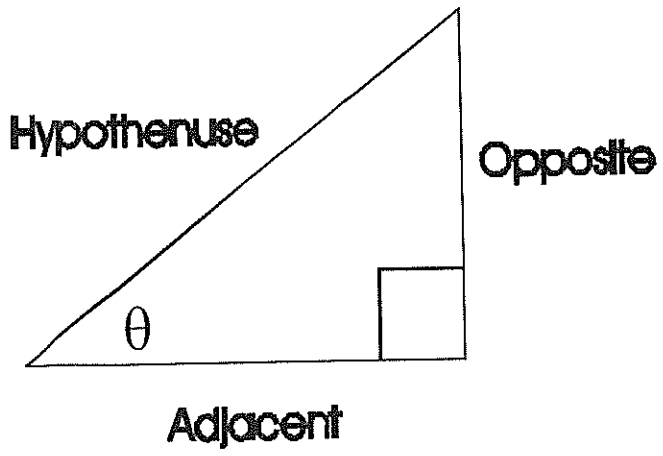
Determine the inverses of the following:

a) $y = \frac{3x-2}{2x-3}$ b) $y = \log(2x-3)$ c) $y = \log(x^2 - 3x + 5)$

d) $y = 2x - 3$ e) $y = \frac{1}{4-t}$

1.4 TRIGONOMETRY

Consider the right-angled triangle shown below:



The following are the trigonometric functions defined from the triangle:

$\sin\theta = \frac{opposite}{hypotenuse}$ $\qquad$ $cosec\theta = \frac{1}{\sin\theta}$

$\cos\theta = \frac{adjacent}{hypotenuse}$ $\qquad$ $\sec\theta = \frac{1}{\cos\theta}$

$\tan\theta = \frac{opposite}{adjacent}$ $\qquad$ $\cot\theta = \frac{1}{\tan\theta}$

TRIGONOMETRIC IDENTITIES	
$\sin^2\theta + \cos^2\theta = 1$	$1 + \tan^2\theta = \sec^2\theta$
$1 + \cot^2\theta = \cos ec^2\theta$	
$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$	$\sin 2A = 2\sin a\cos A$
$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$	$\cos 2A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$
$\sin^2 A = \frac{1}{2}[1 - \cos 2A]$	$\cos^2 A = \frac{1}{2}[1 + \cos 2A]$

1.4.1. Inverse trigonometric functions

Sometimes we want θ such that $\sin\theta = \frac{1}{2}$. To do that we use the inverse sine (sin^{-1}) or arcsine of $\frac{1}{2}$, defined as follows:

$$\sin\theta = \frac{1}{2} \rightarrow \theta = \sin^{-1}\left(\frac{1}{2}\right) \text{ or } \theta = arcsin\left(\frac{1}{2}\right)$$

In essence, $\sin^{-1}\left(\frac{1}{2}\right)$ is an angle whose sine value is $\frac{1}{2}$. This should not be confused with the cosecant function, a ratio that is a reciprocal of the sine function. Generally we use θ between and including $-90°$ and $90°$ in order for the inverse to be a function. This leads to the following definition of the arcsine.

$$\theta = \sin^{-1} x \leftrightarrow \sin\theta = x, -90° \leq \theta \leq 90°$$

We know that the angles for which sine is $\frac{1}{2}$ are $30°$ and $150°$. However, because of the restriction $-90° \leq \theta \leq 90°$ we accept the solution

$$\sin\theta = \frac{1}{2} \rightarrow \sin^{-1}\left(\frac{1}{2}\right) = 30°$$

Similarly, we define the inverse cosine and inverse tangent as follows:

$\theta = \cos^{-1} x \leftrightarrow \cos\theta = x,\ 0° \leq \theta \leq 180°$

$\theta = \tan^{-1} x \leftrightarrow \tan\theta = x,\ -90° \leq \theta \leq 90°$

Examples

1. $\sin\theta = \frac{\sqrt{3}}{2} \leftrightarrow \theta = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = 60°$
2. $\tan\theta = -1{,}810 \leftrightarrow \theta = \tan^{-1}(-1{,}810) = -61{,}08°$

Suppose we want to determine $cosec^{-1}y$

Let $x = cosec^{-1}y \leftrightarrow cosec\ x = y \leftrightarrow \frac{1}{y} = \sin x \therefore x = \sin^{-1}\left(\frac{1}{y}\right)$

Hence $cosec^{-1}y = \sin^{-1}\left(\frac{1}{y}\right)$

Similarly, $\sec^{-1} y = \cos^{-1}\left(\frac{1}{y}\right)$ and $\cot^{-1} y = \tan^{-1}\left(\frac{1}{y}\right)$

Example

1. $cosec\ x = 1{,}245$

$x = cosec^{-1}(1{,}245) = \sin^{-1}\left(\frac{1}{1{,}245}\right) = \sin^{-1}(0{,}80321) =$
$53{,}4377° \sim 53{,}44°$

Exercise 1.4.1

Determine the value of the unknown:

a) $x = \sec^{-1}(2)$ b) $t = cosec\ (cosec^{-1}2)$

c) If $t = \sin^{-1}(\alpha)$, determine $\sin 2t$ in terms of α

d) $x = \sec 75° - cosec\ 60° + \cot 115°$

1.5. BINOMIAL EXPANSION

Binomial expansion involves the algebraic expansion of powers of a binomial, like $(x+y)^n$. When n is large and/or rational, then the binomial theorem is applied.

1.5.1. Binomial Theorem/Formula

The binomial theorem makes it possible to expand the power $(x+y)^n$ into a sum involving terms of the form ax^by^c

$$(a+b)^n = a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^3 \ldots$$

Factorial notation

The factorial expression can be generalised to the factorial of any arbitrary natural number n as:

$$n! = n \times (n-1) \times (n-2) \times \ldots . \times 2 \times 1$$

Thus $3! = 3 \times 2 \times 1 = 6$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

1.5.2. Some properties of the Binomial Formula

For n a positive integer, the series is finite with $n+1$ terms.

For n a fraction or a negative integer, the series is an infinite series.

The sum of the powers of a and b in each term is equal to n, and the powers of a decrease while the powers of b increase to n.

a) For n a positive integer

$$(a+b)^n = a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^3 \ldots + b^n$$

Examples

$$(x+y)^4 = x^4 + 4x^3y + \frac{4\times3}{2!}x^2y^2 + \frac{4\times3\times2}{3!}x^1y^3 + y^4$$

$$= x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$$

$$(5x-2y)^4 = (5x)^4 + 4(5x)^3(-2y) + \frac{4\times3}{2!}(5x)^2(-2y)^2 + \frac{4\times3\times2}{3!}(5x)(-2y)^3 + (-2y)^4$$

$$625x^4 - 1000x^3y + 600x^2y^2 - 160xy^3 + 16y^4$$

$(x^3 + 3y^2)^3 = x^9 + 9x^6y^2 + 27x^3y^4 + 27y^6$

b) For negative integers or fractional n

$$(a + bx)^n = a^n + na^{n-1}bx + \frac{n(n-1)}{2!}a^{n-2}b^2x^2 + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^3x^3 \ldots$$

The series is valid for $|bx| < a$

Examples

1. Use the binomial theorem to expand $\frac{1}{(3-5x)^2}$ in ascending powers of x up to the term in x^3. Identify the restrictions on x for the series to be valid.
2. Expand $(1 - 5x)^{-\frac{1}{3}}$ in ascending powers of x up to the term in x^4. Identify the restrictions on x for the series to be valid.

Solutions

$\frac{1}{(3-5x)^2}$ $\quad -\frac{3}{5} < x < \frac{3}{5}$

1. $\frac{1}{(3-5x)^2} = (3 - 5x)^{-2} = 3^{-2} - 2(3)^{-3}(-5x) + \frac{(-2)(-3)}{2!}(3)^{-4}(-5x)^2 + \frac{(-2)(-3)(-4)}{3!}(3)^{-5}(5x)^3 \ldots$

   $= \frac{1}{9} + \frac{10x}{27} + \frac{25x^2}{27} + \frac{500x^3}{243}$ The series is valid if $|-5x| < 3$, i.e $-\frac{3}{5} < x < \frac{3}{5}$

2. $(1 - 5x)^{-\frac{1}{3}} = 1 + \frac{5x}{3} + \frac{50x^2}{9} + \frac{1750x^3}{81} + \frac{21875x^4}{243} + \ldots$

   $(1-5x)^{-\frac{1}{3}} = \frac{1}{(1-5x)^{\frac{1}{3}}}$

   The series is valid if $|-5x| < 1$, $-\frac{1}{5} < x < \frac{1}{5}$

   $-\frac{1}{5} < x < \frac{1}{5}$

1.5.3. Finding a particular term in binomial expansion

Given the series, $(a + b)^n$, the kth term in the expansion is give as:

$$T_k = \frac{n(n-1)(n-2)\ldots[n-(k-2)]}{(k-1)!}a^{n-(k-1)}b^{k-1}$$

Examples

Write down the indicated term in the binomial expansion of the following:

a) $(2 - 3x)^6$ $\quad 4^{th}$ term

b) $\left(1 - \frac{x}{2}\right)^7$ $\quad 5^{th}$ term

c) $\left(1 + \frac{2x}{5}\right)^{10}$ $\quad 3^{rd}$ term

Solutions

a) $T_4 = \frac{6\times5\times4}{3!}(2)^3(-3x)^3 = -4320x^3$

b) $T_5 = \frac{7\times6\times5\times4}{4!}\left(-\frac{x}{2}\right)^4 = \frac{35x^4}{16}$

c) $T_3 = \frac{10\times9}{2!}\left(\frac{2x}{5}\right)^2 = \frac{36x^2}{5}$

The kth term could also be used to determine coefficients. If the two elements of the binomial being expanded do not involve the same variable, then the kth term can be determined by considering the pattern of the exponents.

Example 1

Determine the coefficient of x^5 in the expansion of $(5-x)^9$

Solution

Note that powers of $(-x)$ in the expansion range from $(-x)^0$ to $(-x)^9$. This means that the term with $(-x)^5$ is the 6th term.

$\therefore T_6 = \frac{9\times8\times7\times6\times5}{5!}(5)^4(-x)^5 = -78750x^5$

Hence the required coefficient is -78750

In other cases the elements of the binomial being expanded may contain the same variable, like the example below.

Example 2

Determine the coefficient of p^5 in the expansion $\left(p-\frac{3}{p}\right)^{11}$

Solution

Method 1: **Using the expansion of terms**

The terms in the expansion of $\left(p-\frac{3}{p}\right)^{11}$ are of the form $p^m\left(-\frac{3}{p}\right)^n$.

To obtain p^5,the following simultaneous equations need to be satisfied:

$m+n=11$ and $m-n=5$.

Solving the equations simultaneously lead to $m=8$ and $n=3$. Since m decreases from 11 to 8 (or n increases from 0 to 3), the expansion leads to the 4th term.

$$T_4 = \frac{11\times10\times9}{3!}p^8\left(-\frac{3}{p}\right)^3 = -4455p^5$$

Therefore the required coefficient is -4455.

Method 2: **Using the kth term**

$$T_k = \frac{n(n-1)(n-2)....[n-(k-2)]}{(k-1)!}a^{n-(k-1)}b^{k-1}$$

From the series $\left(p-\frac{3}{p}\right)^{11}$ we have:

$$T_k = \frac{11\times10\times9\times...[11-(k-2)]}{(k-1)!}p^{11-(k-1)}\left(-\frac{3}{p}\right)^{k-1}$$

$$= \frac{11\times10\times9\times....[11-(k-2)]}{(k-1)!}(-3)^{k-1}p^{11-2k+2}$$

Now equating the exponents of p, we have $11-2k+2=5 \therefore k=4$

$$T_4 = \frac{11\times10\times9}{3!}p^8\left(-\frac{3}{p}\right)^3 = -4455p^5$$

Therefore the required coefficient is -4455.

Method 3: **Counting the terms**

Use the pattern of the terms $p^m\left(-\frac{3}{p}\right)^n$ to establish where their product is p^5

Term	$(p)^m(-\frac{3}{p})^n$	**Resulting power** p^r
T_1	$(p)^{11}(-\frac{3}{p})^0$	p^{11}
T_2	$(p)^{10}(-\frac{3}{p})^1$	p^9
T_3	$(p)^9(-\frac{3}{p})^2$	p^7

$$\therefore T_4 = \frac{11\times10\times9}{3!}p^8\left(-\frac{3}{p}\right)^3 = -4455p^5$$

Hence the required coefficient is $= -4455p^5$

Example 3

Determine the coefficient of p^6 in the expansion of $\left(p+\frac{5}{p^2}\right)^{12}$

Solution: using the kth term

$T_k = \frac{12\times11\times....}{(k-1)!}p^{12-(k-1)}\left(\frac{5}{p^2}\right)^{k-1}$

$\therefore 12-k+1-2k+2=6 \quad \therefore k=3$

$T_3 = \frac{12\times11}{2!}p^{10}\left(\frac{5}{p^2}\right)^2 = 1650p^6$ $\therefore$ the required coefficient is 1650

Now using the method of counting terms, check the table below:

Term	$(p)^m(\frac{5}{p^2})^n$	**Resulting power** p^r
T_1	$(p)^{12}(\frac{5}{p^2})^0$	p^{12}
T_2	$(p)^{11}(\frac{5}{p^2})^1$	p^9
T_3	$(p)^{10}(\frac{5}{p^2})^2$	p^6

$T_3 = \frac{12\times11}{2!}p^{10}\left(\frac{5}{p^2}\right)^2 = 1650p^6$ $\therefore$ the required coefficient is 1650

Exercise 1.5

1. Expand the following, using the binomial theorem

a) $(2+x)^5$ b) $(2x-3y)^4$ c) $(2x^4-y)^3$

d) $(1-2x)^{\frac{1}{2}}$ Write down only the first four terms.

e) $(3x+5y)^{-2}$ Write down only the first four terms.

2. Write down the term indicated in the binomial expansions of the following:

a) $(1-4x)^7$ 3rd term x^3

b) $(2-x)^{15}$ 12th term

c) $\left(1+\frac{x}{2}\right)^{10}$ 6th term

d) $\left(3-\frac{x}{2}\right)^{8}$ 6th term

e) $(1-2x)^{-2}$ 4th term

f) $\left(-x^2+\frac{y}{3}\right)^{10}$ middle term

3. Expand $(1+x)^{-3}$ in ascending powers of x up to the fourth term, using the binomial theorem.

4. Determine the coefficient of p^6 in $\left(p+\frac{2}{p}\right)^{12}$

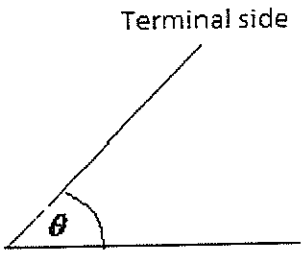
5. Write down the first four terms of the binomial expansion of $\left(2-\frac{x}{2}\right)^{-7}$ simplifying all the coefficients and state the limits of x for which the series is valid.

NB!!

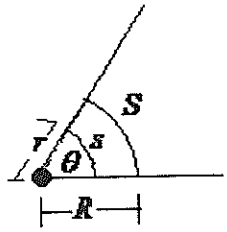
1.6 RADIAN MEASURE OF AN ANGLE

Consider a positive angle θ. There are two ways by which the angle can be measured, as the following diagrams show:

Using a Protractor



Using arclength and radius



s=small arc
r=small radius
S=big arc
R=big radius

$$\theta=\frac{s}{r}=\frac{S}{R}$$

The first way of measuring the angle is quite familiar, while the second one uses the principle that:

$\theta=\frac{arc\ length\ (s)}{radius(r)}$ where θ is the angle subtended by an arc (s) of a circle with radius r.

If the length of the arc is the **same as the radius**, angle θ is said to measure **1 radian**. Note that, strictly speaking, a radian is not a unit of measurement of angle θ,

but an indication of how the angle is measured. The ratio $\theta = \frac{arc\ length}{radius}$ does not have units, since the units of the numerator and denominator cancel out. When using a scientific calculator, a radian measure is a measure of an angle with the calculator in the 'radian mode'. If, for instance, we want $\sin 1$ it is understood to mean we determine $\sin 1$ with the calculator in the *radian* mode. If the calculator is in the *degree* mode, it will give us the answer to $\sin 1°$.

Note that: $\mathbf{\sin 1 = 0,8414709}$ **and** $\mathbf{\sin 1° = 0,017452406}$

1.6.1 Converting angles from degrees to radian measure

Consider the following equation (stated earlier):

$\theta = \frac{arc\ length}{radius}$.

Now for one revolution, $\theta = \frac{2\pi r}{r} = 2\pi$

But to make one revolution we sweep an angle of $360°$

Hence we have $360° = 2\pi$

$\therefore x° = \frac{\pi}{180} \times x\ radians$ and $y\ rad = \frac{180°}{\pi} \times y$

Examples

1. Convert the following angles from degrees to radians.

   a. $90°$ b. $75°$ c. $43°$

Solutions

$a)\ 90° = 90 \times \frac{\pi}{180} = \frac{\pi}{2}$

$b)\ 75° = 75 \times \frac{\pi}{180} = \frac{5\pi}{12}$

$c)\ 43° = 43 \times \frac{\pi}{180} = 0,750491578$

2. Convert the following angles from radians to degrees:

   a. $\frac{\pi}{3}$ b. $\frac{3\pi}{4}$ c. 3.4

Solutions

$a)\ \frac{\pi}{3} = \frac{180°}{\pi} \times \frac{\pi}{3} = 60°$

$b)\ \frac{3\pi}{4} = \frac{180\,°}{\pi} \times \frac{3\pi}{4} = 135\,°$

$c)\ 3.4 = \frac{180\,°}{\pi} \times 3.4 = 184.8057\,°$

Exercises 1.6

1.Express the given angles in radian measure as a multiple of π.

a)$35°$ b) $135°$ c) $225°$ d) $-240°$ e) $120°$ f) $300°$

2. Express the given angles in radians

a) $\frac{2\pi}{3}$ b) $-\frac{\pi}{12}$ c) $\frac{2\pi}{5}$

1.6.2 Applications of radian measure

Note: angles used in this section must be in radians.

a) Arc length

$$\theta = \frac{arc\ length\ (s)}{radius(r)} \quad \text{or} \quad \theta = \frac{s}{r} \rightarrow s = \theta r$$

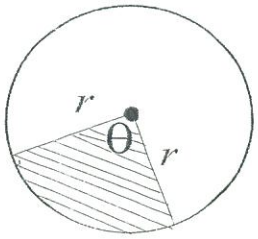
Example

Given the radius of a circle as $3{,}3\ cm$. Determine the length of the arc which subtends an angle $\theta = \frac{\pi}{3}$.

Solution

$$s = \theta r = \frac{\pi}{3} \times 3{,}3\ cm = 3{,}45575\ cm$$

b) Area of a sector (A_{st})

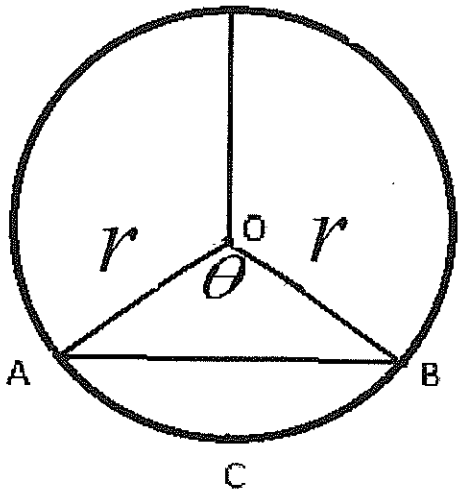


The ratio of the area of a sector to the circle area is the same of the ratio of the angle to 2π, i.e.

$$\frac{A_{st}}{\pi r^2} = \frac{\theta}{2\pi} \therefore A_{st} = \frac{1}{2} r^2 \theta$$

$\frac{A_{st}}{\pi r^2} = \frac{\theta}{2\pi} \therefore A_{st} = \frac{1}{2}r^2\theta$

a) Area of a segment (A_{sm})

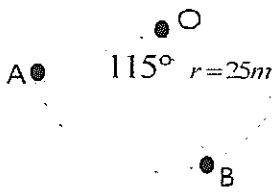


Given: Circle centre O
Radius r
Central angle θ
Segment ABC

$A_{sm} = A_{st} - Area\ of\ \Delta AOB = \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin\theta = \frac{1}{2}r^2(\theta - \sin\theta)$

Examples

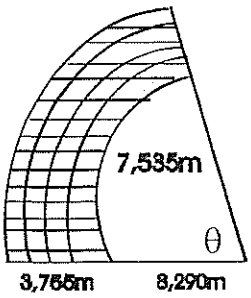
1. A lawn sprinkler can water up to a distance of $25m$. It turns through an angle of $115°$ as shown below. What area can it water?



Solution

$A_{st} = \frac{1}{2}r^2\theta = \frac{1}{2}(25m)^2\left(115 \times \frac{\pi}{180}\right) = 627{,}2277\ m^2$

2. What is the floor area of the hallway shown below? The outside and inside of the hallway are circular arcs.



Solution

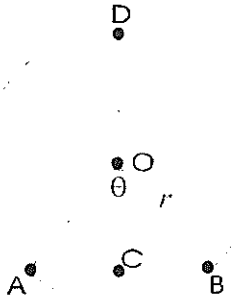
We first determine the angle θ

$\theta = \frac{arc\ length}{radius} = \frac{7{,}535\ m}{8{,}290\ m} = 0{,}908926417$

Note that the radius of inner circle is 8,290 m and radius of outer circle is 12,045 m.

$\therefore$ Floor area $= \frac{1}{2}[(12{,}045m)^2(0{,}908926417) - (8{,}290m)^2(0{,}908926417)] =$
$34{,}701\ m^2$

3. Determine the shaded area:



DC=20cm, AB=12cm
$A\hat{O}B = \theta$

Solution

Considering ΔBOC

$OC = (20 - r)$ and $CB = 6$

Using the Pythagoras theorem, we have

$r^2 = (20 - r)^2 + 6^2$ This simplifies to $r = 10{,}9$

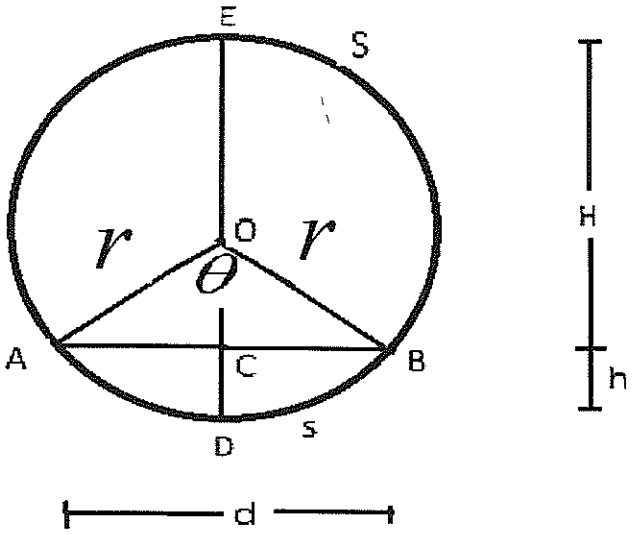
$\therefore \cos\frac{\theta}{2} = \frac{OC}{r} = \frac{9,1}{10,9} = 0,83486$

$\theta = 2\cos^{-1}(0,83486) = 1,16583$

$\therefore$ the shaded area $= \frac{1}{2}(10,9)^2[1,16583 - \sin(1,16583)] = 14,6561cm^2$

Exercise 1.7

Refer to the diagram below for questions 1 and 2



d = length of chord AB
h = length of minor height CD
H = length of major height EC
S = archlength AEB
s = archlength ADB

$\theta = A\hat{O}B$
r = radius of the circle

1. For an arc length S, area of sector A_{st} and central angle θ of a circle of radius r, determine the indicated quantity for the given values:
   a) $s = 1010\ mm,\ \theta = 136,0°$ $\quad r = ?$
   b) $s = 0,3913\ km\ \ r = 0,9449\ km$ $\quad A_{st} = ?$
   c) $r = 4,9\ cm\ \ \theta = 2,443$ $\quad A_{st} = ?$
   d) $\theta = 300°\ \ A_{st} = 0,0119\ m^2$ $\quad r = ?$
   e) $A_{st} = 16,5\ m^2,\ \ r = 4,02\ m$ $\quad s = ?$
2. For an arc length S, area of sector A_{st}, and central angle θ of a circle of radius r, determine the area of the segment (A_{sm}) in each case.
   a) $s = 100\ m,\ \ r = 50\ m$
   b) $d = 60\ m\ \ \theta = 120°$
   c) $A_{st} = 10\ m^2,\ \ r = 3\ m$
   d) $r = 10\ m,\ \ h = 2\ m$

REVISION EXERCISE: UNIT 1

1.Expand completely using the binomial theorem:

a) $(2x-y)^3$ b) $\left(3+\frac{x}{2}\right)^4$ c) $(a-3b)^5$

2. Determine the first four terms in each of the following:

a) $(1-2x)^{-1}$ b)$(3+x)^{-2}$ c) $\sqrt{\left(1-\frac{2x}{3}\right)}$

3. a) Use the binomial theorem to determine the first four terms in $\sqrt{\left(1+\frac{x}{2}\right)}$. Simplify all the terms and hence estimate the value of $\sqrt{1{,}005}$,giving your answer in 4 decimal places.

b) Use the binomial theorem to determine the first four terms in $\sqrt[3]{(1-x)^2}$. Simplify all the terms and hence estimate the value of $\sqrt[3]{(0{,}75)^2}$, giving your answer in 4 decimal places.

4. Write down the term indicated in each of the following series:

a) $(2x+3y)^7$ 4th term

b) $\left(2-\frac{x}{2}\right)^8$ middle term

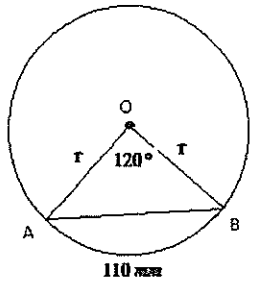
c) $(1-3x)^{-2}$ 5th term

5. Determine the coefficient of p^2 in each of the following:

a) $\left(p+\frac{3}{p}\right)^8$ b) $\left(p-\frac{2}{p^2}\right)^{14}$

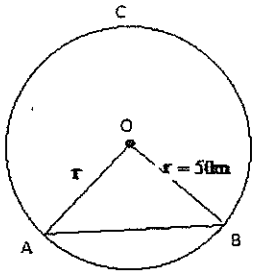
6. Determine the area of the shaded regions

a)

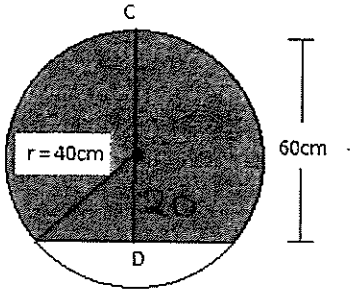


$A\hat{O}B = 120°$, Arclength AB = 110mm

b) Given that triangle ABC is equilateral and the radius is 50m



c) Given $CD = 60\ cm$ and radius= $40\ cm$



UNIT 2

DIFFERENTIATION

2.1 INTRODUCTION

This unit on differential calculus deals with finding the rate of change of one quantity with respect to the other, for example the rate of change of displacement with respect to time or the rate of change of electric charge with respect to time.

After completing this unit the learner should be able to:

- Evaluate limits of functions
- Determine the derivative of a polynomial function from first principles
- Determine the derivatives of a function using basic differentiation rules
- Determine higher order derivatives
- Determine the derivatives of implicit functions
- Differentiate products, quotients and powers of functions using logarithm properties
- Apply differentiation techniques to sketch the graph of a function by determining maxima and minima
- Use differentiation to solve velocity and acceleration problems for linear and circular motions
- Approximate roots of equations using the Newton-Raphson method.

2.2. LIMIT OF A FUNCTION

There are situations where a function is not defined at a particular value of x, say $x = x_0$, however it may be defined for values of x that are very close to x_0.

Let us look at the following functions.

(a) $f(x) = \frac{x^2-1}{x-1}$ (b) $f(x) = \frac{\sin x}{x}$

In (a), the function is not defined at $x = 1$, because at that point the denominator is zero. However we can see that $f(x) = \frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x + 1$ (provided $x \neq 1$). Which means we still cannot permit $x = 1$, but we can say that as the value of x approaches 1, the value of $f(x)$ approaches 2. Clearly the value of $f(x)$ never actually attains the value 2 but it does get as close to it as we select a value of x which is sufficiently close to 1. We then say that the limit of $\frac{x^2-1}{x-1}$ as x approaches 1 is 2, and written as $\lim_{x\to 1} \frac{x^2-1}{x-1} = 2$

Similarly, in (b), $x = 0$ renders the function undefined. However if we select a value of x which is sufficiently close to zero, then the function approaches 1.

This is demonstrated in the table below.

x	$\frac{\sin x}{x}$
1	0,8414709
0,1	0,9983341
0,01	0,9999833
0,001	0,9999998
0,0001	1
0,00001	1
0,0000001	1

As $x \to 0$ (i.e. as n becomes exceedingly small), $\frac{\sin x}{x} \to 1$. We write $\lim_{x \to 0} \frac{\sin x}{x} = 1$.

Note that $\frac{\sin x}{x}$ is not defined at $x = 0$.

a) Evaluating limits using algebra

If f is a real-valued (or complex-valued) function, then taking the limit is compatible with the algebraic operations, provided the limits on the right sides of the equations below exist (the last identity only holds if the denominator is non-zero). This fact is often called the algebraic limit theorem.

$$\begin{aligned}
\lim_{x \to p} (f(x) + g(x)) &= \lim_{x \to p} f(x) + \lim_{x \to p} g(x) \\
\lim_{x \to p} (f(x) - g(x)) &= \lim_{x \to p} f(x) - \lim_{x \to p} g(x) \\
\lim_{x \to p} (f(x) \cdot g(x)) &= \lim_{x \to p} f(x) \cdot \lim_{x \to p} g(x) \\
\lim_{x \to p} (f(x)/g(x)) &= \lim_{x \to p} f(x) / \lim_{x \to p} g(x)
\end{aligned}$$

In each case above, when the limits on the right do not exist, or, in the last case, when the limits in both the numerator and the denominator are zero, nonetheless the limit on the left, called an indeterminate form, may still exist—this depends on the functions f and g.

The following are the steps followed in evaluating limits of functions:

Step 1: If direct substitution of x by the input value p leads to zero in the denominator, simplify the function if possible. If the function cannot be simplified, then the limit does not exist, unless the numerator is also zero, in which case the limit *may* exist, and if it does, can be evaluated using L'Hopital's Rule, which is not going to be covered in this unit.

Step 2: Substitute the input value into the simplified form of the function.

Examples

Determine the following limits:

$1. \lim_{x \to 2}(x-1)$

Solution

The function $f(x) = x - 1$ is in its simplest form. So go straight to step 2:

$$\lim_{x \to 2}(x-1) = (2-1) = 1$$

$2. \lim_{x \to 3} \frac{x^2 - 9}{x - 3}$

Solution

Step 1:

$$\lim_{x \to 3} \frac{x^2 - 9}{x - 3} = \lim_{x \to 3} \frac{(x+3)(x-3)}{x-3} = \lim_{x \to 3}(x+3)$$

Step 2 **:**

$$\lim_{x \to 3}(x+3) = (3+3) = 6$$

$3. \lim_{x \to 2} \sqrt{1-2x}$

Solution

Step 1:

The function $f(x) = \sqrt{1-2x}$ cannot be simplified further.

Step 2 :

Direct substitution leads to a square root of a negative number which is undefined for real numbers. Therefore $\lim_{x\to 2}\sqrt{1-2x}$ does not exist (for a real-valued function).

$$4.\lim_{x\to 3}\frac{x^2-4}{x+2}$$

Solution

Direct substitution does not lead to indeterminate form. So simplifying is not necessary. Step 2 leads to the following solution.

$$\lim_{x\to 3}\frac{x^2-4}{x+2}=\frac{3^2-4}{3+2}=1$$

$$5.\lim_{x\to 0}\frac{\sin x-x^2}{x}=\lim_{x\to 0}(\frac{\sin x}{x}-x)=\lim_{x\to 0}\frac{\sin x}{x}-\lim_{x\to 0}x=1-0=1$$

$$6.\lim_{x\to 0}\frac{\cos x\sin x}{x}=\lim_{x\to 0}\frac{\sin 2x}{2x}=1 \text{ or } \lim_{x\to 0}\frac{\cos x\sin x}{x}=\lim_{x\to 0}\cos x\times\lim_{x\to 0}\frac{\sin x}{x}=\cos 0\times 1=1$$

b) Limits to infinity

The rule for limits to infinity is based on observations demonstrated in the following table:

$\frac{1}{10}=0{,}1$	$\frac{1}{10000}=0{,}0001$
$\frac{1}{100}=0{,}01$	$\frac{1}{10000000}=0{,}0000001$

As the denominator increases, the fraction becomes smaller and smaller. Here we have $\lim_{x\to\infty}\frac{1}{x}=0$

To determine the limit of a rational polynomial at infinity, divide both the numerator and denominator by the highest power in the denominator and then apply the limit algebra.

Examples

1. $\lim_{x\to\infty}\dfrac{x^2-x+3}{x^3-5}=\lim_{x\to\infty}\dfrac{\dfrac{x^2}{x^3}-\dfrac{x}{x^3}+\dfrac{3}{x^3}}{\dfrac{x^3}{x^3}-\dfrac{5}{x^3}}=\lim_{x\to\infty}\dfrac{\dfrac{1}{x}-\dfrac{1}{x^2}+\dfrac{3}{x^3}}{1-\dfrac{5}{x^3}}=\dfrac{0-0+0}{1-0}=0$

2. $\lim_{x\to\infty}\dfrac{e^{-x}-x^2+4x}{x^2-5}=\lim_{x\to\infty}\dfrac{\dfrac{1}{e^x x^2}-\dfrac{x^2}{x^2}+\dfrac{4x}{x^2}}{\dfrac{x^2}{x^2}-\dfrac{5}{x^2}}=\lim_{x\to\infty}\dfrac{\dfrac{1}{e^x}\times\dfrac{1}{x}-\dfrac{x^2}{x^2}+\dfrac{4x}{x^2}}{1-\dfrac{5}{x^3}}=\dfrac{0\times 0-1+0}{1-0}=1$

Exercises 2.1

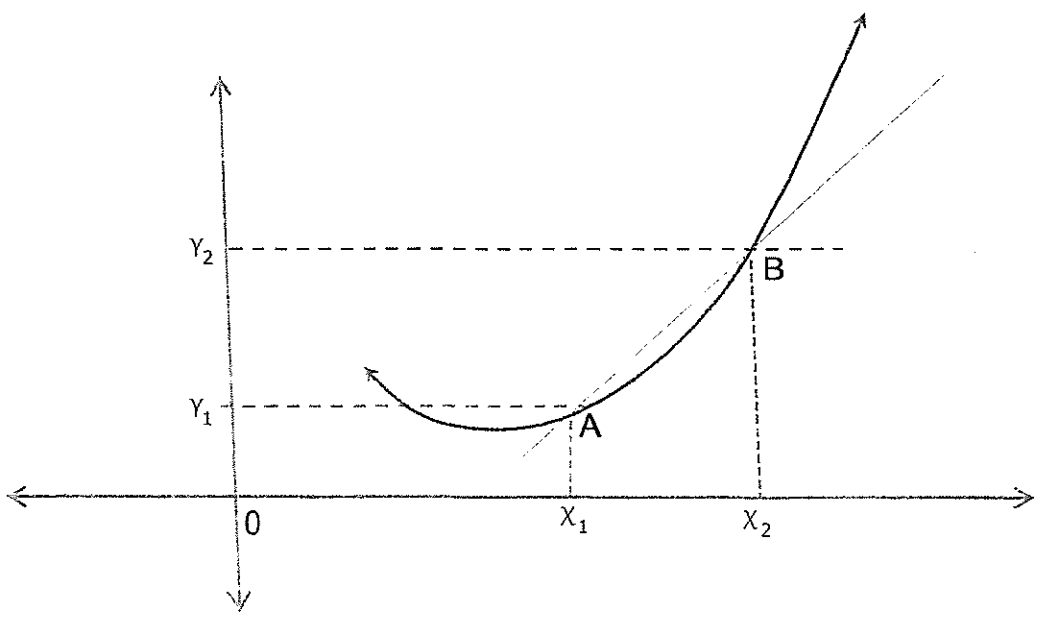
1. Determine the following limits if they exist:

a) $\lim_{x\to 2}\dfrac{1}{x-5}$ b) $\lim_{x\to\infty}\dfrac{2x^2-3}{3x^2+x+5}$ c) $\lim_{x\to\frac{1}{2}}\dfrac{3x-2}{2x-3}$ $d)\lim_{x\to -1}\dfrac{x^3+1}{x+1}$

$e)\lim_{x\to 5}\dfrac{\sqrt{x+4}-3}{x-5}$

2.3. DIFFERENTIATION FROM FIRST PRINCIPLES

Remember, the gradient of a line joining two points $A(x_1,y_1)$ *and* $B(x_2,y_2)$ on a curve $y=f(x)$ is given as $m_{AB}=\frac{y_1-y_2}{x_1-x_2}$



If we allow the point B to be closer to A, the slope of AB will be more closely approximate the slope of a tangent drawn to the curve at A. *The slope of the tangent, often referred to as the slope of the curve, is the limiting value of the slope of the secant line AB as B approaches A.*

Hence we could find the slope of a tangent to a curve at a point $(x_1, f(x_1))$ by calculating the limit (provided it exists) of the difference $f(x_1 + h) - f(x_1)$ divided by h as h→ 0. We can write this as $m_{tan} = \lim_{h\to 0} \frac{f(x_1+h)-f(x_1)}{h}$

The limit on the right is defined as the derivative of $f(x)$ at x_1.

From the above we have the **derivative** of the function $f(x)$ as:

$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ provided the limit exists.

This is the formula for determining the derivative of a function from first principles.

Example

1. Determine the derivative of each of the following functions from first principles:

   a) $f(x) = -3x^2$ b) $f(x) = \sqrt{x}$

Solution:

a) $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{-3(x+h)^2+3x^2}{h} = \lim_{h\to 0} \frac{-3x^2-6xh-3h^2+3x^2}{h}$

$= \lim_{h\to 0} \frac{h(-6x-3h)}{h} = -6x$

b) $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h} = \lim_{h\to 0} \frac{\sqrt{x+h}-\sqrt{x}}{h} \cdot \left(\frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}}\right)$

$$= \lim_{h\to 0} \frac{(x+h)-x}{h\left(\sqrt{x+h}+\sqrt{x}\right)} = \lim_{h\to 0} \frac{1}{\sqrt{x+h}+\sqrt{x}} = \frac{1}{2\sqrt{x}}$$

Exercise 2.2

Determine the derivatives of the following functions from first principles:

a) $f(x) = -5x^3$ b) $f(x) = \frac{1}{x}$ c) $f(x) = 2\sqrt{x}$ d) $f(x) = x^2 + x$

e) $f(x) = \frac{-2}{x^2}$

Notation:

The derivative of a function $y = f(x)$ can be represented as $y', f'(x), \frac{dy}{dx}$, Dy or $Df(x)$

2.4. BASIC DIFFERENTIATION RULES

2.4.1. Derivatives of polynomials

If $y = k$, where k is any constant, then $\frac{dy}{dx} = 0$

If $y = kx$, where k is a constant, then $\frac{dy}{dx} = k$

For example if $y = 4$, then $\frac{dy}{dx} = 0$ also if $y = -9x$, then $\frac{dy}{dx} = -9$

If $y = x^n$ where n is any real number, positive, negative or fraction,
Then $\frac{dy}{dx} = nx^{n-1}$
Thus the general rule for differentiating x^n is to multiply the term by the index or exponent and decrease the index or exponent by one.

2.4.2. Derivative of a sum or difference

The derivative of a sum or difference of functions is the sum or difference of the derivatives of functions. The sum or difference is thus differentiated term by term. i.e.

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}[f(x)] \pm \frac{d}{dx}[g(x)]$$

That is, if $y = 3x^2 - \frac{1}{x} + 4x$, then $\frac{dy}{dx} = \frac{d}{dx}(3x^2) - \frac{d}{dx}(x^{-1}) + \frac{d}{dx}(4x) = 6x + \frac{1}{x^2} + 4$

Example

Differentiate with respect to x

1) $y = 4x^3 - 3x + 5$ 2) $y = \frac{2}{3}x^3 - \frac{2}{x^2}$ 3) $y = -3x^4 - \frac{1}{x} - 3x$

Solution

1) $y = 4x^3 - 3x + 5$

$\frac{dy}{dx} = 12x^2 - 3$

2) $y = \frac{2}{3}x^3 - \frac{2}{x^2} = \frac{2}{3}x^3 - 2x^{-2}$

$\frac{dy}{dx} = 2x^2 + \frac{4}{x^3}$

3) $y = -3x^4 - \frac{1}{x} - 3x$

$\frac{dy}{dx} = -12x^3 + \frac{1}{x^2} - 3$

Exercise 2.3

3. Determine the derivatives of each of the following:

a) $y = 3x^4 - x^3$

b) $y = \frac{2}{\sqrt{x}} - x^{-1} + 2x - 5$

c) $y = 2\sqrt{x} + x^{-2} - \frac{5}{x}$

d) $f(x) = \pi x^2 + x^{\pi} - \pi + \frac{\pi}{x}$

e) $D_x[(1-x)^2]$

f) $f(x) = \sqrt{x^3} + \frac{2}{x^4}$

2.4.3. Product rule

If $y = u(x)v(x)$ where both u and v are differentiable, then by the product rule,
$\frac{dy}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$

The derivatives of functions such as $2x^4(x-2)$ can be obtained using the product rule.

Examples

1. Determine $\frac{dy}{dx}$ in each of the following:

a) $y = (x^2+4)(2x-3)$

b) $y = (x^2-1)(3x-2)$

Solution

a) $y = (x^2+4)(2x-3)$

Let $u = x^2 + 4 \quad u' = 2x$

$v = 2x - 3 \quad v' = 2$

$\frac{dy}{dx} = (2x-3)(2x) + (x^2+4)(2)$

$= 4x^2 - 6x + 2x^2 + 8$

$= 6x^2 - 6x + 8$

b) $y = (x^2-1)(3x-2)$

Let $u = x^2 - 1 \quad u' = 2x$

$v = 3x - 2 \quad v' = 3$

$$\frac{dy}{dx} = (3x-2)(2x) + (x^2-1)(3)$$
$$= 6x^2 - 4x + 3x^2 - 3$$
$$= 9x^2 - 4x - 3$$

Exercise 2.4

4. Determine the derivative of each of the following:

a) $y = (x-2)(1-2x^2)$
b) $y = x^{-1}(x^2-1)$
c) $y = x^2(1-2x)$
d) $f(x) = (2+2\sqrt{x})(2x^2-1)$
e) $f(x) = (3-2\sqrt{x})(3+2\sqrt{x})$

2.4.4. Quotient rule

The quotient rule is applied to functions of the type $y = \frac{u(x)}{v(x)}$ where both functions are differentiable and $v(x) \neq 0$

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2} \qquad [Quotient\ Rule]$$

The derivatives of functions such as $\frac{x-5}{(x^2-1)}$ or $\frac{x+1}{x^2+1}$ can be obtained by using the quotient rule.

Examples

1. Determine the derivatives of:

a) $y = \frac{1-x^2}{1+x^2}$ b) $y = \frac{1+x}{1-2x^2}$ c) $y = \frac{x-1}{1+x}$

Solution

a) $y = \frac{1-x^2}{1+x^2}$ Let $u = 1 - x^2$, then $\frac{du}{dx} = -2x$ and $v = 1 + x^2$ then $\frac{dv}{dx} = 2x$

$$\frac{dy}{dx} = \frac{(1+x^2)(-2x)-(1-x^2)(2x)}{(1+x^2)^2} = \frac{-4x}{(1+x^2)^2}$$

b) $y = \frac{1+x}{1-2x^2}$ Let $= 1 + x$,then $\frac{du}{dx} = 1$ and let $v = 1 - 2x^2$ then $\frac{dv}{dx} = -4x$

$$\frac{dy}{dx} = \frac{(1-2x^2)(1)-(1+x)(-4x)}{(1-2x^2)^2} = \frac{2x^2+4x+1}{(1-2x^2)^2}$$

c) $y = \frac{x-1}{1+x}$ Let $u = x - 1$,then $\frac{du}{dx} = 1$ and let $v = 1 + x$ then $\frac{dv}{dx} = 1$

$$\frac{dy}{dx} = \frac{(1+x)(1)-(x-1)(1)}{(x+1)^2} = \frac{-2}{(x+1)^2}$$

Exercise 2.5

1. Find the derivative of each of the following:

a) $y = \frac{3x+1}{3x-1}$ d) $y = \frac{x^2+1}{2x-1}$

b) $y = \frac{x^2-5}{x+1}$ e) $y = \frac{3x^2-1}{x^3-1}$

c) $y = \frac{1-2x}{x-3}$ f) $f(x) = \frac{2+2\sqrt{x}}{2x^2-1}$

2.4.5. The chain rule

The chain rule states that if y is a function of u and u is a function of x then

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

The chain rule is also known as the **function of a function** rule.

$(a-bx)^n$

Example

1. Use the chain rule to differentiate with respect to x

a) $y = (5x+2)^3$ b) $y = (3-5x)^{-3}$ c) $y = (4x^3-5x)^5$
d) $\sqrt{x-4x^2}$

$-3(3-5x)^{-4}(-5)$

Solution

a) $y = (5x+2)^3$ Let $u = 5x+2$ =>$y = u^3$

$n(a-bx)^{n-1}(-b)$

$\frac{du}{dx} = 5$ and $\frac{dy}{du} = 3u^2$

$\frac{dy}{dx} = \frac{dy}{du}.\frac{du}{dx} = 15u^2 = 15(5x+2)^2$

where

b) $y = (3-5x)^{-3}$ Let $u = 3-5x$ then $y = u^{-3}$

$\frac{du}{dx} = -5$ and $\frac{dy}{du} = -3u^{-4}$

$\frac{dy}{dx} = \frac{dy}{du}.\frac{du}{dx} = 15u^{-4} = 15(3-5x)^{-4}$

c) $y = (4x^3-5x)^5$ Let $u = 4x^3-5x$ then $y = u^5$

$\frac{du}{dx} = 12x^2 - 5$ and $\frac{dy}{du} = 5u^4$

$\frac{dy}{dx} = \frac{dy}{du}.\frac{du}{dx} = 5(12x^2 - 5)u^4 = 5(12x^2 - 5)(4x^3 - 5x)^4$

d) $f(x) = \sqrt{x - 4x^2}$ Let $u = x - 4x^2$ then $y = u^{\frac{1}{2}}$

$\frac{du}{dx} = 1 - 8x$ and $\frac{dy}{du} = \frac{1}{2}u^{\frac{-1}{2}}$

$\frac{dy}{dx} = \frac{dy}{du}.\frac{du}{dx} = \frac{1}{2}(1 - 8x)u^{\frac{-1}{2}} = \frac{1}{2}(1 - 8x)(x - 4x^2)^{\frac{-1}{2}} = \frac{1-8x}{2\sqrt{x-4x^2}}$

Exercise 2.6

1. Differentiate each of the following:

a) $y = (3x^2 - 5)^4$

b) $y = (3 - x^2)^{-3}$

c) $y = (\sqrt{x} + 1)^5$

d) $y = \sqrt{3x^2 - 1}$

e) $y = \frac{1}{5x+2}$

2.5. DERIVATIVES OF TRIGONOMETRIC FUNCTIONS

NOTE: $\lim_{h\to 0} \frac{\sin h}{h} = 1$ and $\lim_{h\to 0} \frac{1-\cos h}{h} = 0$

If $f(x) = \sin x$

Then $f'(x) = \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h}$

$$\boldsymbol{f'(x)} = \lim_{h\to 0} \frac{\sin x\cos h + \cos x\sin h - \sin x}{h}$$

$$= \lim_{h\to 0} \frac{\sin x(\cos h - 1) + \cos x\sin h}{h}$$

$$= \lim_{h\to 0} \left[-\sin x \frac{1-\cos h}{h} + \cos x \frac{\sin h}{h}\right]$$

$$= (-\sin x)(0) + (\cos x)(1) = \cos x$$

Similarly, using $\cos(x + h) = \cos x\cos h - \sin x\sin h$, we may find the derivative of $y = \cos x$.

If $y = \cos x$, then $\frac{dy}{dx} = -\sin x$

You will not be required to find the derivatives of $\sin x$ and $\cos x$ from first principles.

If $y = \sin ax$, then $\frac{dy}{dx} = a\cos ax$, and if $y = \cos ax$, then $\frac{dy}{dx} = -a\sin ax$

In general, if $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x)\cos f(x)$, using the chain rule.

For example, $y = \sin(3x - 1)$

Let $u = 3x - 1$ then $y = \sin u$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$\frac{du}{dx} = 3$ and $\frac{dy}{du} = \cos u$

then $\frac{dy}{dx} = 3\cos(3x - 1)$

Example

1. Determine the derivatives of the following:

a) $y = \sin 3x + \cos 3x$ b) $y = 2\cos(3x^2 - 1)$
c) $y = \sin(\sqrt{x}) - 2\cos(x^2 - 3)$

Solution

a) $y = \sin 3x + \cos 3x$

$\frac{dy}{dx} = 3\cos 3x - 3\sin 3x$

b) $y = 2\cos(3x^2 - 1)$

$\frac{dy}{dx} = -12x\sin(3x^2 - 1)$

c) $y = \sin(\sqrt{x}) - 2\cos(x^2 - 3)$

$\frac{dy}{dx} = \frac{1}{2\sqrt{x}}\cos(\sqrt{x}) + 4x\sin(x^2 - 3)$

Exercise 2.7 NBM. check up!

Determine the derivative of each of the following:

a) $y = x^3\cos 2x$ b) $y = \sin^2 x$ c) $y = \sin x^2 - \cos(x^2 + 2)$ d) $y = \frac{\cos x}{1 - \sin x}$

e) $y = \tan x = \frac{\sin x}{\cos x}$

2.6. DERIVATIVES OF EXPONENTIAL FUNCTIONS

If $y = e^x$ then $\frac{dy}{dx} = e^x$

If $y = e^{ax}$ (where a is any constant) then $\frac{dy}{dx} = ae^{ax}$

In general, if $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = a^x$ (where a is any constant) then $\frac{dy}{dx} = a^x \ln a$

In general, if $y = a^{f(x)}$, then $\frac{dy}{dx} = f'(x)a^{f(x)} \ln a$

Examples

Determine $\frac{dy}{dx}$

1. $y = e^x - e^{3x-1} + e^{0.5x}$ $\qquad \frac{dy}{dx} = e^x - 3e^{3x-1} + \frac{1}{2}e^{0.5x}$

2. $y = \frac{1}{3}e^{2x} + e^{2\pi x} - e^{\sin x}$ $\qquad \frac{dy}{dx} = \frac{2}{3}e^{2x} + 2\pi e^{2\pi x} - \cos x\, e^{\sin x}$

3. $y = 5^{2x}$ $\qquad \frac{dy}{dx} = 2(5^{2x}) \ln 5$

4. $y = 3^{x^2}$ $\qquad \frac{dy}{dx} = 2x(3^{x^2}) \ln 3$

5. $y = 2^{5x+1}$ $\qquad \frac{dy}{dx} = 5(2^{5x+1}) \ln 2$

Exercise 2.8

Obtain $\frac{dy}{dx}$ in each of the following:

1. $y = e^{-3x} + 2e^{0.5x}$ 2. $y = e^{x^2} + \frac{2}{3}e^{-3x} - e^{2ax}$ 3. $y = 3^{2x-1} + 5^{3ax}$

4. $y = 7^{x^2} + 5^{\sqrt{x}}$ 5. $y = 3^{\cos 2x}$

2.7. DERIVATIVES OF LOGARITHMIC FUNCTIONS

If $y = \ln x$, then $\frac{dy}{dx} = \frac{1}{x}$

If $y = \log_a x$, then $\frac{dy}{dx} = \frac{1}{x \ln a}$

In general,

If $y = \ln[f(x)]$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$, and

If $y = \log_a f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x) \ln a}$

Example

1. Determine the derivatives of :

a) $y = \log_2 x$ b) $y = ln(2x^2 + 3)$ c) $y = \log(\sin x)$

Solution

a) $y = \log_2 x$

$\frac{dy}{dx} = \frac{1}{xln2}$

b) $y = ln(2x^2 + 3)$

$\frac{dy}{dx} = \frac{4x}{2x^2+3}$

c) $y = \log(\sin x)$

$\frac{dy}{dx} = \frac{\cos x}{\sin x(\ln 10)} = \frac{\cot x}{\ln 10}$

Exercise 2.9

Determine the derivatives of each of the following:

a) $y = \log_3(x^2 + 5)$ b) $y = x^2 \log_5 3x^2$ c) $y = 3x \log(1 - x)$

d) $f(x) = x^3 \ln(5x - 1)$ e) $f(x) = \sin 2\pi x \ln(x^2 + 5)$

Differentiation Rules

The table below gives the differentiation rules of the functions covered

$y = f(x)$	$\frac{dy}{dx}$
k (any constant)	0
ax^n	nax^{n-1}
$e^{f(x)}$	$f'(x)e^{f(x)}$
$a^{f(x)}$	$f'(x)a^{f(x)} \ln a$
$\ln\lvert f(x)\rvert$	$\frac{f'(x)}{f(x)} [f(x) > 0]$
$\log_a f(x)$	$\frac{f'(x)}{f(x) \ln a}$
$[f(x)]^n$	$n[f(x)]^{n-1} f'(x)$
$\sin f(x)$	$f'(x) \cos f(x)$
$\cos f(x)$	$-f'(x) \sin f(x)$
$\tan f(x)$	$f'(x)sec^2 f(x)$
$cosecf(x)$	$-f'(x)cosecf(x)cotf(x)$
$secf(x)$	$f'(x)secf(x)tanf(x)$
$cotf(x)$	$-f'(x)cosec^2 f(x)$

NB!!

2.8. HIGHER ORDER DERIVATIVES

If $y = f(x)$, then $\frac{dy}{dx}$ will be a function of x. The result of differentiating $\frac{dy}{dx}$ with respect to x is called the second derivative of y. This is denoted by $\frac{d^2y}{dx^2}$.

Example

1.Let $y = \frac{1}{2}x^4 - 3x^2 + 1$

$\frac{dy}{dx} = 2x^3 - 6x$

$\frac{d^2y}{dx^2} = 6x^2 - 6$ and $\frac{d^3y}{dx^3} = 12x$

2. Determine $\frac{d^2R}{dt^2}$ if $R(t) = \frac{t+1}{t}$

Solution:

$R(t) = \frac{t+1}{t} = 1 + \frac{1}{t} = 1 + t^{-1}$

$\therefore \frac{dR}{dt} = -t^{-2}$ and $\frac{d^2R}{dt^2} = 2t^{-3}$

Exercise 2.10

1. Determine $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ if:

a) $y = 2x^{-3} - 3x^2 + 5x$
b) $y = cos3x - sin3x$
c) $y = e^{2x}\sin(2\pi x)$
d) $y = 2e^x - \ln x$
e) $y = ax^2 - be^x - c\ln x$
f) $s(t) = 2^t - 3t + \log_2 t$

2. The displacement, x, of a particle is given by $x = te^{-t}$. Determine the velocity, $v = \frac{dx}{dt}$ and the acceleration, $a = \frac{d^2x}{dt^2}$.

2.9. DERIVATIVES OF IMPLICIT FUNCTIONS

An implicit function in two variables x and y, is one in which neither variable can be easily expressed in terms of the other. For example, the equation $2x^2 + y^3 - 5xy = 12$ is an implicit function since y cannot be easily isolated on one side of the equation. The method of finding the derivative of implicit functions is called implicit differentiation.

To differentiate such an implicit function:

1. Differentiate it term by term to give an equation in $\frac{dy}{dx}$, x and y;
2. Make $\frac{dy}{dx}$ the subject of the equation;
3. It is important to remember that:

$$\frac{d}{dx}[y] = \frac{dy}{dx}\ ;\ \frac{d}{dx}[y^2] = 2y\frac{dy}{dx}\ ;\ \frac{d}{dx}[2y^3] = 6y^2\frac{dy}{dx}\,[by\ chain\ rule]$$

Or in general $\frac{d}{dx}[g(y)] = g'(y)\frac{dy}{dx}$ where $g'(y)$ is the derivative of $g(y)$ with respect to y.

Example

1. Determine $\frac{dy}{dx}$, if:

a) $2x^2 + y^3 - 5xy = 12$ b) $x^2y^2 + 4xy - x\ln 3 = e^2$

Solution

a) $2x^2 + y^3 - 5xy = 12$

$$4x + 3y^2\frac{dy}{dx} - 5y - 5x\frac{dy}{dx} = 0$$

$$\frac{dy}{dx}(3y^2 - 5x) = 5y - 4x$$

$$\frac{dy}{dx} = \frac{5y-4x}{3y^2-5x}$$

b) $x^2y^2 + 4xy - x\ln 3 = e^2$

$$2xy^2 + 2x^2y\frac{dy}{dx} + 4y + 4x\frac{dy}{dx} - \ln 3 = 0$$

$$\frac{dy}{dx}(2x^2y + 4x) = \ln 3 - 4y - 2xy^2$$

$$\frac{dy}{dx} = \frac{\ln 3-4y-2xy^3}{2x^2y+4x}$$

Exercise 2.11

Find $\frac{dy}{dx}$ in each of the following

$a)\ x^3 + y^3 = 4$ $b)\ (x-y)^2 = x + y - 1$ $c)\ y = \sin(3x + 4y)$

$d)\ e^{xy} = e^{4x} - e^{5y}$ e) $x^2 - 2x^3y + y^2 = \ln 5$

1. Evaluate $\frac{dy}{dx}$ at $(3\,;1)$, given that $x^2 - xy + 2y^2 = 9$

2. Given the function $e^{sinx} + e^{siny} = 2$, evaluate $\frac{dy}{dx}$ at the point $(0\,,\pi)$.

2.10. LOGARITHMIC DIFFERENTIATION

Logarithmic differentiation is a means of differentiating algebraically complicated functions or functions for which the ordinary rules of differentiation do not apply, for example $y = x^{(2x+3)}$, $y = a^{f(x)}$, $y = \frac{f(x)g(x)}{r(x)s(x)}$

In order to find the derivative of any of such functions, it is usually advisable to take the logarithm of both sides of the function first and then differentiate implicitly. This process is known as Logarithmic differentiation.

Examples

1.Determine $\frac{dy}{dx}$ in each of the following:

a) $y = 5^{2x}$ b) $y = \frac{x^5 e^{3x}}{(1-5x)\sqrt{x^2+2}}$ c) $y = (3x+2)^{\ln x}$

Solution

$a)\ y = 5^{2x}$

$\ln y = 2x \ln 5$

$\frac{1}{y}\frac{dy}{dx} = 2\ln 5$

$\frac{dy}{dx} = 2y \ln 5$

But $y = 5^{2x}$

$\therefore \frac{dy}{dx} = 2 \cdot 5^{2x} \cdot \ln 5$ $[do\ not\ multiply\ 2\ by\ 5\ since\ \ 2 \cdot 5^{2x} \neq 10^{2x}]$

$b)\ y = \frac{x^5 e^{3x}}{(1-5x)\sqrt{x^2+2}}$ $[take\ \boldsymbol{ln}\ on\ both\ sides]$

$\ln y = \ln \frac{x^5 e^{3x}}{(1-5x)\sqrt{x^2+2}}$

$= 5\ln x + 3x \ln e - \ln(1-5x) - \frac{1}{2}\ln(x^2+2)$

$\frac{1}{y}\frac{dy}{dx} = \frac{5}{x} + 3 + \frac{5}{(1-5x)} - \frac{x}{(x^2+2)}$

$\frac{dy}{dx} = y\left[\frac{5}{x} + 3 + \frac{5}{(1-5x)} - \frac{x}{(x^2+2)}\right]$

$c)\ y = (3x+2)^{\ln x}$

$\ln y = \ln(3x+2)^{\ln x}$

$\ln y = (\ln x)\ln(3x+2)$

$$\frac{1}{y}\frac{dy}{dx} = \frac{\ln(3x+2)}{x} + \frac{3\ln x}{(3x+2)}$$

$$\frac{dy}{dx} = y\left[\frac{\ln(3x+2)}{x} + \frac{3\ln x}{(3x+2)}\right]$$

Exercise 2.12

1.Find the derivatives of the following using logarithmic differentiation

$a)\ y = 5^{x^2}$ $\quad b)\ y = e^{\sin x}$ $\quad c)\ y = \ln\frac{x(x^2+1)^2}{\sqrt{2x^3-1}}$

$d)\ y = \frac{e^{3x}sin^4x}{x^2\sqrt{x^3-1}}$ $\quad e)\ y = (x^2-3)^{\cos x}$

2.11. APPLICATIONS OF DIFFERENTIATION

In this section we will look at some problems to which the techniques of differentiation can be applied. The following concepts will be dealt with in this section: *Curve sketching, velocity and acceleration and finding roots of equations.*

2.11.1. Curve sketching (polynomials and rational functions only)

When sketching a polynomial, first determine the turning points and distinguish between these turning points (i.e. state whether they are maximum or minimum points).

A turning point is a point on a curve where $\frac{dy}{dx} = 0$

Procedure for finding and distinguishing turning points

i) Given $y = f(x)$, determine the first and second derivatives (i.e $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$)

ii) Solve $\frac{dy}{dx} = 0$ for values of$x$

iii) Substitute the x values found in ii) into the original equation to determine the corresponding y coordinates.

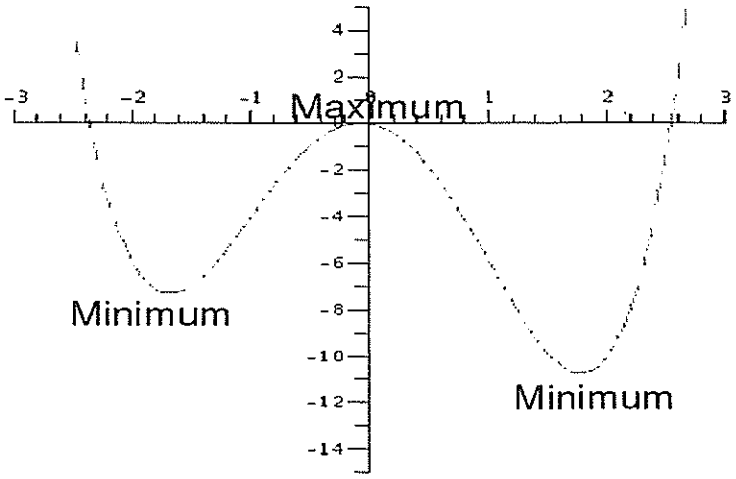
iv) Use the second derivative to distinguish between the turning points.

Substitute the values of x found in ii) above into $\frac{d^2y}{dx^2}$

If $\frac{d^2y}{dx^2} > 0$ (i.e. positive) then the point $(x, f(x))$ is a minimum point.

If $\frac{d^2y}{dx^2} < 0$ (i.e. negative) then the point $(x, f(x))$ is a maximum point.

In the sketch below, the maximum and the minimum points are shown



Example

1. A curve is defined by the function $y = x^3 - 6x^2 - 15x - 1.$

a) Determine the turning points and distinguish between them (i.e whether maximum or minimum).

b) Sketch the graph of the function.

Solution

$y = x^3 - 6x^2 - 15x - 1.$

$\frac{dy}{dx} = 3x^2 - 12x - 15$

At the turning point $\frac{dy}{dx} = 0 \;\; => 3x^2 - 12x - 15 = 0$

$=> x^2 - 4x - 5 = 0 \;\; => (x-5)(x+1) = 0 \therefore x = 5 \;\; or \;\; x = -1$

Therefore the turning points occur at $x = 5 \;\; and \; x = -1$

When $x = 5$, $y = 5^3 - 6(5)^2 - 15(5) - 1 = -101$

When $x = -1, y = (-1)^3 - 6(-1)^2 - 15(-1) - 1 = 7$

The second derivative $\frac{d^2y}{dx^2} = 6x - 12$

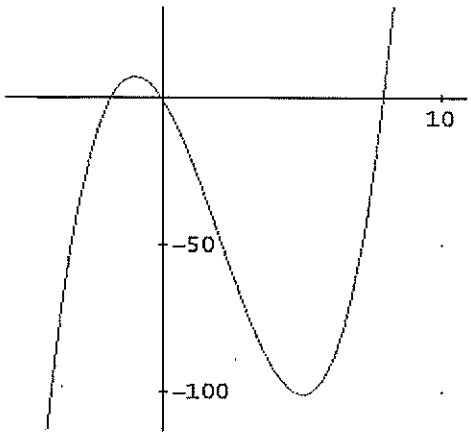
To distinguish between maximum and minimum turning points, we substitute

$x = 5\ and\ \ x = -1$ into the second derivative.

At $x = 5$, $\frac{d^2y}{dx^2} = 6(5) - 12 = 18 > 0$. Thus the function has a minimum turning point at $x = 5$. Hence minimum point is $(5\,; -101)[sometimes\ referred\ to\ as\ local\ minimum]$

At $= -1, \frac{d^2y}{dx^2} = 6(-1) - 12 = -18 < 0$. Thus the function has a maximum turning point at $x = -1$. Hence maximum point is $(-1\,; 7)[sometimes\ referred\ to\ as\ local\ maximum]$

Sketch



Exercise 2.13

1.Let $f(x) = \frac{1}{2}x^4 - x^2 + 2$

a) Determine the turning points of the function f

b) Use the second derivative to determine the minimum and the maximum turning points.

c) Sketch the graph of the function f

2.11.2. Velocity and acceleration (linear and circular motion)

Consider a particle in motion such that its displacement from a fixed point is s metres after time t seconds, then:

a) Displacement , $s = f(t)$

b) Velocity, $v = \frac{ds}{dt}$

c) Acceleration, $a = \frac{dv}{dt}$

Note: a) if $v = 0$, then the particle is at rest or momentarily at rest.

If $v < 0$ (or negative), then the particle is moving in the opposite direction.

If $a = 0$, then the velocity of the particle is constant.

If $a > 0$ (or positive), then the particle is accelerating.

If $a < 0$ (or negative), then the particle is retarding or decelerating.

For example, if $s = 5t^2 + 2t - 4$, $v = \frac{ds}{dt} = 10t + 2$ and $a = \frac{dv}{dt} = 10$

Thus given an expression for displacement s in terms of t, we can determine the velocity, v by differentiating the expression for s with respect to t and if we know the velocity v, we can find the acceleration, a.by differentiating the expression for v with respect to t.

Examples

1. The displacement, s, of a particle after t seconds, is given by $s(t) = t^3 - 2t^2 + t + 10$. Determine

a) Its velocity after 2 seconds

b) Its acceleration

c) When the particle is momentarily at rest.

2. An object projected vertically upwards satisfies the relation $h = 27t - 3t^2$, where h is the height after t seconds.

a) Find the time it takes to reach the highest point

b) How high does it go?

3. The displacement, s, of a particle is given by $s = (1 - t)e^{-t}$

a) Find the time t for the velocity $v = 0$

b) Find the time t for the acceleration $a = 0$

Solution

1. $s(t) = t^3 - 2t^2 + t + 10.$

a) $v = \frac{ds}{dt} = 3t^2 - 4t + 1$

When $t = 2$, $v = 3(2)^2 - 4(2) + 1 = 5$

b) $a = \frac{dv}{dt} = 6t - 4$

c) The particle is momentarily at rest when $v = 0$,

Thus $3t^2 - 4t + 1 = 0$

$(3t - 1)(t - 1) = 0$

$t = \frac{1}{3}$ or $t = 1$

The particle is momentarily at rest when $t = \frac{1}{3} or\ t = 1\ sec$

2. $h = 27t - 3t^2$,

a) at the highest point $v = \frac{dh}{dt} = 0$

$\frac{dh}{dt} = 27 - 6t$

$27 - 6t = 0 \therefore t = 4.5\ secs$

b) the height h at $t = 4.5$ is

$h = 27(4.5) - 3(4.5)^2 = 60.75$

3. $s = (1 - t)e^{-t}$

a) $v = \frac{ds}{dt} = -(1 - t)e^{-t} + e^{-t}(-1)$

$= e^{-t}(t - 2)$

For $v = 0$

$e^{-t}(t - 2) = 0 \therefore t = 2$

b) $a = \frac{dv}{dt} = -(t - 2)e^{-t} + e^{-t}$

$= e^{-t}(3 - t)$

For $a = 0$

$e^{-t}(3 - t) = 0 \therefore t = 3$

Exercise 2.14

1. A projectile is fired vertically upwards with height h given by $h = 25t - 4.9t^2$ where $0 \leq t \leq 5$

a) Find an expression for the velocity and acceleration.

b) Determine the maximum height reached by the projectile.

2. The displacement, θ, of a flywheel at time t is given by $\theta = t^3 - 80\pi t$

Determine the velocity, ω, and the acceleration, α, at $t = 1$

3. The displacement, θ, of a particle in in circular motion is given by

$$\theta = \frac{1}{4}\sin\left(2t + \frac{\pi}{6}\right) (0 \leq t \leq \pi)$$

a) Find an expression for acceleration, α.

b) Determine the first time, t, for $\alpha = 0$

2.11.3. Approximation of roots of equations: Newton-Raphson method

The Newton-Raphson method is a procedure used to find a root of non-linear equations of the form $f(x) = 0$

The first approximation, x_0, to the solution of the equation $f(x) = 0$ may be found by graphing or just guessing.

This initial approximation, x_0, is used to get a second, the second to get the third, and so on, using the formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \text{ where } f'(x_n) \neq 0 n = 0, 1, 2, \cdots$$

You need to know the above formula. Its application to solving equations of the form $f(x) = 0$, as demonstrated below, is called the **Newton Raphson method.** This iterative process follows a set guideline to approximate one root considering the function, its derivative and an initial x-value.

Example

1. Use Newton Raphson method to find a root of the equation $x^3 - x - 1 = 0$ which is closest to 1.
2. Solve $\cos x - 2x = 0$ to 5 decimal places, with an initial value $x_0 = 0.5$

Solution

1. In the above $f(x) = x^3 - x - 1$, so $f'(x) = 3x^2 - 1$ and the initial value $x_0 = 1$

Using the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 1 - \frac{1^3 - 1 - 1}{3(1)^2 - 1} = 1.5$$

$$x_2 = 1.5 - \frac{(1.5)^3 - 1.5 - 1}{3(1.5)^2 - 1} = 1.34783$$

Carrying on, we find that $x_3 = 1.32520$, $x_4 = 1.32472$ and $x_5 = 1.32472$.

We can stop when two successive values agree to the required degree of accuracy. We conclude that the root is approximately 1.32472 to 5 decimal places.

2. This is equivalent to solving $f(x) = 0$, where $f(x) = \cos x - 2x$ so

$f'(x) = -\sin x - 2$

$$[NB: \textbf{makesure your calculator is in radian mode}]$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Using the above formula, we obtain

$$x_0 = 0.5,\ x_1 = 0.45063,\ x_2 = 0.45018,\ x_3 = 0.45018$$

with no further changes in the digits, to five decimal places. Therefore, to this degree of accuracy, the root is $x = 0.45018$.

Exercise 2.15

1. Newton's equation $y^3 - 2y - 5 = 0$ has a root near $y = 2$. Starting with $y_0 = 2$, compute y_1, y_2 and y_3, the next three Newton-Raphson estimates for the root.
2. Use Newton-Raphson method to find the roots of the following equations, correct to three decimal places. An initial value x_0 is provided in each case.

   a) $x - \cos x = 0$ $\quad x_0 = 0.7$
   b) $2\sin x - x^2 = 0$ $\quad x_0 = 1.2$
   c) $e^x - 2x - 5 = 0$ $\quad x_0 = -2$

REVISION EXERCISE UNIT 2

1. Obtain the derivative of each of the following from first principles:

a). $f(x) = x^2 + 2x$ b). $f(x) = (x+2)^2$ c). $f(x) = \frac{-3}{x^2}$

2. Determine $\frac{dy}{dx}$ in each of the following:

a) $y = (4x-5)^6$ b) $y = e^{\sin 2x}$ c) $y = cos^3(3x)$

d) $y = \log(2x-1)$ e) $y = x^2 \tan x$ f) $y = e^{5x}(3x+1)^2$ g) $y = \frac{\sin 2x}{2x+5}$

h) $y = \frac{\sqrt{x+1}}{x}$

3. Determine $\frac{dy}{dx}$ in each of the following:

a) $x^2 + 2xy + 3y^2 = \ln 4$ b) $x^3 + y^3 - 3xy^2 = 8$ c) $\sin x + \sin y = 1$

d) $y = \ln\left(\frac{e^{2x}(\sin 4x)^2}{\sqrt{2-x^2}}\right)$ e) $y = \frac{(x-1)^4 \cos 5x}{x\sqrt{\tan 3x}}$ f) $y = (3x^2 - 5)^{\sin x}$

4. Sketch the graphs of the following functions:

a) $y = x^3 - 2x^2 - x + 2$ b) $y = \frac{x^3}{3} - \frac{x^2}{2} - 2x + 5$ c) $y = \frac{1}{4}x^4 - \frac{2}{3}x^3 - \frac{1}{2}x^2 + 2x + 1$

d) $y = x^4 - 2x^2 - 3$

5. Determine the turning points on a curve $y = 4x + \frac{1}{x}$ and distinguish between them. Sketch the curve.
6. A projectile is fired vertically upwards with a height h given by $h = 5t - 4t^2$

a) determine an expression for the velocity and acceleration

b) determine the maximum height reached by the projectile.

7. The displacement of a particle at time t seconds is given by $y = e^{-t}\sin 2t$.

a) Determine an expression for its velocity. b) Calculate the initial velocity.

8. A particle moves s metres in t seconds according to the equation

$$s = \cos 3t + 2\sin 3t.$$

a) determine an expression for the velocity of the particle.

b) when does the particle come to rest for the first time?

c) determine the acceleration of the particle at $t = \frac{\pi}{6}$.

9. A body P, moves in a straight line so that its distance from a fixed point O after t seconds is s metres, where $s = \frac{1}{3}t^3 - \frac{3}{2}t^2 + 2t$.

Show that the body is at rest at two different times and find these times.
Determine the acceleration of the body at these times.

10. When viewed through a microscope a bacterium is seen to move in accordance with the equation
$s = (4t + 6t^2) \times 10^{-6}$.

Determine the velocity after 2 seconds.

11. The displacement s meters of a particle from a fixed point at a time t seconds is given by $s = 10 - 5te^{-0.t}$

a) Determine an expression for velocity.

b) Determine the value of t at which the velocity is zero.

12. The distance s meters moved by a body in t seconds is given by
$s = 2t^3 - 13t^2 + 24t + 10$, determine

a) the velocity when t =4 seconds

b) the value of t when the body comes to rest

c) the value of t when the acceleration is 10 m/s^2

13. Apply Newton–Raphson method to determine the roots of the following equations correct to 3 decimal places. An initial value x_0 is provided in each case.

a) $f(x) = x^3 - 3x - 4$ $\quad x_0 = 2$

b) $f(x) = x^3 + 2x^2 5x - 1$ $\quad x_0 = 1.5$

c) $x - \cos x = 0$ $\quad x_0 = 0.7$

14. The pressure, P, and volume, V, of a gas in a cylinder are related by $PV^n = C$ (where C is constant). Show that $\frac{d^2P}{dV^2} = \frac{n(n+1)P}{V^2}$.

15. The density, ρ, is defined as $\rho = \frac{1}{V}$ where V represents volume of unit mass. Find an expression for $\frac{dV}{d\rho}$

16. The voltage , V, across a capacitor of capacitance C, in series with a resistor of resistance R, is given by $V = Ee^{-t/RC}$ where $E > 0$ is constant. Determine i where $i = C\frac{dV}{dt}$

17. The distribution function, F, of a set of components is given by $F = 4 - \frac{20}{5+\frac{t}{3}}$ where t is in years. Determine the density function $\frac{dF}{dt}$

18. The reliability function $R(t)$, of a component is given by $R(t) = e^{-\sqrt{t}}$ $(t \geq 0)$. Determine $\frac{d}{dt}[1 - R(t)]$.

UNIT 3

INTEGRATION

3.1. INTRODUCTION

Integration is an important concept in mathematics and, together with its inverse, differentiation, is one of the two main operations in calculus.

Integrals and derivatives became the basic tools of calculus, with numerous applications in science and engineering.

In this Unit, we are going to focus mainly on the definite and indefinite integrals by using different rules.

By the end of this unit the learner should be able to:

- Evaluate both indefinite and definite integrals using different rules and properties of integration,
- Integrate by using the substitution method,
- Determine the area between a curve and the X-axis by using the definite integrals,
- Determine the area between two curves,
- Use Simpson's rule to determine the area between a curve and X-axis.

3.2. INDEFINITE INTEGRALS

Definition : If the function $f(x)$ is a derivative, then the set of all antiderivatives of f is called the indefinite integral of f, denoted by $\int f(x)dx$.

The symbol $\int$ is called an integral sign and the function f is the integrand of the integral and x is the variable of integration.

Once we have found an antiderivative $F(x)$ of a function $f(x)$, the other antiderivatives of f differ from F only by a constant. We indicate this in the new notation by writing

$$\int f(x)dx = F(x) + C \qquad (1)$$

The constant C is called the constant of integration.

Example : Evaluate $\int x^2 dx$

Solution: $\int x^2 dx = \frac{x^3}{3} + C$

The constant of integration

antiderivative

Rules of integration

1. $\int k dx = kx + C$
2. $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \qquad (n \neq -1)$
3. $\int kf(x) = k \int f(x)$ (provided k is a constant)
4. $\int (f(x) \pm g(x)) \, dx = \int f(x)dx \pm \int g(x)dx$
5. $\int -f(x)dx = -\int f(x)dx$ (special case rule)
6. $\int \sin x dx = -\cos x + C$
7. $\int \cos x dx = \sin x + C$

Examples

1. $\int 3x dx = 3 \int x dx$

$$= 3\frac{x^{1+1}}{1+1} + C = \frac{3}{2}x^2 + C$$

2. $\int \left(\pi + x^3 - \frac{1}{2x^2}\right) dx = \int \pi dx - \int x^3 \, dx - \frac{1}{2}\int \frac{1}{x^2} dx$

$$= \pi x + \frac{x^4}{4} - \frac{1}{2}\left(\frac{x^{-1}}{-1}\right) + C \qquad \text{(since} \frac{1}{x^2} = x^{-2})$$

$$= \pi x + \frac{x^4}{4} + \frac{1}{2x} + C$$

Exercise 3.1

Evaluate the following integrals:

1. $\int x^3 dx$ 2. $\int 3\sqrt{x} dx$ 3. $\int x^{-\frac{1}{3}} dx$ 4. $\int (6 - 6x) dx$ 5. $\int \frac{4}{x^5} dx$

6. $\int (2x^4 - 3x^2 + ax) dx$ 7. $\int \left(\frac{x^2}{2} + \frac{x^3}{3}\right) dx$ 8. $\int (1 + x)^3 dx$ 9. $\int \frac{7}{\sqrt[4]{x}} dx$

10. $\int (1 - 4x^{-3} + \pi^2) dx$

3.2.1. Integration by substitution – Running the chain rule backward

A change of variable can often turn an unfamiliar integral into one we can evaluate. The method of doing this is called the substitution method of integration. It is the principal method by which integrals are evaluated.

The generalized power rule in integral form

When u is a differential function of x and is a rational number different from -1, the chain rule tells us that

$$\frac{d}{dx}\left(\frac{u^{n+1}}{n+1}\right) = u^n \frac{du}{dx} \qquad (2)$$

This same equation, from another point of view, says that $\frac{u^{n+1}}{n+1}$ is one of the antiderivatives of $u^n \frac{du}{dx}$. The set of all antiderivatives of $u^n \frac{du}{dx}$ is therefore

$$\int \left(u^n \frac{du}{dx}\right) dx = \frac{u^{n+1}}{n+1} + C \qquad (3)$$

The integral on the left hand side of this equation is usually written in a simpler "differential" form

$$\int u^2 \, du \qquad (4)$$

Obtained by treating dx's as differentials that cancel. Combining (3) and (4) then gives the following rule.

$$\int u^n du = \frac{u^{n+1}}{n+1} + C \qquad (5)$$

Examples

1. (a)Evaluate $\int (x+2)^5 \, dx$

Solution

Let $x + 2 = u, \frac{du}{dx} = 1, du = dx$.

$\therefore \int (x+2)^5 \, dx = \int u^5 \, du$ (substitute $u = x + 2$ $and du = dx$)

$= \frac{u^6}{6} + C$ (integrate using equation (4) with n = 5

$= \frac{(x+2)^6}{6} + C$ (replace u by $x + 2$)

(b) the same function in (a) can be solved by using the following rule:

$\int f'(x)[f(x)]^n \, dx = \frac{[f(x)]^{n+1}}{n+1} + C$, then

$\int (x+2)^5 \, dx \rightarrow f(x) = x + 2, f'(x) = 1$

$\therefore \int (x+2)^5 \, dx = \frac{(x+2)^6}{6} + C$

2. $\int (4x-1)^3 \, dx = \int u^3 \frac{1}{4} du$ (substitute $u = 4x - 1, du = 4dx, \frac{1}{4} du = dx$)

$= \frac{1}{4} \int u^3 \, du = \frac{1}{4} \frac{u^4}{4} + C = \frac{1}{16}(4x-1)^4 + C$

3. $\int \frac{3x^2}{(2x^3+1)^2} dx = \int 3x^2 (2x^3 + 1)^{-2} \, dx$

$= \int 3u^{-2} \frac{du}{6}$ (subst. $u = 2x^3 + 1, du = 6x^2 dx, \frac{du}{6} = x^2 dx$)

$$= \frac{1}{2}\int u^{-2}\,du \;= \frac{1}{2}\frac{u^{-1}}{-1} + C \;= -\frac{1}{2u} + C \;= -\frac{1}{2(2x^3+1)} + C$$

Applying the same rule as in 1.(b) we have;

$$\int \frac{3x^2}{(2x^3+1)^2}dx = \int 3x^2(2x^3+1)^{-2}\,dx$$

$$f(x) = 2x^3 + 1, f'(x) = 6x^2$$

$$\int 3x^2(2x^3+1)^{-2}\,dx = 3.\frac{1}{6}\int 6x^2\,(2x^3+1)^{-2}\,dx$$

$$= \frac{1}{2}\frac{(2x^3+1)^{-1}}{-1} + C = -\frac{1}{2(2x^3+1)} + C$$

Exercise 3.2

Evaluate the following integrals:

1. $\int (3x^2 - 6x + 1)^2\,(x-1)dx$ 2. $\int \frac{2tdt}{\sqrt{t^2-1}}$ 3. $\int 28(7x-2)^3\,dx$

4. $\int \frac{dy}{2\sqrt{y}(1+\sqrt{y})^4}$ 5. $\int (s^3 + 2s^2 - 5s + 6)^2\,(3s^2 + 4s - 5)ds$ 6. $\int \frac{x^3dx}{\sqrt[3]{x^4+3}}$

3.2.2. Integrals involving trigonometric functions

Rules:

1. $\int f'(x)\sin f(x)\,dx = -\cos f(x) + C$
2. $\int f'(x)\cos f(x)dx = \sin f(x) + C$
3. $\int f'(x)\sec^2 f(x)dx = \tan f(x) + C$
4. $\int f'(x)\sec f(x)\,\tan f(x)dx = \sec f(x) + C$
5. $\int f'(x)\operatorname{cosec}^2 f(x)dx = -\cot f(x) + C$
6. $\int f'(x)\operatorname{cosec} f(x)\cot f(x)dx = -\operatorname{cosec} f(x) + C$

Examples

1. Evaluate $\int \sin 3x dx$

Solution: let $f(x) = 3x$, then $f'(x) = 3$

$$\therefore \int \sin 3x dx = \frac{1}{3}\int 3\sin 3x\,dx = -\frac{1}{3}\cos 3x + C$$

2. $\int \cos(4x-1)dx = \frac{1}{4}\int 4\cos(4x-1)dx$ (since $f(x) = 4x - 1$ and $f'(x) = 4$)

$$= \frac{1}{4}\sin(4x+1) + C$$

3. $\int x^3\sec^2 x^4 dx = \frac{1}{4}\int 4x^3\sec^2 x^4\,dx \;= \frac{1}{4}\tan x^4 + C$

4. $\int \frac{\sin x dx}{(3+\cos x)^2} = \int \sin x(3+\cos x)^{-2}\,dx \;= -\int -\sin x(3+\cos x)^{-2}\,dx$

$= -\frac{(3+\cos x)^{-1}}{-1} + C = \frac{1}{3+\cos x} + C$

Exercise 3.3

Evaluate the following integrals:

1. $\int \sec^2 \frac{x}{4} dx$ 2. $\int \sqrt{\text{Tan}x} \sec^2 x dx$ 3. $\int (1-\cos 3x)\sin 3x dx$ 4. $\int \text{Sin}^3\theta \cos\theta d\theta$

5. $\int \frac{\sin\sqrt{x}}{\sqrt{x}} dx$ 6. $\int \frac{8\sin t}{\sqrt{5-4\cos t}} dt$ 7. $\int \sin 2t(3-\cos 2t)^{\frac{3}{2}} dt$ 8. $\int \frac{\sin 2x}{\cos^4 2x} dx$

9. $\int \sec\left(\theta + \frac{\pi}{2}\right) \tan\left(\theta + \frac{\pi}{2}\right) d\theta$ 10. $\int \frac{1}{x^2} \text{cosec}\left(\frac{1}{x}\right) \cot\left(\frac{1}{x}\right) dx$

3.2.3. Integrals involving exponential and logarithmic functions

Rules

1. $\int f'(x) a^{f(x)}\, dx = \frac{a^{f(x)}}{\ln a} + C \qquad (a > 0, a \neq 1)$
2. $\int f'(x) e^{f(x)}\, dx = \frac{e^{f(x)}}{\ln e} + C = e^{f(x)} + C$
3. $\int \frac{f'(x)}{f(x)} dx = \ln f(x) + C \qquad (f(x) > 0, f(x) \neq 1)$

$\int (x^2+1) e^{x^3+3x}$

$\int (3x^2+3) e^{(x^3+3x)}$

$\int 3(x^2+1) e^{x^3+3x}$

$\int \frac{1}{3} \cdot 3(x^2+1) e^{x^3+3x}$

$\frac{1}{3} \int 3(x^2+1) e^{x^3+3x}$

$\frac{1}{3} e^{x^3+3x} + C$

Examples

1. Evaluate $\int 2.3^{2x} dx$

Let $f(x) = 2x$ then $f'(x) = 2 \qquad \therefore \int 2.3^{2x} dx = \frac{3^{2x}}{\ln 3} + C$

2. $\int 2^{4x+1}\, dx = \frac{1}{4}\int 4.2^{4x+1}\, dx \qquad$ (since $f(x) = 4x+1$ *and* $f'(x) = 4$)

$= \frac{2^{4x+1}}{4\ln 2} + C$

3. $\int (x^2+1)\, e^{x^3+3x}\, dx = \frac{1}{3}\int 3(x^2+1) e^{x^3+3x}\, dx \qquad (f(x) = x^3+3x \text{ and} f'(x) = 3x^2+3)$

$= \frac{e^{x^3+3x}}{3} + C$

4. $\int \sec^2 2x e^{\tan 2x} dx = \frac{1}{2}\int 2\sec^2 2x e^{\tan 2x} dx = \frac{e^{\tan 2x}}{2} + C$
5. $\int \frac{dx}{x+1} = \ln(x+1) + C$
6. $\int \frac{2x dx}{x^2-2} = \ln(x^2-2) + C$

Exercise 3.4

Evaluate the following integrals:

1. $\int 2^{x^2} x dx$ 2. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$ 3. $\int 4x^2 a^{x^3-1} dx \ (a > 1)$ 4. $\int \frac{e^x cose^x}{sine^x} dx$

5. $\int \frac{x^3}{x^4+1} dx$ 6. $\int \frac{3sec^2 2x}{1-tan2x} dx$ 7. $\int \frac{4x-4}{2x^2-4x+1} dx$

3.3. DEFINITE INTEGRALS

Definition: if f is continuous on $[a, b]$ and F is any antiderivative of f on $[a, b]$, then the definite integral of f is defined by

$$\int_a^b f(x)dx = F(x)\Big|_a^b = F(b) - F(a)$$

Examples

1. Evaluate $\int_0^1 x^3\, dx$

Solution

$$\int_0^1 x^3\, dx = \frac{x^4}{4}\Big|_0^1 = \left[\frac{1^4}{4} + C\right] - \left[\frac{0^4}{4} + C\right] = \frac{1}{4} - 0 = \frac{1}{4}$$

Note: When evaluating definite integrals, the constant of integration can be omitted since it cancels out in the arithmetic.

2. $\int_1^2 (2x + 5)\, dx = x^2 + 5x\Big|_1^2 = [2^2 + 5(2)] - [1^2 + 5(1)] = 14 - 6 = 8$

Exercise 3.5.

Use the definite integrals to evaluate the following integrals

1. $\int_0^3 (4 - x^2)\, dx$ 2. $\int_1^{32} \frac{1}{\sqrt[5]{x^6}}\, dx$ 3. $\int_0^{\pi} (1 + cosx)\, dx$ 4. $\int_4^9 \frac{1-\sqrt{u}}{\sqrt{u}}\, du$

5. $\int_0^{\pi/3} 4secxtanx\, dx$

3.4. APPLICATIONS OF INTEGRATION

3.4.1 Area and the definite integral

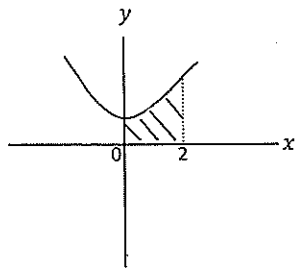
If $y(x)$ is non-negative and integrable over a closed interval $[a, b]$, then the integral of f from a to b is the **area** of the region between the graph of f and the X-axis from a to b. We sometimes call this number the area under the curve $y = f(x)$ from a to b. The formula is given by

$A = \int_a^b f(x)dx$ when $f(x) \geq 0$ or $A = -\int_a^b f(x)dx$ when $f(x) \leq 0$

Examples

1. Calculate the area enclosed by the function $y = x^2 + 1$ and the X-axis on the interval [0,2]

Solution



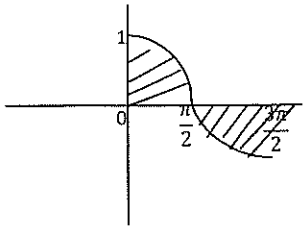
$$\int_0^2 (x^2 + 1)\, dx = \frac{x^3}{3} + x\Big|_0^2 = \left[\frac{2^3}{3} + 2\right] - 0 \quad = \frac{14}{3}$$

2. Calculate the area enclosed by the function $y = \cos x$ on the interval $\left[0; \frac{3\pi}{2}\right]$

Solution: Area $= \int_0^{3\pi/2} \cos x\, dx = \int_0^{\pi/2} \cos x\, dx - \int_{\pi/2}^{3\pi/2} \cos x\, dx$

$$= \sin x\Big|_0^{\pi/2} - \sin x\Big|_{\pi/2}^{3\pi/2}$$

$$= 1 - (-2) = 3$$



Exercise 3.6

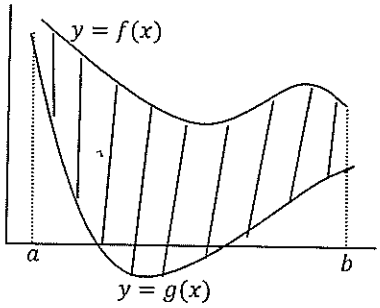
Sketch the given region and determine the area

1. Between the X – axis and $y = x^2 - 1$ on the interval $[0; 1]$
2. Between the X – axis and $y = \cos x$ on the interval $\left[\frac{\pi}{2}; \frac{3\pi}{2}\right]$
3. Between the X – axis and $y = x^3$ on the interval $[-2; 2]$

3.4.2.Area between Curves

In this section we are going to look at finding the area between two curves. There are two cases that we are going to be looking at

- In the first case we want to determine the area between $y = f(x)$ and $y = g(x)$ on the interval $[a, b]$, we are also going to assume that $f(x) \geq g(x)$.
  Consider the following sketch:

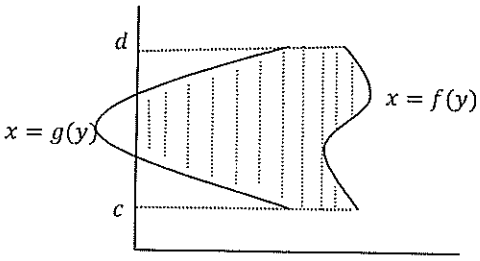


The area between the two curves on the figure above can be calculated as:

$$\mathrm{A} = \int_a^b [f(x) - g(x)]\, dx \text{ or } \mathrm{A} = \int_a^b [(\text{upper function}) - (\text{lower function})]\, dy$$

- The second case is almost identical to the first case. Here we are going to determine the area between $x = f(y)$ and $x = g(y)$ on the interval $[c, d]$ with $f(y) \geq g(y)$

Consider the following sketch:



In this case the formula is:

$$\mathrm{A} = \int_c^d [f(y) - g(y)]\, dy \text{ or } \mathrm{A} = \int_a^b [(right\ function) - (left\ function)]\, dy$$

Examples

1. Determine the area of the region enclosed by $y = x^2$ and $y = \sqrt{x}$

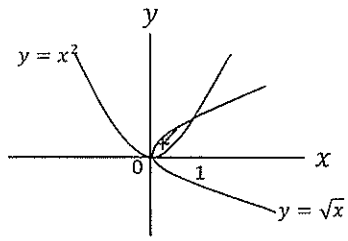
The limits of integration will be the intersection points of the two curves which we obtain by equating the two functions, then the roots.

Finding the limits:

$x^2 = \sqrt{x}$

$x = x^4$

$x - x^4 = 0 \ \therefore x = 0 \text{ or } x = 1$



So the integral that we'll need to compute to find the area is:

$A = \int_a^b [(upper\ function) - (lower\ function)]\, dx$

$= \int_0^1 (\sqrt{x} - x^2)\, dx$

$= \frac{2}{3}x^{\frac{3}{2}} - \frac{1}{3}x^3 \Big|_0^1$

$= \frac{1}{3}\text{units}^2$

2. Determine the area of the region bounded by $y = 2x^2 + 10$ and $y = 4x + 16$

Solution

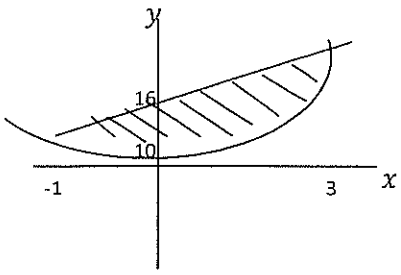
Finding the limits:

$2x^2 + 10 = 4x + 16$

$2x^2 - 4x + 10 - 16$

$(x^2 - 2x - 3) = 0$

$x = 3 \text{ or } x = -1$



$A = \int_{-1}^{3} [(4x + 16) - (2x^2 + 10)]dx$

$= \int_{-1}^{3} (-2x^2 + 4x + 6)dx$

$= \left(-\frac{2}{3}x^3 + 2x^2 + 6x\right) \Big|_1^3 = \frac{64}{3}$

Exercise 3.7

Determine the area between the following curves:

1. $y = x^2 - 1$ and $y = 2 - 2x^2$

2. $y = x^2 - 3x$ and $y = 2x$
3. $y = 4 - x^2$and $y = x - 2$
4. $x = -y^2 + 10$ and $x = (y - 2)^2$
5. $x = \frac{1}{2}y^2 - 3$ and $y = x - 1$

3.4.3.Numerical Integration

As we have seen in the previous sections, the ideal way to evaluate a definite integral $\int_a^b f(x)dx$ is to find a formula $F(x)$ for one of the antiderivatives of $f(x)$ and calculate the number $F(b) - F(a)$. But some antiderivatives are hard to find and still others like $\int \frac{\sin x}{x} dx$ and $\int \sqrt{1 + x^4}\, dx$, have no elementary formula.

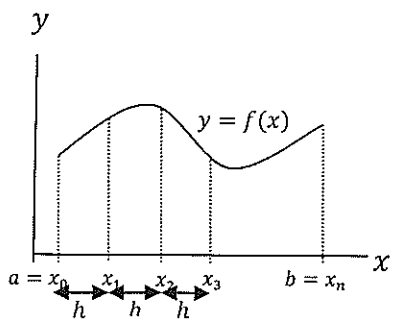
So when we cannot evaluate a definite integral with an antiderivative, we turn to numerical methods, like Simpson's rule. This rule enables us to estimate an integral's value to as many decimal places as we please. It also enables us to calculate integrals with reasonable accuracy from numerical tables of function values.

The Simpson's rule

In Simpson's rule the curve $y = f(x)$ is approximated by a number of quadratic curves. The formula is:

$$\int_a^b y\, dx \approx \frac{h}{3}[y_0 + 4(y_1 + y_3 + y_5 + \cdots) + 2(y_2 + y_4 + y_6 + \cdots) + y_n] \qquad (*)$$

Where $h$is the uniform width ,$y_0, y_1, \ldots, y_n$ are the ordinates shown in the figure below and n is the number of blocks, which must be even. (The formula for h is given by $h = \frac{b-a}{n}$, n is even)



Formula (*) can also be written as:

$$Area \approx \frac{h}{3}[(\text{first ordinates}) + 4(\text{sum of odd ordinates}) + 2(\text{sum of even ordinates}) + (\text{last ordinates})]$$

Examples

1. Evaluate $\int_0^1 x^2\, dx$ using Simpson's rule with four intervals.

Solution

Since we have an even number (4) of intervals, putting $n = 4$ into Simpson's rule gives:

$$\int_0^1 x^2 dx \approx \frac{h}{3}[y_0 + 4(y_1 + y_3) + 2y_2 + y_4] \qquad [1]$$

the niform width h is given by $h = \frac{1-0}{4} = 0.25$

we also need to find the $x - coordinates$ by the sub-intervals $a = x_0 < x_1 < \cdots < x_{n+1} = b$, with$x_{i+1} - x_i = h$

in this case we have: $x_0 = 0$ $\quad y_0 = 0^2 = 0$

$x_1 = 0.25$ $\quad y_1 = 0.25^2 = 0.0625$

$x_2 = 0.5$ $\quad y_2 = 0.5^2 = 0.25$

$x_3 = 0.75$ $\quad y_3 = 0.75^2 = 0.5625$

$x_4 = 1$ $\quad y_4 = 1^2 = 1$

Putting all these into [1] gives

$$\int_0^1 x^2\, dx \approx \frac{h}{3}[y_0 + 4(y_1 + y_3) + 2y_2 + y_4]$$

$$= \frac{0.25}{3}[0 + 4(0.0625 + 0.5625) + 2(0.25) + 1] = \frac{1}{3}$$

2. Find the approximated value of $\int_0^{\pi/2} \sqrt{\cos x}\ dx$ by using Simpson's rule with 2 intervals

Solution:

$$\int_0^{\pi/2} \sqrt{\cos x}\ dx \approx \frac{h}{3}[y_0 + 4y_1 + y_2]$$

$$h = \frac{\frac{\pi}{2} - 0}{2} = \frac{\pi}{4}$$

We can find the y values as follows:

$x_0 = 0$ $\quad y_0 = \sqrt{\cos(0)} = 1$

$x_1 = \frac{\pi}{4}$ $\quad y_1 = \sqrt{\cos\left(\frac{\pi}{4}\right)} = 0.8409$

$x_2 = \frac{\pi}{2}$ $\quad y_2 = \sqrt{\cos\frac{\pi}{2}} = 0$

Substituting the y values in the formula, we have

$$\int_0^{\pi/2} \sqrt{\cos x}\ dx \approx \frac{h}{3}[y_0 + 4y_1 + y_2] \quad = \frac{\pi/4}{3}[1 + 4(0.8409) + 0]$$
$$= 1.055$$

Exercise 3.8.

Do all calculations accurate to at least 5 decimal places and give the final answer to 3 decimal places.

1. Approximate $\int_0^1 \frac{1}{1+x^2}\ dx$ using Simpson's rule with 6 intervals
2. Determine the area between the function $y = x^2 + 1$ and the x –axis on the interval [2,4] using Simpson's rule with 2 intervals. Compare your answer using the definite integral.
3. Evaluate $\int_0^{\pi/2} \sin t\ dt$ using Simpson's rule with 8 intervals
4. Approximate the area between $y = e^{x^2}$ and the x –axis on the interval [2,4] using Simpson's rule with 4 intervals.

REVISION EXERCISE FOR UNIT 3

1. Determine the following integrals

a) $\int \left(x^3 + \frac{1}{x^2} + \sqrt{x}\right) dx$ b) $\int \left(\frac{2x^3-3x}{4x}\right) dx$ c) $\int 3\cos\left(4x - \frac{\pi}{2}\right) dx$

d) $\int (sec^2 2x - cosec^2 4x)dx$ e) $\int \left(5^x + \frac{1}{x}\log_3 x\, dx\right)$ f) $\int x^2\sqrt{10 - x^3}dx$

g) $\int \frac{5x\, dx}{2x^2-1}$

2. Evaluate the following

a) $\int_1^3 (3x^2 + 6x + 8)dx$ b) $\int_{-2}^2 (x^2 - 2)\,(2x - 1)dx$ c) $\int_1^3 5x\sqrt{2x^2 + 7}dx$

d) $\int_0^{\pi} \frac{2\sin 3\theta}{2-\cos 3\theta} d\theta$ e) $\int_0^1 2e^{3x+1}dx$

3. a) Calculate the area of the region enclosed by the curve $y = x^3 - 4x^2 + 3x$ and the x-axis between $x = 0$ and $x = 4$.

b) Determine the area enclosed by $y = 3x^2 + 4$, the x-axis and the ordinates $x = 1$ and $x = 4$

c) On the same axes sketch and label clearly the graphs $y = x^2 - 5x$ and $y = 5x - x^2$. Determine the area of the region enclosed by the two curves.

d) Determine the coordinates of the points of intersection of the curves $y = x^2$ and $y^2 = 8x$. Sketch the curves on the same axes and calculate the area enclosed by the two curves.

4. Use Simpson's rule to evaluate the following with the given subintervals

a) $\int_0^2 x\sqrt{x^2+1}dx$ $\quad n = 4$

b) $\int_2^4 \frac{1}{x^2+1} dx$ $\quad n = 6$

c) $\int_0^{0.5\pi} \sqrt{\cos\theta}\, d\theta$ $\quad n = 2$

d) $\int_1^2 \frac{dx}{1+x^3}$ $\quad n = 4$

e) $\int_1^2 xe^x dx$ $\quad n = 2$

5. The distance travelled by a train on a straight track in the first two seconds is given by $S = \int_0^2 20(1 - e^{-t})dt$. Determine S.

UNIT 4

MATRICES AND VECTORS

4.1 INTRODUCTION

In this unit we will define a matrix and focus on some elementary operations of a 2 x 2 matrix. Finding the determinant of a 2 x 2 matrix is also covered in this unit. The unit continues with vectors. Definition of a vector and how a vector can be represented are given. Other aspects on vectors discussed in this unit include the magnitude, unit vector and resolution of a vector into its components. The scalar and cross products are also covered in this unit.

After completing this unit the learner should be able to:

- Perform elementary operations on 2 x 2 matrices, such as addition and subtraction, scalar multiplication and multiplication of matrices
- Calculate the determinant of a 2 x 2 matrix
- Perform elementary operations with vectors, such as scalar multiplication, addition and subtraction of vectors
- Determine parallel vectors,
- Determine the magnitude (length) of a vector and resolve a vector into components,
- Determine the scalar product and use it to determine the angle between two vectors and to calculate work done,
- Determine the vector product and calculate the moment of a vector.

4.2. MATRICES

A matrix is defined as a rectangular array of numbers. The numbers in the matrix are called elements/entries. The order of a matrix is the number of rows x the number of columns. For example an m x n matrix has m rows and n columns. A matrix with m rows and m columns is called a square matrix, e.g. $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

The numbers a and b are called the elements of the first row and the numbers a and c are the elements of the first column. The numbers a and d are the elements of the principal (leading) diagonal and the numbers b and$c$ are the elements of the secondary diagonal.

In this section, we will discuss elementary operations and the determinant of a

2 X 2 matrix.

4.2.1. Elementary operations of 2 x 2 Matrices

a) Addition and subtraction

Two matrices can be added (or subtracted) if they are of the same order.

Example

Let $\boldsymbol{A} = \begin{pmatrix} 3 & -4 \\ 5 & 2 \end{pmatrix}$ and $\boldsymbol{B} = \begin{pmatrix} -5 & 2 \\ 1 & -2 \end{pmatrix}$,

Determine $\boldsymbol{A} + \boldsymbol{B}$ and $\boldsymbol{A} - \boldsymbol{B}$

Solution

$$\boldsymbol{A} + \boldsymbol{B} = \begin{pmatrix} 3 & -4 \\ 5 & 2 \end{pmatrix} + \begin{pmatrix} -5 & 2 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} -2 & -2 \\ 6 & 0 \end{pmatrix}$$

$$\boldsymbol{A} - \boldsymbol{B} = \begin{pmatrix} 3 & -4 \\ 5 & 2 \end{pmatrix} - \begin{pmatrix} -5 & 2 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 8 & -6 \\ 4 & 4 \end{pmatrix}$$

The commutative law of addition applies to matrices, thus $\boldsymbol{A} + \boldsymbol{B} = \boldsymbol{B} + \boldsymbol{A}$

b) Scalar multiplication

To multiply a matrix by a scalar (number), we multiply each element of the matrix by the scalar. Thus given the scalar (number) k and the matrix $\boldsymbol{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

then $k\boldsymbol{A} = k\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ka & kb \\ kc & kd \end{pmatrix}$

Example $3\begin{pmatrix} -1 & 0 \\ 2 & -2 \end{pmatrix} = \begin{pmatrix} -3 & 0 \\ 6 & -6 \end{pmatrix}$

c) Multiplication of matrices

Two matrices can be multiplied together if they are **compatible** (i.e. if the number of columns of the first matrix is the same as the number of rows of the second matrix). NB!!

Example

Given the matrices $\boldsymbol{A} = \begin{pmatrix} 4 & -2 \\ 1 & 5 \end{pmatrix}$, $\boldsymbol{B} = \begin{pmatrix} 2 & 3 \\ -3 & 5 \end{pmatrix}$ *and* $\boldsymbol{C} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

$$\boldsymbol{AB} = \begin{pmatrix} 4 & -2 \\ 1 & 5 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ -3 & 5 \end{pmatrix} = \begin{pmatrix} 8+6 & 12-10 \\ 2-15 & 3+25 \end{pmatrix} = \begin{pmatrix} 14 & 2 \\ -13 & 28 \end{pmatrix}$$

$$\boldsymbol{BA} = \begin{pmatrix} 2 & 3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} 4 & -2 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} 8+3 & -4+15 \\ -12+5 & 6+25 \end{pmatrix} = \begin{pmatrix} 11 & 11 \\ -7 & 31 \end{pmatrix}$$

Note that $\boldsymbol{AB} \neq \boldsymbol{BA}$

$$\boldsymbol{AC} = \begin{pmatrix} 4 & -2 \\ 1 & 5 \end{pmatrix}\begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 13 \end{pmatrix}$$

Note that $\boldsymbol{AC}$ is defined but $\boldsymbol{CA}$ is not defined.

4.2.2 Determinant of a 2 x 2 matrix

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then the determinant of $A = det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

We can find determinants of square matrices only.

The determinant of a 2 x 2 matrix, which is a number, is obtained by taking the product of the elements of the leading diagonal and subtracting the product of the elements of the secondary diagonal.

Example

Calculate the determinant of $\boldsymbol{A} = \begin{pmatrix} -3 & 2 \\ 4 & -1 \end{pmatrix}$

Solution

$detA = \begin{vmatrix} -3 & 2 \\ 4 & -1 \end{vmatrix} = (-3 \text{ x } -1) - (4 \text{ x } 2) = -5$

Exercise 4.1

1. Calculate the determinant of each of the following matrices:
   $\boldsymbol{E} = \begin{pmatrix} 3 & -1 \\ 4 & 2 \end{pmatrix}$ $\boldsymbol{F} = \begin{pmatrix} 0 & 5 \\ -3 & 6 \end{pmatrix}$ $\boldsymbol{G} = \begin{pmatrix} -2 & -1 \\ 5 & 3 \end{pmatrix}$
2. If $\boldsymbol{A} = \begin{pmatrix} 2 & 6 \\ 4 & 8 \end{pmatrix}$, $\boldsymbol{B} = \begin{pmatrix} 3 & -5 \\ 4 & 8 \end{pmatrix}$, $\boldsymbol{C} = \begin{pmatrix} -2 & 3 \\ 1 & -1 \end{pmatrix}$

Evaluate the determinants of the following:

i) $\boldsymbol{A}$ ii) $\boldsymbol{B} - 3\boldsymbol{C}$ iii) $\boldsymbol{A} + \boldsymbol{C}$ iv) $3(\boldsymbol{C} - \boldsymbol{B})$
v) $\boldsymbol{AB}$ vi) $\boldsymbol{BC}$ vii) $\boldsymbol{AC}$

4.3. VECTORS

One of the most important concepts in mathematics is the vector. We define a vector by a direction and a magnitude. For example, a force applied at a point is a vector: it is completely determined by the magnitude of the force and the direction in which it is applied. Graphically, we represent a vector as a **directed line segment** from an initial point A to a terminal point B.

4.3.1. Notation

Free vectors are denoted by a pair of block letters and an arrow. For example, $\overrightarrow{AB}$ denotes the directed line segment representing a movement from A to another point B. Position vectors are denoted by small letters printed in bold type, for example, $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{m}, \boldsymbol{n}, \boldsymbol{p}, \boldsymbol{q}$ *etc*. In hand writing, we underline the small letters to represent position vectors as $\underline{a}$.

4.3.2. Representation

Vectors can be represented in 2 dimensions or in 3 dimensions by column or component form:

$\overrightarrow{OA} = \begin{pmatrix} a \\ b \end{pmatrix}$ or $\overrightarrow{OA} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ or $\overrightarrow{OA} = (a, \quad b, \quad c)$

The Cartesian co-ordinate form:

$\overrightarrow{OA} = \boldsymbol{a} = x\boldsymbol{i} + y\boldsymbol{j}$ $\quad (2\ dimensions)$

$\overrightarrow{OA} = \boldsymbol{a} = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$ $\quad (3\ dimensions)$

If $A(x, y)$ and $B(p, q)$ are two points in a plane, then $\overrightarrow{AB}$ is the vector that joins A and B. It has length equal to the length of the line segment that joins A and B and direction parallel to an arrow pointing from A to B. We assign coordinates to $\overrightarrow{AB}$ by subtracting the respective coordinates of A from those of B.

The position vector of a point A is the vector $\overrightarrow{OA}$ where O is the origin relative to which A has coordinates. So A and $\overrightarrow{OA}$ expressed as coordinates look the same.

4.3.3. Magnitude/ length/ modulus of a vector

If $\boldsymbol{a} = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$, then the magnitude (or the length) of $\boldsymbol{a}$ is denoted by $|a| = \sqrt{x^2 + y^2 + z^2}$

Examples

1. If $\boldsymbol{b} = -4\boldsymbol{i} + 3\boldsymbol{j}$ and $\boldsymbol{c} = 4\boldsymbol{i} - 6\boldsymbol{j}$ Determine $|\boldsymbol{b}|$ and $|c|$.
2. Given the vector $\overrightarrow{AB} = 2\boldsymbol{i} + 5\boldsymbol{j} - \boldsymbol{k}$, determine $|\overrightarrow{AB}|$.

Solutions:

1. $|b| = \sqrt{(-4)^2 + (3)^2} = \sqrt{25} = 5$
   $|c| = \sqrt{(4)^2 + (-6)^2} = \sqrt{52} = 2\sqrt{13} = 7{,}211103$

2. $|\overrightarrow{AB}| = \sqrt{(2)^2 + (5)^2 + (-1)^2} = \sqrt{30} = 5{,}477226$

Exercise 4.2

Given the vectors $\boldsymbol{a} = 2\boldsymbol{i} - 5\boldsymbol{j} + 3\boldsymbol{k}$, $\boldsymbol{m} = 3\boldsymbol{i} + \boldsymbol{j} - 2\boldsymbol{k}$, $\boldsymbol{n} = -4\boldsymbol{i} + 3\boldsymbol{j} - 2\boldsymbol{k}$ determine $|\boldsymbol{a}|$; $|\boldsymbol{n}|$, $|\boldsymbol{m}|$

4.3.4. Resolution of a vector

The process of splitting a vector into various parts or components is called resolution of vector. These parts of a vector may act in different directions and are called components of vector.

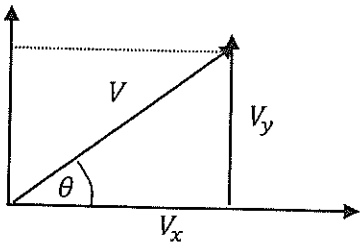
We can resolve a vector into a number of components. Usually we use 3 components of a vector, namely;

Component along X-axis called x-component

Component along Y-axis called y-component

Component along Z-axis called z-component.

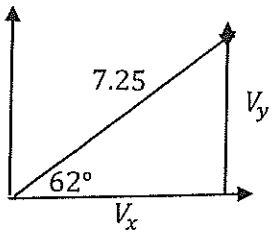
In this section we will use only two components, the x-component and the y-component which are perpendicular to each other. These components are called rectangular components of vector.



If v is the magnitude of the vector $\boldsymbol{v}$ and θ is the acute angle the vector makes with the x-axis or horizontal axis then the x-component of the vector is $vcos\theta$ and the y-component is $vsin\theta$. Thus the vector$\boldsymbol{v} = \begin{pmatrix} vcos\theta \\ vsin\theta \end{pmatrix}$

Example.

Find the x- and y-components of the vector$\boldsymbol{A}$ whose magnitude is 7.25 and makes an angle of 62° with the positive x-axis.



$\cos 62° = \frac{A_x}{|A|}$

$A_x = 7.25 \cos 62° = 3.40$

$\sin 62° = \frac{A_y}{|A|}$

$A_y = 7.25 \sin 62° = 6.40$

Exercise 4.3

1. Determine the x- and y-components of the following vectors. The magnitude is shown first, followed by the direction as an angle in the standard position

a) 5.40 N, $\theta = 75°$ b) 83.6 m/s, $\theta = 215°$ c) 673 N, $\theta = 285.2°$

4.3.5. Unit Vectors

A unit vector is a vector whose magnitude is 1. The unit vector in the direction of a vector $\boldsymbol{a}$ is denoted by $\hat{\boldsymbol{a}}$.
The unit vector of any vector is given by the formula:
unit vector= $\frac{vector}{magnitude}$ i.e. $\hat{\boldsymbol{a}} = \frac{a}{|a|}$

Example

Determine the unit vectors in the directions of the vector $\boldsymbol{a} = 5\boldsymbol{i} + 12\boldsymbol{j}$ and $\boldsymbol{m} = 2\boldsymbol{i} - \boldsymbol{j} + 5\boldsymbol{k}$

Solution

The magnitude of a is $|\boldsymbol{a}| = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$
Therefore the unit vector is, $\hat{\boldsymbol{a}} = \frac{1}{13}(5\boldsymbol{i} + 12\boldsymbol{j}) = \frac{5}{13}\boldsymbol{i} + \frac{12}{13}\boldsymbol{j}$

The magnitude of $\boldsymbol{m}$ is $|m| = \sqrt{4 + 1 + 25} = \sqrt{30}$
The unit vector is $\hat{m} = \frac{1}{\sqrt{30}}(2\boldsymbol{i} - \boldsymbol{j} + 5\boldsymbol{k}) = \frac{2}{\sqrt{30}}\boldsymbol{i} - \frac{1}{\sqrt{30}}\boldsymbol{j} + \frac{5}{\sqrt{30}}\boldsymbol{k}$

Exercise 4.4

Determine the unit vectors of each of the following.

1. $\boldsymbol{a} = 3\boldsymbol{i} - 4\boldsymbol{j}$ 2. $\boldsymbol{b} = 2\boldsymbol{i} + \boldsymbol{j}$ 3. $\boldsymbol{c} = \boldsymbol{i} + 3\boldsymbol{j} - 2\boldsymbol{k}$ 4. $\boldsymbol{d} = -2\boldsymbol{i} + \boldsymbol{j} - \boldsymbol{k}$

4.3.6. Direction vectors

A vector which is used to specify the direction of another vector is called a **direction vector**.
For example if a vector $\boldsymbol{u}$ of magnitude 10 units is said to be in the direction of the vector $3\boldsymbol{i} + 4\boldsymbol{j}$, then $3\boldsymbol{i} + 4\boldsymbol{j}$ is the direction vector of $\boldsymbol{u}$.

The unit direction vector; $\hat{\boldsymbol{u}} = \frac{1}{5}(3\boldsymbol{i} + 4\boldsymbol{j})$
Hence $\boldsymbol{u} = |u| \times (unit\ direction\ vector)$
$= 10 \times \frac{1}{5}(3\boldsymbol{i} + 4\boldsymbol{j}) = 6\boldsymbol{i} + 8\boldsymbol{j}$

Example

Determine the components of the force, $\boldsymbol{F}$, which have a combined magnitude of 28 kN in the direction of the vector $6\boldsymbol{i} - 2\boldsymbol{j} - 3\boldsymbol{k}$
Solution

unit vector $= \frac{1}{7}(6\boldsymbol{i} - 2\boldsymbol{j} - 3\boldsymbol{k})$

$\boldsymbol{F} = 28 \text{ x } \frac{1}{7}(6i - 2j - 3k) = (24\boldsymbol{i} - 8\boldsymbol{j} - 12\boldsymbol{k})$ kN

Exercise 4.5

1. If a vector has magnitude 10 units and is in the direction of the vector $3\boldsymbol{i} - 4\boldsymbol{j} - 5\boldsymbol{k}$; determine the vector.
2. A force $\boldsymbol{F}$ of magnitude 36 kN is applied in the direction of the vector$\overrightarrow{AB}$, where $A = (7, 5, 14)$ and $B = (-1, 1, 15)$. Determine the force $\boldsymbol{F}$.
3. $\boldsymbol{F}_1$ has a magnitude of 26N in the direction of $12i + 5k$ and $\boldsymbol{F}_2$ has a magnitude of 18N in the direction of $2\boldsymbol{i} - 2\boldsymbol{j} + \boldsymbol{k}$. Determine the resultant force $\boldsymbol{F}_1 + \boldsymbol{F}_2$

4.3.7. The Negative or inverse of a vector

If $\overrightarrow{\boldsymbol{OB}} = \binom{3}{4}$ then $\overrightarrow{\boldsymbol{BO}} = -\overrightarrow{\boldsymbol{OB}} = \binom{-3}{-4}$

These can also be expressed as: $\overrightarrow{\boldsymbol{OB}} = 3\boldsymbol{i} + 4\boldsymbol{j}$

$\overrightarrow{\boldsymbol{BO}} = -3\boldsymbol{i} - 4\boldsymbol{j}$

If $\overrightarrow{\boldsymbol{AB}} = \begin{pmatrix} -2 \\ 5 \\ -1 \end{pmatrix}$ then $\overrightarrow{\boldsymbol{BA}} = \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$ or $\overrightarrow{\boldsymbol{AB}} = -2\boldsymbol{i} + 5\boldsymbol{j} - \boldsymbol{k}$ and $\overrightarrow{\boldsymbol{BA}} = 2\boldsymbol{i} - 5\boldsymbol{j} + \boldsymbol{k}$

4.3.8. Multiplication of a vector by a scalar (or number)

If $\overrightarrow{\boldsymbol{AB}} = \binom{x}{y}$ then $k\overrightarrow{\boldsymbol{AB}} = \binom{kx}{ky}$ where k is a scalar or number which can be negative or positive whole number or fraction. When k is positive, it implies that the vectors are parallel and in the same direction. When k is negative, it implies the vectors are parallel but in opposite directions.

Example

If $\boldsymbol{a} = -3\boldsymbol{i} + 5\boldsymbol{j}$ then $-2\boldsymbol{a} = 6\boldsymbol{i} - 10\boldsymbol{j}$ and $3\boldsymbol{a} = -9\boldsymbol{i} + 15\boldsymbol{j}$

4.3.9. Parallel Vectors

Two vectors are said to be parallel if one is a scalar multiple of the other. If the scalar multiple is positive, the vectors are in the same direction and if it is negative then the vectors are in opposite directions.

Example

Given the vector $\boldsymbol{a} = 2\boldsymbol{i} - 6\boldsymbol{j}$ the following vectors are parallel to $\boldsymbol{a}$:

$\boldsymbol{b} = 4\boldsymbol{i} - 12\boldsymbol{j}$ $\boldsymbol{c} = -\boldsymbol{i} + 3\boldsymbol{j}\boldsymbol{d} = 3i - 9j$

because $\boldsymbol{b} = 2\boldsymbol{ac} = -\frac{1}{2}\boldsymbol{a}$ and $\boldsymbol{d} = \frac{3}{2}\boldsymbol{a}$

Exercise 4.6

Which of the following is parallel to the vector $\boldsymbol{a} = 20\boldsymbol{i} + 8\boldsymbol{j}$?
$\boldsymbol{b} = 5\boldsymbol{i} + 2\boldsymbol{j}$, $\boldsymbol{c} = 10\boldsymbol{i} - 4\boldsymbol{j}$, $\boldsymbol{d} = -5\boldsymbol{i} - 2\boldsymbol{j}$, $\boldsymbol{e} = 4\boldsymbol{i} + 2\boldsymbol{j}$

4.3.10. Addition and subtraction of vectors

In addition or subtraction of vectors we add or subtract corresponding components

Example

If $\boldsymbol{a}$ and $\boldsymbol{b}$ are two vectors such that $\boldsymbol{a} = 3\boldsymbol{i} + \boldsymbol{j} - 4\boldsymbol{k}$ and $\boldsymbol{b} = 4\boldsymbol{i} + 5\boldsymbol{j} - \boldsymbol{k}$
then $\boldsymbol{a} + \boldsymbol{b} = 7\boldsymbol{i} + 6\boldsymbol{j} - 5\boldsymbol{k}$ (adding corresponding components)
$\boldsymbol{a} - \boldsymbol{b} = -\boldsymbol{i} - 4\boldsymbol{j} - 3\boldsymbol{k}$ (subtracting corresponding components)

Exercise 4.7

Given the vectors $\boldsymbol{a} = -3\boldsymbol{i} + 5\boldsymbol{j} + 2\boldsymbol{k}, \boldsymbol{b} = 3\boldsymbol{i} + \boldsymbol{j} - 4\boldsymbol{k}$, $\boldsymbol{c} = 4\boldsymbol{i} + 5\boldsymbol{j} - \boldsymbol{k}$ and
$\boldsymbol{d} = -\boldsymbol{i} - 4\boldsymbol{j} - 3\boldsymbol{k}$; do the following:
i) $2\boldsymbol{a} - \boldsymbol{b}$ ii) $3\boldsymbol{a} - 2\boldsymbol{c}$ iii) $\boldsymbol{b} + 2\boldsymbol{d}$ iv) $\boldsymbol{a} + (\boldsymbol{b} - 2\boldsymbol{c})$ v) $3\boldsymbol{c} - (\boldsymbol{a} + \boldsymbol{b})$

4.3.11. Scalar Product (Dot Product)

The scalar or dot product of two non-zero vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ is defined as
$\boldsymbol{a}.\boldsymbol{b} = a_1b_1 + a_2b_2 + a_3b_3$,
and geometrically as
$\boldsymbol{a}.\boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|cos\theta$, where θ is the angle between $\boldsymbol{a}$ and $\boldsymbol{b}$.

Example

Given the vectors $= 2i - j + 3k$; $\boldsymbol{b} = -i + 2j + 5k$ and $\boldsymbol{c} = 3i + 2j - k$, evaluate:
(i) $\boldsymbol{a} \cdot \boldsymbol{c}$ (ii) $\boldsymbol{b} \cdot \boldsymbol{c}$ (iii) $(\boldsymbol{a} + \boldsymbol{b}) \cdot \mathbf{c}$

Solution

i) $\boldsymbol{a} \cdot \boldsymbol{c} = (2 \times 3) + (-1 \times 2) + (3 \times -1) = 1$
ii) $\boldsymbol{b} \cdot \boldsymbol{c} = (-1 \times 3) + (2 \times 2) + (5 \times -1) = -4$
iii) $(\boldsymbol{a} + \boldsymbol{b}) \cdot \boldsymbol{c} = (1 \times 3) + (1 \times 2) + (5 \times -1) = 0$

Exercise 4.8

Calculate the dot (scalar) product for the following pairs of vectors

i) $\boldsymbol{m} = 3\boldsymbol{i} - \boldsymbol{j} - \boldsymbol{k}$, $\boldsymbol{n} = \boldsymbol{i} - \boldsymbol{j} - \boldsymbol{k}$

ii) $\boldsymbol{a} = 6\boldsymbol{i} + 2\boldsymbol{j} - 6\boldsymbol{k}, \quad \boldsymbol{b} = 9\boldsymbol{i} + 6\boldsymbol{j} + \boldsymbol{k}$

iii) $\boldsymbol{c} = 18\boldsymbol{i} - 6\boldsymbol{j} + \boldsymbol{k}, \quad \boldsymbol{d} = 8\boldsymbol{i} + 4\boldsymbol{j} + \boldsymbol{k}$

4.3.12. Angle between two vectors

When calculating the angle between two vectors, we use the equation

$\boldsymbol{a} \cdot \boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}| \cos\theta$

where θ is the angle between the vectors $\boldsymbol{a}$ and $\boldsymbol{b}$.

Note: the two vectors must be directed away from a common point or directed towards a common point.

θ can be an acute or obtuse angle. If θ is obtuse then $\boldsymbol{a} \cdot \boldsymbol{b}$ is negative.

Special cases

If the vectors are parallel; then $\theta = 0°$ and $\boldsymbol{a} \cdot \boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|$, because $\cos 0° = 1$

If the vectors are perpendicular; then $\theta = 90°$ and $a \cdot b = 0$ because $\cos 90° = 0$

Example

Given the vectors $\boldsymbol{m} = \boldsymbol{i} + 2\boldsymbol{j} + 3\boldsymbol{k}$ and $\boldsymbol{n} = 2\boldsymbol{i} - 3\boldsymbol{j} + \boldsymbol{k}$, determine the angle between $\boldsymbol{m}$ and $\boldsymbol{n}$.

Solution:

Let θ be the angle between the vectors.

From the equation $\boldsymbol{a}.\boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}| \cos\theta$, it follows that $\cos\theta = \frac{\boldsymbol{a}.\boldsymbol{b}}{|\boldsymbol{a}||\boldsymbol{b}|}$

$\therefore \cos\theta = \frac{\boldsymbol{m}.\boldsymbol{n}}{|\boldsymbol{m}||\boldsymbol{n}|}$. Thus $\boldsymbol{m}.\boldsymbol{n} = -1$, $|\boldsymbol{m}| = \sqrt{14}$ and $|\boldsymbol{n}| = \sqrt{14}$

$\cos\theta = \frac{-1}{14} = -0.0714$

$\therefore \theta = 180° - 85{,}904° = 94{,}096°$

Exercise 4.9

1. Determine the angle between the following pairs of vectors.

a) $\boldsymbol{F} = 3\boldsymbol{i} - \boldsymbol{j} - \boldsymbol{k}, \quad \boldsymbol{r} = \boldsymbol{i} - \boldsymbol{j} - \boldsymbol{k}$

b) $\boldsymbol{F} = 6\boldsymbol{i} + 2\boldsymbol{j} - 6\boldsymbol{k}, \quad \boldsymbol{r} = 9\boldsymbol{i} + 6\boldsymbol{j} + \boldsymbol{k}$

2. The velocity, $\boldsymbol{v}$, and acceleration, $\boldsymbol{a}$, of a particle are given by

$\boldsymbol{v} = 18\boldsymbol{i} - 6\boldsymbol{j} + \boldsymbol{k}$

$\boldsymbol{a} = 8\boldsymbol{i} + 4\boldsymbol{j} + \boldsymbol{k}$

Determine the angle between the velocity and the acceleration.

3. Determine the value(s) of x for which the angle between the two vectors $\boldsymbol{a} = 2\boldsymbol{i} + 2\boldsymbol{j} + \boldsymbol{k}$ and $\boldsymbol{b} = x\boldsymbol{i} + \boldsymbol{j} - \boldsymbol{k}$ is $90°$

4. Determine the value(s) of x if the vectors $\boldsymbol{a} = x\boldsymbol{i} - 2\boldsymbol{j} - 5\boldsymbol{k}$ and $\boldsymbol{b} = x\boldsymbol{i} + 3\boldsymbol{j} + 2\boldsymbol{k}$ are perpendicular.

4.3.13. Work done

If a constant force $\boldsymbol{F}$ moves a particle through a displacement vector $\boldsymbol{r}$, then the work done by the force $\boldsymbol{F}$ is given by
Work done= $\boldsymbol{F} \cdot \boldsymbol{r}$
NB: J =joule is the SI unit for work done.

Example

Determine the work done by a force, $\boldsymbol{F}$, in moving an object through a displacement $\boldsymbol{r}$ in the following where $\boldsymbol{F} = \boldsymbol{i} - 4\boldsymbol{j} + 3\boldsymbol{k}$, and $\boldsymbol{r} = 2\boldsymbol{i} - \boldsymbol{j} + 3\boldsymbol{k}$

Solution

$\boldsymbol{F} = \boldsymbol{i} - 4\boldsymbol{j} + 3\boldsymbol{k}$ and $\boldsymbol{r} = 2\boldsymbol{i} - \boldsymbol{j} + 3\boldsymbol{k}$

Work done $= \boldsymbol{F} \cdot \boldsymbol{r}$
$= (1 \times 2) + (-4 \times -1) + (3 \times 3)$
$= 15J$

Exercise 4.10

1. Determine the work done by a force, $\boldsymbol{F}$, in moving an object through a displacement $\boldsymbol{r}$ in each of the following:
   a) $\boldsymbol{F} = 3\boldsymbol{i} - \boldsymbol{j} - \boldsymbol{k}$, $\boldsymbol{r} = \boldsymbol{i} - \boldsymbol{j} - \boldsymbol{k}$
   b) $\boldsymbol{F} = 6\boldsymbol{i} + 2\boldsymbol{j} - 6\boldsymbol{k}$, $\boldsymbol{r} = 9\boldsymbol{i} + 6\boldsymbol{j} + \boldsymbol{k}$

4.3.14. Vector Product (Cross product)

The vector or cross product of two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ is defined geometrically as $\boldsymbol{a} \times \boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}| \sin\theta\hat{n}$, where θ is the angle between the vectors and $\hat{n}$ is the unit vector perpendicular to both $\boldsymbol{a}$ and $\boldsymbol{b}$.
Note: Rather than using the above formula to find the vector product, we can use determinant as a device for evaluating the vector product.

Thus if $\boldsymbol{p} = a\boldsymbol{i} + b\boldsymbol{j} + c\boldsymbol{k}$ and $\boldsymbol{q} = d\boldsymbol{i} + e\boldsymbol{j} + f\boldsymbol{k}$

Then $\boldsymbol{p} \times \boldsymbol{q} = det\begin{bmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ a & b & c \\ d & e & f \end{bmatrix} = \boldsymbol{i}\left[det\begin{pmatrix} b & c \\ e & f \end{pmatrix}\right] - \boldsymbol{j}\left[det\begin{pmatrix} a & c \\ d & f \end{pmatrix}\right] + \boldsymbol{k}\left[det\begin{pmatrix} a & b \\ d & e \end{pmatrix}\right]$

Example

1. Obtain the vector product of $\boldsymbol{m} = \boldsymbol{i} + 2\boldsymbol{j} + 3\boldsymbol{k}$ and $\boldsymbol{n} = 2\boldsymbol{i} - 3\boldsymbol{j} + \boldsymbol{k}$:

Solution:

$$\boldsymbol{m} \times \boldsymbol{n} = det\begin{bmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ 1 & 2 & 3 \\ 2 & -3 & 1 \end{bmatrix} = \boldsymbol{i}\left[det\begin{pmatrix} 2 & 3 \\ -3 & 1 \end{pmatrix}\right] - \boldsymbol{j}\left[det\begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}\right] + \boldsymbol{k}\left[det\begin{pmatrix} 1 & 2 \\ 2 & -3 \end{pmatrix}\right]$$
$$= 11\boldsymbol{i} + 5\boldsymbol{j} - 7\boldsymbol{k}$$

Exercise 4.11

Given the three vectors $\boldsymbol{a} = 2\boldsymbol{i} - \boldsymbol{j} + \boldsymbol{k}$; $\boldsymbol{b} = 2\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$ and $\boldsymbol{c} = -\boldsymbol{i} + 2\boldsymbol{j} - \boldsymbol{k}$, determine:

i) $\boldsymbol{a} \times \boldsymbol{c}$

ii) $\boldsymbol{a}.(\boldsymbol{c} \times \boldsymbol{a})$

iii) $(\boldsymbol{a} \times \boldsymbol{c}) \times \boldsymbol{b}$

4.3.15. Moment of a vector

The moment of a vector is a generalization of the moment of a force. A moment of a force is the tendency of a force to turn or rotate an object. The moment vector, $\boldsymbol{M}$, of a force, $\boldsymbol{F}$, about the origin O is defined as $\boldsymbol{M} = \boldsymbol{r} \times \boldsymbol{F}$ where $\boldsymbol{r}$ is the position vector from O of any point on the line of action of $\boldsymbol{F}$.

Example

Determine the moment vector, $\boldsymbol{M}$, about the origin, O, of a force $\boldsymbol{F} = 10\boldsymbol{i} - \boldsymbol{j} + 2\boldsymbol{k}$, passing through the point with position vector $\boldsymbol{r} = 2\boldsymbol{i} - 3\boldsymbol{j} + 5\boldsymbol{k}$.

Solution:

The moment vector, $\boldsymbol{M}$, of a force, $\boldsymbol{F}$, about the origin O is defined as $\boldsymbol{M} = \boldsymbol{r} \times \boldsymbol{F}$ where $\boldsymbol{r}$ is the position vector from O of any point on the line of action of $\boldsymbol{F}$.
Hence

$$\boldsymbol{r} \times \boldsymbol{F} = det\begin{bmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ 2 & -3 & 5 \\ 10 & -1 & 2 \end{bmatrix} = \boldsymbol{i}\left[det\begin{pmatrix} -3 & 5 \\ -1 & 2 \end{pmatrix}\right] - \boldsymbol{j}\left[det\begin{pmatrix} 2 & 5 \\ 10 & 2 \end{pmatrix}\right] + \boldsymbol{k}\left[det\begin{pmatrix} 2 & -3 \\ 10 & -1 \end{pmatrix}\right]$$
$$= -\boldsymbol{i} + 46\boldsymbol{j} + 28\boldsymbol{k}$$

Exercise 4.12

1. Determine the moment vector, $\boldsymbol{M}$, about the origin, O, of a force $\boldsymbol{F} = 5\boldsymbol{i} - \boldsymbol{j} + 7\boldsymbol{k}$ passing through the point with position vector $\boldsymbol{r} = \boldsymbol{i} + 3\boldsymbol{j} + \boldsymbol{k}$.

2. Let A and B have position vectors $\boldsymbol{a} = 7\boldsymbol{i} - 4\boldsymbol{j} - \boldsymbol{k}$ and $\boldsymbol{b} = 2\boldsymbol{i} - 3\boldsymbol{j} + \boldsymbol{k}$ respectively. The force $\boldsymbol{F} = 3\boldsymbol{i} - \boldsymbol{j} + 5\boldsymbol{k}$ passes through the point A. Determine the vector moment of $\boldsymbol{F}$ about the point B.

REVISION EXERCISE: UNIT 4

1. Given that $\boldsymbol{a} = 2\boldsymbol{i} + 3\boldsymbol{j} - 5\boldsymbol{k}$, $\boldsymbol{b} = \boldsymbol{i} - 4\boldsymbol{j} + 6\boldsymbol{k}$ and $\boldsymbol{c} = -3\boldsymbol{i} - 2\boldsymbol{j} + 4\boldsymbol{k}$, evaluate:

$a)\ 3\boldsymbol{a} - 4\boldsymbol{c}$ $\quad b)\ 2\boldsymbol{b} + 3\boldsymbol{c}$ $\quad c)\boldsymbol{a} + \boldsymbol{c}$

2. Given the vectors $\boldsymbol{a} = 3\boldsymbol{i} - 4\boldsymbol{j}\boldsymbol{b} = \boldsymbol{i} + 2\boldsymbol{j}$ and $\boldsymbol{c} = 2\boldsymbol{i} - \boldsymbol{j}$

If $\boldsymbol{a} = k\boldsymbol{b} + m\boldsymbol{c}$ where k and m are scalars, find the values of k and m.

3. $\boldsymbol{c} = 24\boldsymbol{i} + x\boldsymbol{j}$ and $\boldsymbol{d} = 16\boldsymbol{i} + 20\boldsymbol{j}$. Find x if $\boldsymbol{c}$ is parallel to $\boldsymbol{d}$.

4. The following forces have units of kN:
   $\boldsymbol{F}_1 = 4\boldsymbol{i} - 6\boldsymbol{j}$
   $\boldsymbol{F}_2 = 7\boldsymbol{i} - 12\boldsymbol{j}$

Determine the magnitude of the force $\boldsymbol{F} = 3\boldsymbol{F}_1 - 3\boldsymbol{F}_2$.

5. Determine the unit vector and the magnitude of each of the following:
$\boldsymbol{a} = 2\boldsymbol{i} - \boldsymbol{j} - 3\boldsymbol{k}$, $\boldsymbol{b} = 7\boldsymbol{i} - 7\boldsymbol{j} - 7\boldsymbol{k}$ and $\boldsymbol{c} = 9\boldsymbol{i} - 6\boldsymbol{j} + \boldsymbol{k}$.

6. The magnitude and direction of a force $\boldsymbol{F}$, are given by 375.4 and 205.32° respectively. Resolve the force $\boldsymbol{F}$ into its x- and y- components.

7. The tension $\boldsymbol{T}$ in a cable is 85.0 N. If the cable makes an angle of 53.5° with the horizontal, determine the horizontal and vertical components of the tension.

8. If the position vectors of the points P and Q are $\boldsymbol{i} + 3\boldsymbol{j} - 7\boldsymbol{k}$ and $5\boldsymbol{i} - 2\text{j} + 4\boldsymbol{k}$ respectively, find $\overrightarrow{PQ}$ and determine its length.

9. Determine the scalar products of the following:
   i) $(\boldsymbol{i} + 3\boldsymbol{j} + \boldsymbol{k}) \cdot (5\boldsymbol{i} - \boldsymbol{j} + 7\boldsymbol{k})$
   ii) $(-\boldsymbol{i} + \boldsymbol{j} + \boldsymbol{k}) \cdot (3\boldsymbol{i} + 2\boldsymbol{j} - 3\boldsymbol{k})$
   iii) $(10\boldsymbol{i} - 15\boldsymbol{j} + \boldsymbol{k}) \cdot (-21\boldsymbol{i} - \boldsymbol{k})$

10. Let the velocity, $\boldsymbol{v}$, and acceleration, $\boldsymbol{a}$, be given by: $\boldsymbol{v} = -4\boldsymbol{i} - 8\boldsymbol{j} + 4\boldsymbol{k}$ and

    $\boldsymbol{a} = 2\boldsymbol{i} + \boldsymbol{j} + 4\boldsymbol{k}$. Determine the angle between the velocity and the acceleration.

11. A force $\boldsymbol{F} = \boldsymbol{i} - \boldsymbol{j} + 3\boldsymbol{k}$ moves an object through a displacement $\boldsymbol{s} = 15\boldsymbol{i} + 20\boldsymbol{j} + 7\boldsymbol{k}$. Calculate the work done by the force $\boldsymbol{F}$.

12. Calculate the work done by the force $\boldsymbol{F} = 6\boldsymbol{i} - 3\boldsymbol{j} + 2\boldsymbol{k}$ in moving an object from point $(1\,;4\,;-2)$ to the point$(3\,;2\,;\ 5)$.

13. If $\boldsymbol{r} = 5\boldsymbol{i} + \boldsymbol{j} + 3\boldsymbol{k}$ and $\boldsymbol{s} = 2\boldsymbol{i} - 3\boldsymbol{j} - 4\boldsymbol{k}$, determine the vector products $\boldsymbol{r} \times \boldsymbol{s}$ and then $\boldsymbol{s} \times \boldsymbol{r}$. What can you say about your answers?

14. Determine the moment vector, M, about the origin, O, of a force $\boldsymbol{F} = 2\boldsymbol{i} - \boldsymbol{j} + 2\boldsymbol{k}$ passing through the point with position vector $\boldsymbol{r} = \boldsymbol{i} + 3\boldsymbol{j} + \boldsymbol{k}$.

15. If the angle between the vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ is 90°, determine the value(s) of x where $\boldsymbol{a} = 2\boldsymbol{i} + \boldsymbol{j} + 2x\boldsymbol{k}$ and $\boldsymbol{b} = 3\boldsymbol{i} - 2\boldsymbol{j} + 6\boldsymbol{k}$.

16. If the vectors $\boldsymbol{a} = -2\boldsymbol{i} + x\boldsymbol{j} + 2\boldsymbol{k}$ and $\boldsymbol{b} = 3\boldsymbol{i} + x\boldsymbol{j} - 5\boldsymbol{k}$ are perpendicular, determine the value(s) of x.

UNIT 5

COMPLEX NUMBERS

5.1. INTRODUCTION

In solving equations such as $x^2 + 25 = 0$ there are no real numbers which satisfy such polynomial equation. In order to permit solutions of this and similar equations the set of complex numbers, is introduced.

By the end of this unit the learner should be able to:

- Solve polynomial equations,
- Define a complex number and distinguish it from a real number,
- Perform algebraic operations with complex numbers in rectangular form,
- Represent complex numbers in an Argand diagram,
- Represent complex numbers in a polar form,
- Use De Moivre's Theorem to simplify complex numbers and to calculate roots of numbers.

5.2. REPRESENTATION OF A COMPLEX NUMBER

By a complex number we mean an ordered pair $(a\,;\,b)$ of real numbers, where a is called the real part and b is called the imaginary part.

From now on we adopt the system $\{a + bj,\ a, b \in R,\ j^2 = -1\}$. In this system we write $z = a + bj$. We call a the real part of z $(Re\{z\})$ and b the imaginary part of z $(Im\{z\})$.

Since there are no real numbers which satisfy the solution of equations such as $x^2 = -25$, we must define a number system to include square roots of negative numbers. In the above equation, $= \sqrt{-25}$. We then write it as $x = \sqrt{25} \cdot \sqrt{-1}$. The symbol $\sqrt{-1}$ is the imaginary unit and is denoted by the j. Hence $j = \sqrt{-1}$ =>$j^2 = -1$ and $j^4 = 1$

Now with this system we can solve quadratic equations as in the following examples.

Example

1. Solve the quadratic equations: a) $x^2 + 25 = 0$ b) if $x^2 + 2x + 10 = 0$

Solution

a) $x^2 + 25 = 0$

$x^2 = -25$

$\therefore x = \pm\sqrt{-25} = \pm\sqrt{25} \cdot \sqrt{-1} = \pm 5j$

b) $x^2 + 2x + 10 = 0$

Using the quadratic formula, we have

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{-36}}{2}$

$= \frac{-2 \pm 6j}{2} = -1 \pm 3j$

Exercise 5.1

Solve the following quadratic equations

a) $x^2 + 4 = 0$ b) $2x^2 + 3x + 5 = 0$ c) $x^2 + 2x + 10 = 0$ d) $3x^2 - 5x + 4 = 0$

e) $x^2 = 3x - 4$

5.3. ADDITION AND SUBTRACTION OF COMPLEX NUMBERS

In addition or subtraction of complex numbers, we add (or subtract) real parts and then add or subtract imaginary parts.

Example

1. $(5 + 2j) + (-3 - j) = (5 - 3) + (2 - 1)j = 2 + j$
2. $(3 - 5j) - (4 + 2j) = (3 - 4) + (-5 - 2)j = -1 - 7j$

Exercise 5.2

a) $(9 + 2j) + (8 + 6j)$ b) $(6 + 3j) + (5 - 7j)$ c) $(-4 + 3j) + \left(-1 - \sqrt{-4}\right)$

b) d) $(3 + 4j) - (2 + j)$ e) $(-8 + 4j) - (3 + 10j)$

5.4. MULTIPLICATION OF COMPLEX NUMBERS

Multiplication of complex numbers is achieved by assuming that all quantities are real numbers and using $j^2 = -1$ to simplify.

Example

$(2 - 3j)(2 + 4j) = 4 + 8j - 6j - 12j^2 = 4 + 2j + 12 = 16 + 2j$

Exercise 5.3

$(2 + 5j)(3 - 4j)$ b) $(5 + 3j)^2$ c) $(3 + 4j)(3 - 4j)$ d) $(j - 3)(4 - 2j)$

e) $(j - 3)(4 - 2j)$ f) $(3j + 5)(-3 - j)$ g) $(4 - 3j)^2$

5.5. DIVISION OF COMPLEX NUMBERS

If $z = a + bj$, then $\bar{z} = a - bj$ is called the complex conjugate of z. The conjugate of a complex number is obtained by changing the sign of the imaginary part.

The conjugate of a complex number is particularly useful in division of complex numbers.

Example

If $z = 2 - 3j$ *then* $\bar{z} = 2 + 3j$. Now simplify $\frac{4-j}{2-3j}$ and express your answer in the standard form.

$$\frac{4-j}{2-3j} = \left(\frac{4-j}{2-3j}\right)\left(\frac{2+3j}{2+3j}\right) = \frac{11}{13} + \frac{10}{13}j$$

Exercise 5.4

1. Write down the conjugates of the following complex numbers:

a) $3 + 4j$ b) $3 - 5j$ c) $7j$

2. Simplify and write the answers in the form $a + bj$

a) $\frac{4-3j}{7+2j}$ c) $\frac{2+2j}{3j-4}$

b) $\frac{1}{1+j} - \frac{3}{2-j} + \frac{7}{1-j}$ d) $\frac{j-3}{2j+3}$

5.6. EQUALITY OF COMPLEX NUMBERS

Suppose $z_1 = a + bj$ and $z_2 = x + yj$ are complex numbers. If $z_1 = z_2$, then $a = x$ and $b = y$

Example

Determine a and b if:

a) $a + bj = 3 - 5j$
b) $a + bj = (2 + 5j) - (4 + 3j)$
c) $a + bj = (2 + j)^2$

Solution

a) $a + bj = 3 - 5j$

$=> a = 3 \; and \; b = -5$

b) $a + bj = (2 + 5j) - (4 + 3j)$

$= -2 + 2j$

$=> a = -2$ and $b = 2$

c) $a + bj = (2 + j)^2$

$= 4 + 4j + j^2 = 3 + 4j$

$=> a = 3 \ and \ b = 4$

Exercise 5.5

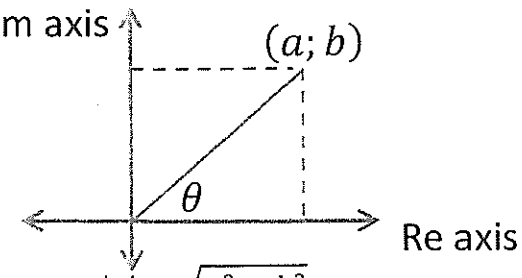
Solve the following equations for $x \ and \ y$:

a) $2(x + yj) = 6 - 3j$ b) $(x - 2yj) + (y - 3xj) = 2 + 3j$ c) $2 - 3j = \sqrt{x + yj}$

d) $(1 + 2j)(-2 - 3j) = x + yj$ e) $x(\overline{3 + j} - 12 = yj - j^5)$

5.7. ARGAND DIAGRAM AND POLAR FORM OF COMPLEX NUMBERS

We have seen that a complex number $z = a + bj$ can be considered as an ordered pair (a, b) of real numbers. This representation facilitates the representation of complex numbers as points in the XY-plane called the complex plane or Argand diagram.



$$r = |z| = \sqrt{a^2 + b^2}$$

$$\cos\theta = \frac{a}{r} => r\cos\theta = a$$

$$\sin\theta = \frac{b}{r} => r\sin\theta = b$$

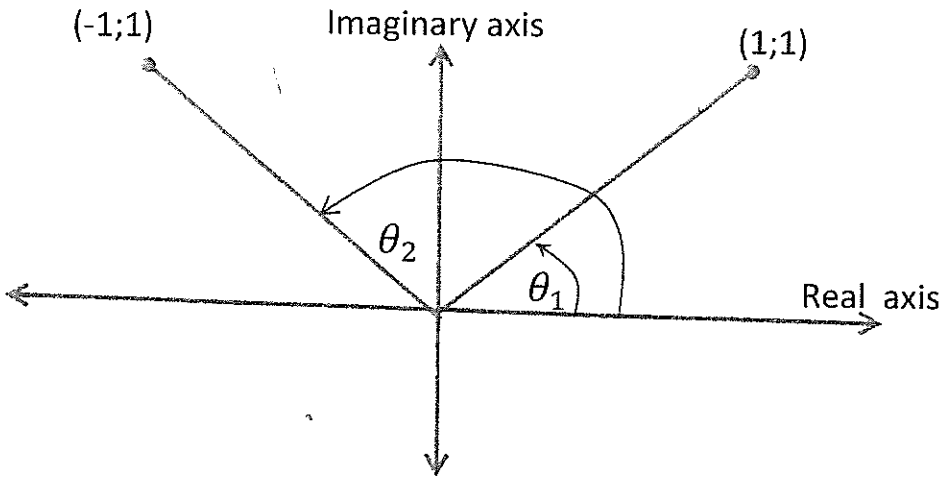
Using polar coordinates $(r, \ \theta)$ in the Argand diagram, we can write $z = (a\,; b) = a + bj = r\cos\theta + jr\sin\theta = r(\cos\theta + j\sin\theta) = rcis\theta$ for short. Thus $r(\cos\theta + j\sin\theta) = rcis\theta$ is the polar form of the complex number $z = a + bj$

$r = |z| = \sqrt{a^2 + b^2}$ is the modulus of z. We call θ the argument of z denote $arg\{z\}$.

$\tan\theta = \frac{Im\{z\}}{Re\{z\}}$, provided $Re\{z\} \neq 0$. However, in general, $\tan^{-1}\left(\frac{Im\{z\}}{Re\{z\}}\right)$

Determines θ only up to multiples of π.

The Argand diagram for $z_1 = 1 + j \; and \; z_2 = -1 + j$



Thus (r, θ) determines z uniquely, that is, (r, θ) determines (a, b) uniquely by $a = r\cos\theta, b = r\sin\theta$.

The unique real number θ in$a = |z|\cos\theta \;\; and \; b = |z|\sin\theta, \; -\pi < \theta \leq \pi$, is called the principal argument of $z, denoted \; by \; arg\{z\}$.

Every complex number $z \neq 0$ can be represented in the form

$z = r(\cos\theta + jsin\,\theta)$, where $r = |z|$and $\theta = arg(z) + 2\pi k, k \in Z$. The argument θ will satisfy the equations $|z|\cos\theta = Re\{z\}, \; |z|\sin\theta = Im\{z\}$ and hence $\tan\theta = \frac{Im\{z\}}{Re\{z\}}$, provided $Re\{z\} \neq 0$. However, in general, $\tan^{-1}\left(\frac{Im\{z\}}{Re\{z\}}\right)$ determines θ only up to multiples of π.

- $arg(1 + j) = \frac{\pi}{4} = 45°$
- $arg(-1 + j) = \frac{3\pi}{4} = 135°$
- $arg(-1 - j) = \frac{-3\pi}{4} = -135°$
- $arg(1 - j) = \frac{-\pi}{4} = -45°$
- $arg(1) = 0$
- $arg(j) = \frac{\pi}{2} = 90°$
- $arg(-1) = \pi = 180°$

- $arg(-j) = \frac{-\pi}{2} = -90°$

Example

Convert the following into polar forms:

1. $z = 1 + j$ 2. $z = -1 + j$

Solution

1. $z = 1 + j$ is in the first quadrant, and so θ will be positive and acute.

$\theta = \tan^{-1} 1 = 45°$ $r = \sqrt{1+1} = \sqrt{2}$

Thus $z = 1 + j = \sqrt{2}(\cos 45° + j \sin 45°)$

2. $z = -1 + j$ is in the second quadrant, and so $\theta = 180° - |\tan^{-1}(-1)| = 135°$

Thus $z = -1 + j = \sqrt{2}(\cos 135° + j \sin 135°)$

Exercise 5.6

Convert the following complex numbers to polar forms:

1. $3 + 4j$ 2. $3 - 4j$ 3. $-2 - 2j$ 4. $\sqrt{3}j + 1$ 5. $j - \sqrt{3}$

5.8. DE MOIVRE'S THEOREM

If$z_1 = r_1 cis\theta_1$ $z_2 = r_2 cis\theta_2$, then $z_1 z_2 = r_1 r_2 cis(\theta_1 + \theta_2)$.

$\frac{z_1}{z_2} = \frac{r_1}{r_2} cis(\theta_1 - \theta_2)$

From the above theorem we also have the following:

If $z = rcis\theta$. Then $z^n = r^n cis\, n\theta$

Example

Determine the following, using De Moivre's theorem, and write your answer in the standard form:

1. $(1 + j)^5$ 2. $(2 - 2j)^{-4}$

Solution

1. $(1 + j)^5 = \left(\sqrt{2}\right)^5 (cis\ 5 \cdot 45)$

$= 4\sqrt{2}(\cos 225° + j \sin 225°)$

$= -4 - 4j$

2. $(2-2j)^{-4} = \left(2\sqrt{2}\right)^{-4}[cis(-4\cdot -45°)] \ = \frac{1}{64}(\cos 180° + j\sin 180°) \ = -\frac{1}{64}$

Exercise 5.7

Determine the following and write your answer in standard form. Round off your answers to two decimals where necessary.

1. $(-1-j)^3$ 2. $(5-2j)^{-2}$ 3. $\left(\frac{1-3j}{j+1}\right)^5$ 4. $(j-1)^2(3-j)^3$

5.9. USING DE MOIVRE'S THEOREM TO CALCULATE ROOTS

De Moivre's theorem can be used to calculate roots of both real and complex numbers.

First the number is expressed in polar form. Then express the root as a fractional exponent, i.e. $(rcis\,\theta)^n = r^n cis\, n\theta$.

Add 360° to θ, $n-1$ times to find the other roots.

$z^{\frac{1}{n}} = r^{\frac{1}{n}}\left[\cos\frac{1}{n}(\theta+2\pi k) + j\sin\frac{1}{n}(\theta+2\pi k)\right]$ where $k = 0, 1, \ldots (n-1)$.

Example

Calculate the following by using De Moivre's theorem and write your answers in the standard form.

1. The cube root of -1.
2. The two square roots of 3j

Solution

1. First we express -1 in polar form

Thus $-1 = 1(\cos 180° + j\sin 180°) = cis\ 180°$

$(-1)^{\frac{1}{3}} = \cos\frac{180°}{3} + j\sin\frac{180°}{3}$

$(-1)^{\frac{1}{3}} = \cos 60° + j\sin 60° = 0.5 + 0.8660j$

For other root we add 360° to $\theta(180°)$

$(-1)^{\frac{1}{3}} = \cos\frac{540°}{3} + \sin\frac{540°}{3}$

$(-1)^{\frac{1}{3}} = \cos 180° + j\sin 180° = -1$

Next we add 360° to 540°

$(-1)^{\frac{1}{3}} = \cos\frac{900°}{3} + j\sin\frac{900°}{3}$

$(-1)^{\frac{1}{3}} = \cos 300° + j\sin 300° = 0.5 - 0.8660j$

Therefore the three cube roots of -1 are -1, $0.5 + 0.8660j$ and $0.5 - 0.8660j$

2. Express $3j$ in polar form

Thus $3j = 3(\cos 90° + j\sin 90°)$

$(3j)^{\frac{1}{2}} = \sqrt{3}(\cos 45° + j\sin 45°) = 1.225 + 1.225j$

For the other root add 360° to 90°

$(3j)^{\frac{1}{2}} = \sqrt{3}(\cos 225° + j\sin 225°) = -1.225 - 1.225j$

Therefore the two square roots of 3j are $1.225 + 1.225j$ and $-1.225 - 1.225j$

Exercise 5.8

Use De Moivre's theorem to calculate all the indicated roots.

a) The square roots of $1 - j$

b) The three cube roots of $-1 + j$

c) The three cube roots of $3 + 4j$

d) The square roots -2

1. Determine all the roots of the given equations.
   a) $x^3 - 8 = 0$ b) $x^4 - 1 = 0$ c) $x^5 + 32 = 0$

REVISION EXERCISE UNIT 5

1. Solve the following quadratic equations. In each case express your answers in the standard form.
   a) $x^2 + 9 = 0$ b) $x^2 + x + 1 = 0$ c) $5x^2 - 6x + 5 = 0$ d) $3x^2 - 4x + 2 = 0$
2. Simplify the following
   a) $(2 + 5j) - (3 + 8j)$ b) $(3 + 4j)(j - 2)$ c) $(5 - j)^2$ d) $(4 - 3j)(4 + 3j)$

3. Determine the following
   a) $\frac{2+3j}{4-3j}$ b) $\frac{1-3j}{j+2}$ c) $\frac{4}{-5j}$ d) $\frac{(1+j)^2}{j-4}$
4. Determine the values of x and y in each of the following equations:
   a) $2x + yj = (1 + 2j)^2$ b) $x + yj = \frac{1}{3-4j}$ c) $x(3 + j) - 12 = yj - j^5$
5. Change each of the following complex numbers to polar form.
   a) $z = 3 + 4j$ b) $z = -5 - 12j$ c) $z = -1 + 3j$ d) $z = \sqrt{3} - j$

6.1 Use De Moivre's theorem to simplify each of the following. Leave your answer in standard form.

a) $(-2-j)^{-3}$ b) $\frac{(-3+j)^3}{(-1-j)^4}$ c) $(-3+j)^3(-1-j)^4$

6.2. If $z = \frac{1}{2+3j} + \frac{1}{1-2j}$. Express z in the form $a + bj$.

7.1 Use De Moivre's theorem to calculate all the indicated roots.

a) The square roots of -1 b) The cube roots of $3j$

7.2 Determine all the roots of the given equations

a) $x^2 - 3 = 0$ b) $x^3 - 2 = 0$ c) $x^3 + 1 = 0$

8. In electrical engineering it is customary to use the complex numbers E, I and Z to designate, respectively, the voltage, current and impedance. If $E = IZ$, find the third quantity (in simplified form) when:

a) $E = 100 + 40j\ volts$ $I = 4 + 3j\ amp$

b) $I = 2 + 3j\ amp$ $Z = 30 - 12j\ ohms$

c) $E = 120\ volts$ $Z = 20 - 20j\ ohms$

UNIT 6

STATISTICS

6.1. INTRODUCTION

Statistics is the study of methods for collecting, organizing, presenting, analysing data and making inferences. Suppose we wish to determine whether there is a relationship between lengths and masses of the male population of a certain city. It is impossible to measure every man and to determine his mass. Hence only a certain number of men will be measured and weighed. We say that we are taking a sample from the total population. There are many ways in which samples may be taken to ensure randomness. The information contained in a sample is called data. Discrete data is usually obtained by counting and continuous data by measuring.

After completion of this unit the leaner should be able to:

- Construct a frequency table of collected data.
- draw a histogram, and cumulative frequency curve (ogive)
- Calculate the mean of a raw data and grouped data.
- Calculate the median and the mode of raw data
- Calculate the standard deviation and the variance of raw and grouped data.
- Determine the equation of a linear regression line and determine the value of the correlation coefficient.

6.2. DATA COLLECTION

Data are collected using various methods. The main data collection methods are interviews, questionnaires and observation. As an example consider the numbers below which represent IQ scores for a random sample of 30 students.

109, 106, 106, 110, 108, 110, 108, 106, 110, 108, 110, 112, 110, 109, 109, 112, 107, 113, 108, 107, 113, 109, 111, 111, 111, 108, 109, 112, 112, 111

Summarizing this data set, it can be transformed into tabular form.

6.3. TABULATION.

Tabulation means we are putting our data into tables. This has the advantage of summarizing large data sets. There are two types of tables which are the frequency table and frequency distribution table.

6.3.1. Frequency Table

This is a tabular arrangement of data by grouping them into classes, together with their corresponding class frequencies. Frequency is the number of times an individual item occurs under a particular class.

An example of a frequency table is shown below

Marks	No. of students(frequency)
0	5
1	8
2	12
3	15

6.3.2. Frequency Distribution Table

A grouped frequency distribution table is used to show the number of values (frequency) for each group as shown in the table below:

Class marks	No. of students (frequency)
$10-19$	2
$20-29$	5
$30-39$	9
$40-49$	15

The range of possible values for a group or a class is called **class interval**. For example $10-19$ in the above table is called a **class interval**. The end marks are called **class limits**. The smaller number and the larger number are the lower and upper class limits respectively.

Class boundaries: Class boundaries separate one class from another.

To find the class boundaries from a grouped data, we follow the steps below:

First find the difference between the upper class limit of a particular class interval and the lower class limit of the next class and divide by 2, then subtract the value obtained from the lower class limit of each class and add (the same) to the upper class limit of each class to obtain the class boundaries.

For example

Class marks	Frequency	Class boundaries
$10-19$	2	$9{,}5-19{,}5$
$20-29$	5	$19{,}5-29{,}5$
$30-39$	9	$29{,}5-39{,}5$
$40-49$	15	$39{,}5-49{,}5$

6.4. DATA REPRESENTATION

This is an arrangement of the data in diagrammatic form in such a way that it creates interest. It is also compact, concise and in simple form but at the same time it can aid further statistical analysis.

6.4.1 Histogram

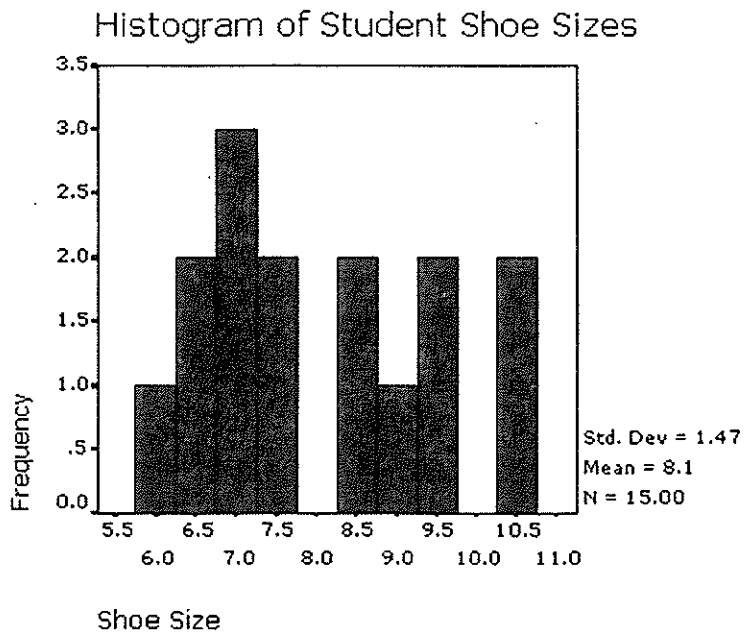
A histogram consists of a set of rectangles where the area of the rectangle is proportional to the frequency having a base on a horizontal axis (x-axis) and there is no gap between the rectangles.

Example

The table below represents the shoe sizes of 15 students. Draw a histogram for the distribution.

Frequency table: Student shoe sizes

Shoe size	6,0	6.5	7,0	7,5	8,0	8,5	9,0	9,5	10,0	10,5
Frequency	1	2	3	2	0	2	1	2	0	2



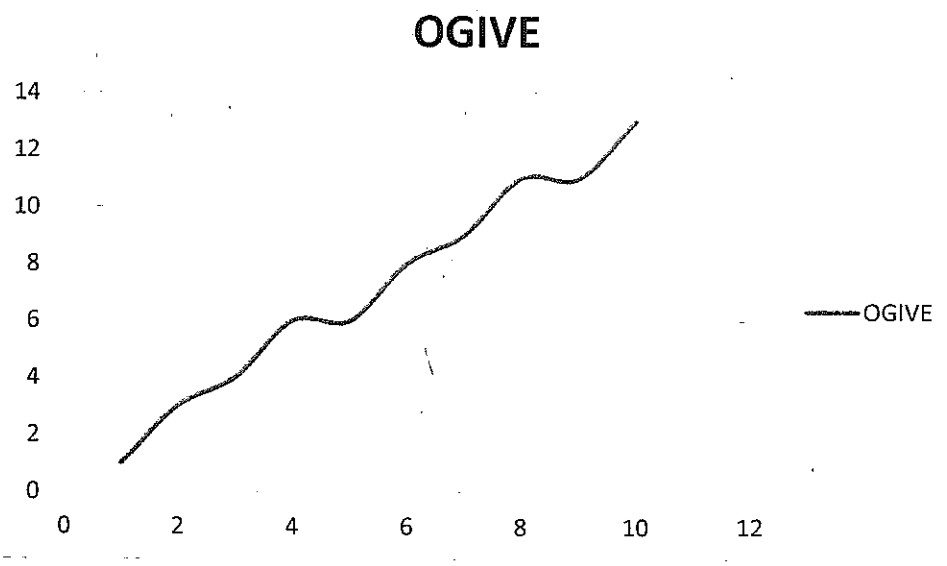
6.4.2. Cumulative Frequency Curve (Ogive)

An Ogive can be drawn from a cumulative frequency table. On the graph, the upper class boundaries are always along the horizontal axis and the cumulative frequencies along the vertical axis. Joining all the consecutive points by the best shape curve is called an ogive. An example of an ogive is shown below from the previous data.

Frequency table: Student shoe sizes

Shoe size	6,0	6.5	7,0	7,5	8,0	8,5	9,0	9,5	10,0	10,5
Frequency	1	2	3	2	0	2	1	2	0	2

Cumulative Frequency	1	3	4	6	6	8	9	11	11	13



Exercise 6.1

Construct a histogram for the frequency distribution for the record high temperatures of the 50 states.

Class Limits	Class Boundaries	Frequency (*f*)
100-104	99.5-104.5	2
105-109	104.5-109.5	8
110-114	109.5-114.5	18
115-119	114.5-119.5	13
120-124	119.5-124.5	7
125-129	124.5-129.5	1
130-134	129.5-134.5	1

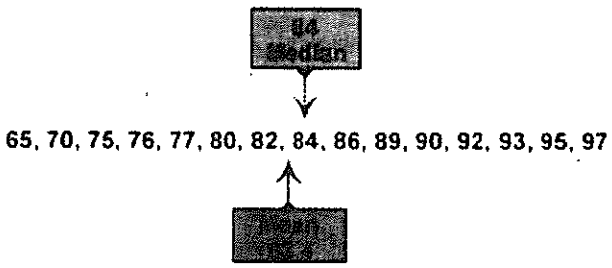
6.5. MEASURES OF CENTRAL TENDENCY.

Data tends to be clustered in the middle and to measure this tendency, we use the three averages, **Mean, median and mode.**

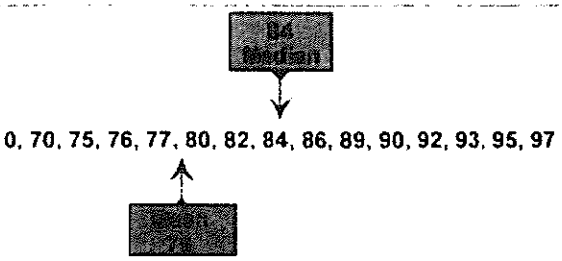
6.5.1. Ungrouped Data

Mean:	**Average.** The sum of a set of data divided by the number of observations. (Do not round your answer unless directed to do so.)
Median:	The **middle** value or the mean of the two middle values, **when the data are arranged in numerical order.** Think of a "median" being in the middle of a highway.
Mode:	The value (number) that appears **most** often. It is possible to have more than one mode, and it is possible to have no mode. If there is no mode-write "no mode", do not write zero (0).

Consider this set of test score values:



Normal listing of scores.



Scores with the lowest score replaced with outlier.

The two sets of scores above are identical except for the first score. The set on the left shows the actual scores. The set on the right shows what would happen if one of the scores was WAY out of range in regard to the other scores. Such a term is called an **outlier.**

With the outlier, the mean changed.
With the outlier, the median did NOT change.

How do I know which measure of central tendency to use?

MEAN

Use the mean to describe the middle of a set of data that *does not* have an outlier.

Advantages:

- **Most** popular measure in fields such as business, engineering and computer science.
- It is unique - there is only one answer.
- Useful when comparing sets of data.

Disadvantages:

- Affected by extreme values (outliers)

MEDIAN

Use the median to describe the middle of a set of data that *does* have an outlier.

Advantages:

- **Extreme** values (outliers) do not affect the median as strongly as they do the mean.
- Useful when comparing sets of data.
- It is unique - there is only one answer.

Disadvantages:

- Not as popular as mean.

MODE

Use the mode when the data is non-numeric or when asked to choose the most popular item.

Advantages:

- Extreme values (outliers) do not affect the mode.

Disadvantages:

- Not as popular as mean and median.
- Not necessarily unique - may be more than one answer
- When no values repeat in the data set, the mode is every value and is useless.
- When there is more than one mode, it is difficult to interpret and/or compare.

What will happen to the measures of central tendency if we add the same amount to all data values, or multiply each data value by the same amount?

	Data	Mean	Mode	Median
Original Data Set:	6, 7, 8, 10, 12, 14, 14, 15, 16, 20	12.2	14	13
Add 3 to each data value	9, 10, 11, 13, 15, 17, 17, 18, 19, 23	15.2	17	16
Multiply 2 times each data value	12, 14, 16, 20, 24, 28, 28, 30, 32, 40	24.4	28	26

When added: Since all values are **shifted** the same amount, the measures of central tendency all shifted by the same amount. If you add 3 to each data value, you will add 3 to the mean, mode and median.

When multiplied: Since all values are affected by the same multiplicative values, the measures of central tendency will feel the same effect. If you multiply each data value by 2, you will multiply the mean, mode and median by 2.

Example

1. Find the mean, median and mode for the following data: 5, 15, 10, 15, 5, 10, 10, 20, 25, 15.
Arranging data in ascending order: **5, 5, 10, 10, 10, 15, 15, 15, 20, 25**

Mean: $\bar{x} = \frac{\sum x_i}{n}$

$$\frac{sum\ of\ data}{number\ of\ data} = \frac{130}{10} = 13$$

Median: 5, 5, 10, 10, **10, 15**, 15, 15, 20, 25

Listing the data in order is the easiest way to find the median.
The **numbers 10 and 15 both** fall in the middle.
Average these two numbers to get the median. $\frac{10 + 15}{2}$ = **12.5**

Mode: Two numbers appear **most often: 10 and 15.**
There are three 10's and three 15's.
In this example there are two answers for the mode.

2. On his first 5 biology tests, Bob received the following scores: 72, 86, 92, 63, and 77. What test score must Bob earn on his sixth test so that his average (mean score) for all six tests will be 80?

Solution:

Set up an equation to represent the situation. Remember to use all 6 test scores:

$$\frac{72 + 86 + 92 + 63 + 77 + x}{6} = 80$$

cross multiply and solve:

$$(80)(6) = 390 + x$$
$$480 = 390 + x$$
$$90 = x$$

Bob must get a 90 on the sixth test.

3. The mean (average) weight of three dogs is 38 pounds. One of the dogs, Sparky, weighs 46 pounds. The other two dogs, Eddie and Sandy, have the same weight. Find Eddie's weight.

Let x = Eddie's weight (they weigh the same, so they are both represented by "x".)
Let x = Sandy's weight

Average: sum of the data divided by the number of data.

$$\frac{x + x + 46}{3(\text{dogs})} = 38$$ cross multiply and solve

$$(38)(3) = 2x + 46$$
$$114 = 2x + 46$$
$$-46 \qquad -46$$
$$\frac{68}{2} = \frac{2x}{2}$$

$34 = x$ Eddie weighs 34 pounds.

Exercise 6.2

1. Find the mean, median, mode and range of this data set: 19, 18, 21, 16, 15, 17, 20, 18

2. Find the mean, median, mode and range of this data set

11, 10, 12, 12, 9, 10, 14, 12, 9

6.5.2. Mean of grouped data

The idea of grouped data can be illustrated by considering the following raw dataset:

Table 1: *Time taken (in seconds) by a group of students to answer a simple math question*

20	25	24	33	13
26	8	19	31	11
16	21	17	11	34
14	15	21	18	17

The above data can be organised into a frequency distribution (or grouped data) in several ways. One method is to use intervals as a basis.

The smallest value in the above data is 8 and the largest is 34. The interval from 8 to 34 is broken up into smaller subintervals (called class intervals). For each class interval, the amount of data items falling in this interval is counted. This number is called the frequency of that class interval. The results are tabulated as a frequency table as follows:

The table below shows the *Frequency distribution of the time taken (in seconds) by the group of students to answer a simple math question*

Class Intervals	Frequency (f)	Midpoint (x)	fx
5 and above, below 10	1	7.5	7.5
10 and above, below 15	4	12.5	50
15 and above, below 20	6	17.5	105
20 and above, below 25	4	22.5	90
25 and above, below 30	2	27.5	55
30 and above, below 35	3	32.5	97.5
TOTAL	**20**		**405**

An estimate, $\bar{x}$, of the **mean** of the population from which the data are drawn can be calculated from the grouped data as:

$$\bar{x} = \frac{\sum f x}{\sum f}.$$

In this formula, x refers to the midpoint of the class intervals, and f is the class frequency. Note that the result of this will be different from the sample mean of the ungrouped data. The mean for the grouped data in the above example, can be calculated as follows:
Thus, the mean of the grouped data is

$$\bar{x} = \frac{\sum f x}{\sum f} = \frac{405}{20} = 20.25$$

Exercise 6.3

The following gives the frequency distribution of the daily commuting time (in minutes) from home to work for all 25 employees of a company.

Daily commuting time	Number of employees
0 to less than 10	4
10 to less than 20	9
20 to less than 30	6
30 to less than 40	4
40 to less than 50	2

Calculate the mean of the daily commuting times.

6.6. MEASURES OF DISPERSION (STANDARD DEVIATION AND VARIANCE)

Standard deviation is one of the measures of dispersion of a set of data and most widely used in statistics. It is based on the mean and shows how widely the data are spread about the mean. The larger its size, the more widely the data are spread.

6.6.1. Ungrouped data

$$s = \sqrt{\frac{n}{n-1}\left[\frac{\sum_{i=1}^{n} x_i^2}{n} - \left(\frac{\sum_{i=1}^{n} x_i}{n}\right)^2\right]}$$

$$= \sqrt{\frac{n}{n-1}\left[\overline{x^2} - (\bar{x})^2\right]}$$

Example

For the set of numbers 3; 6; 9 and 12 the standard deviation is given by:

$$s = \sqrt{\frac{n}{n-1}\left[\frac{\sum_{i=1}^{n} x_i^2}{n} - \left(\frac{\sum_{i=1}^{n} x_i}{n}\right)^2\right]}$$

$$= \sqrt{\frac{n}{n-1}\left[\overline{x^2} - (\bar{x})^2\right]}$$

$$= \sqrt{\frac{4}{3}\left[\frac{270}{4} - \left(\frac{30}{4}\right)^2\right]}$$

$$= \sqrt{\frac{4}{3}\left[67{,}5 - (7{,}5)^2\right]}$$

$$= \sqrt{\frac{4}{3}(11{,}25)}$$

$$= \sqrt{15} = 3{,}873$$

6.6.2 Grouped data

$$s = \sqrt{\frac{n}{n-1}\left[\frac{\sum_{i=1}^{n} f_i x_i^2}{\sum_{i=1}^{n} f_i} - \left(\frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}\right)^2\right]} \text{ where } n = \sum_{i=1}^{n} f_i$$

$$= \sqrt{\frac{n}{n-1}\left[\overline{x^2} - (\bar{x})^2\right]}$$

Example

Suppose the table below depicts the mass of 50 students and frequencies in a Mathematics II class:

Mass (in kg)	Frequency (f_i)	Class marks (x_i)	$f_i x_i$	$f_i x_i^2$
60 – 62	8	61	488	29768
63 – 65	11	64	704	45056
66 – 68	18	67	1206	80802
69 – 71	9	70	630	44100
72 – 74	4	73	292	21316
	$n = \sum = 50$		$\sum = 3320$	$\sum = 221042$

The standard deviation is calculated as follows:

$$s = \sqrt{\frac{n}{n-1}\left[\frac{\sum_{i=1}^{n} f_i x_i^2}{\sum_{i=1}^{n} f_i} - \left(\frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}\right)^2\right]} \text{ where } n = \sum_{i=1}^{n} f_i$$

$$= \sqrt{\frac{50}{49}\left[\frac{221042}{50} - \left(\frac{3320}{50}\right)^2\right]}$$

$$= 3{,}482$$

Exercises 6.4

1. (a) determine the variance and standard deviation of 6,7,10,11,11,13,16,18,25

   (b) determine the variance and standard deviation of 10,16,12,15,9,16,10,17,12,15

   (c) determine the variance and standard deviation of 74,72,83,96,64,79,88,69

   (d) determine the variance standard and deviation of R326, R438, R375, R366, R419, R424

6.7. CORRELATION AND REGRESSION

Correlation concerns the strength of the relationship between two variables, such as the relationship between the height and weight of students in a class.

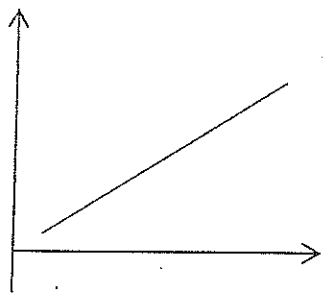
Regression analysis determines the nature of that relationship and enables us to make predictions based on it.

One way to determine if there is a relationship between the two variables is to draw a scatter diagram with the data.

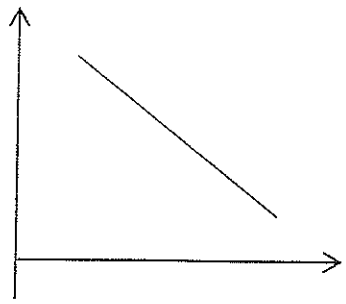
A **scatter diagram** is a dot diagram showing the relationship between two variables. Each pair of values is plotted as a point on the graph. The scatter diagram enables us to see if there is a relationship between the paired values (correlation). The relationship may be linear, quadratic, exponential or anything else. In this unit we will be dealing with the linear case.

Types of correlation

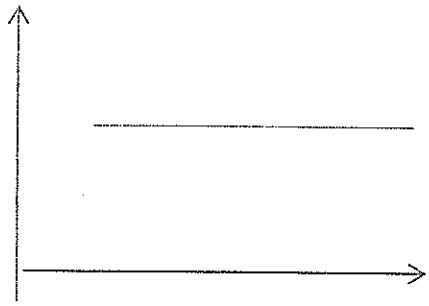
Positive correlation



Negative correlation



No Correlation



6.7.1. Correlation coefficient

The strength of correlation is given by a numerical measure (coefficient). Such a coefficient is called the **correlation coefficient**, usually represented by the letter "r". The correlation coefficient lies between -1 and 1 $i.e. -1 \leq r \leq 1$.

$r = +1$ Implies perfect positive correlation

$r = -1$ Implies perfect negative correlation

$r = 0$ Implies there is no correlation at all

The formula for calculating r is given below, where n represents the number of paired values.

$$r = \frac{n\sum xy - \left(\sum x\right)\left(\sum y\right)}{\sqrt{n\left(\sum x^2\right) - \left(\sum x\right)^2}\sqrt{n\left(\sum y^2\right) - \left(\sum y\right)^2}}$$

Example

Compute the value of the correlation coefficient from the following table:

Subject	Age x	Glucose Level y
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81

Solution:

Subject	Age x	Glucose Level y	xy	x^2	y^2
1	43	99	4257	1849	9801
2	21	65	1365	441	4225
3	25	79	1975	625	6241
4	42	75	3150	1764	5625
5	57	87	4959	3249	7569
6	59	81	4779	3481	6561
Σ	247	486	20485	11409	40022

$$r = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[\,n\Sigma x^2 - (\Sigma x)^2\,]\,[\,n\Sigma y^2 - (\Sigma y)^2\,]}}$$

r =2868 / 5413.27 = 0.529809

Exercise 6.5

With the growth of internet service providers, a researcher decides to examine whether there is a correlation between cost of internet service per month (rounded to the nearest rand) and degree of customer satisfaction (on a scale of 1 - 10 with a 1 being not at all satisfied and a 10 being extremely satisfied). The researcher only includes programs with comparable types of services. A sample of the data is provided below.

Dollars	11	18	17	15	9	5	12	19	22	25
Satisfaction	6	8	10	4	9	6	3	5	2	10

(a) Compute the correlation coefficient.

(b) What does this statistic mean concerning the relationship between amount of money spent per month on internet provider service and level of customer satisfaction?

(c) What percentage of the variability is accounted for by the relationship between the two variables and what does this statistic mean?

6.7.2. The regression line

The least squares regression line is the line that minimises the sum of the squares of the vertical deviation from each data point to the line.

Time series is a collection of data recorded over a period of time (i.e. weekly, monthly, quarterly or yearly). The long term trend of many business series, such as sales, exports and production often approximates a straight line.

The least square regression line for the set of n data points is given by

$$y = ax + b$$

where a and b are given by:

$$a = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - (\sum_{i=1}^{n} x_i)^2}$$

$$b = \frac{1}{n}(\sum_{i=1}^{n} y_i - a\sum_{i=1}^{n} x_i)$$

Examples

1. Consider the following set of points: {(-2 , -1) , (1 , 1) , (3 , 2)}

   a) Find the least square regression line for the given data points.

   b) Plot the given points and the regression line in the same rectangular system of axes.

2. a) Find the least square regression line for the following set of data

{(-1 , 0),(0 , 2),(1 , 4),(2 , 5)}

b) Plot the given points and the regression line in the same rectangular system of axes.

3. The values of y and their corresponding values of y are shown in the table below

x	0	1	2	3	4
y	2	3	5	4	6

   a) Find the least square regression line $y = a\bar{x} + b$.

   b) Estimate the value of y when x = 10.

4. The sales of a company (in million dollars) for each year are shown in the table below.

x (year)	2005	2006	2007	2008	2009
y (sales)	12	19	29	37	45

a) Find the least square regression line y = ax + b.

b) Use the least squares regression line as a model to estimate the sales of the company in 2012.

Solutions

1.a) Let us organize the data in a table.

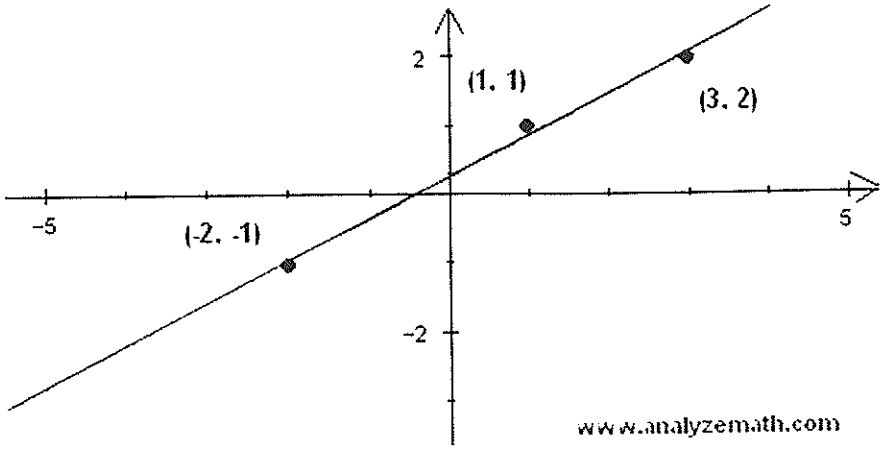
x	Y	x y	x^2
-2	-1	2	4
1	1	1	1
3	2	6	9
$\Sigma x = 2$	$\Sigma y = 2$	$\Sigma xy = 9$	$\Sigma x^2 = 14$

We now use the above formula to calculate a and b as follows

$a = (n\Sigma x\, y - \Sigma x \Sigma y) / (n\Sigma x^2 - (\Sigma x)^2) = (3\times 9 - 2\times 2) / (3\times 14 - 2^2) = 23/38$

$b = (1/n)(\Sigma y - a\, \Sigma x) = (1/3)(2 - (23/38)\times 2) = 5/19$

b) We now graph the regression line given by $y = ax + b$ and the given points.



2.a) We use a table as follows

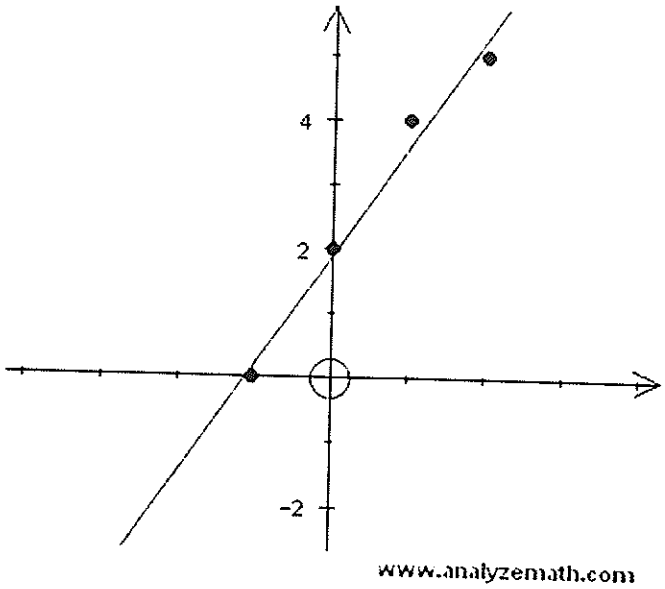
X	Y	x y	x^2
-1	0	0	1
0	2	0	0
1	4	4	1
2	5	10	4
$\Sigma x = 2$	$\Sigma y = 11$	$\Sigma x\, y = 14$	$\Sigma x^2 = 6$

We now use the above formula to calculate a and b as follows

$a = (n\Sigma x\, y - \Sigma x \Sigma y) / (n\Sigma x^2 - (\Sigma x)^2) = (4\times14 - 2\times11) / (4\times6 - 2^2) = 17/10 = 1.7$

$b = (1/n)(\Sigma y - a\, \Sigma x) = (1/4)(11 - 1.7\times2) = 1.9$

b) We now graph the regression line given by $y = ax + b$ and the given points.



(change the star to be a real multiplication)

3.a) We use a table to calculate a and b.

x	Y	x y	x^2
0	2	0	0
1	3	3	1
2	5	10	4
3	4	12	9
4	6	24	16
$\Sigma x = 10$	$\Sigma y = 20$	$\Sigma x\, y = 49$	$\Sigma x^2 = 30$

We now calculate a and b using the least square regression formulas for a and b.

$a = (n\Sigma x\, y - \Sigma x \Sigma y) / (n\Sigma x^2 - (\Sigma x)^2) = (5\times59 - 10\times20) / (5\times30 - 10^2) = 1.9$

$b = (1/n)(\Sigma y - a\, \Sigma x) = (1/5)(20 - 1.9\times10) = 0.2$

b) Now that we have the least square regression line $y = 1.9\, x + 0.2$, substitute x by 10 to find the value of the corresponding y.

$y = 1.9 \times 10 + 0.2 = 19.2$

4.a) We first change the variable x into t such that t = x - 2005 and therefore t represents the number of years after 2005. Using t instead of x makes the numbers smaller and therefore manageable. The table of values becomes.

t (years after 2005)	0	1	2	3	4
y (sales)	12	19	29	37	45

We now use the table to calculate a and b included in the least regression line formula.

t	Y	t y	t^2
0	12	0	0
1	19	19	1
2	29	58	4
3	37	111	9
4	45	180	16
$\Sigma x = 10$	$\Sigma y = 142$	$\Sigma xy = 368$	$\Sigma x^2 = 30$

We now calculate a and b using the least square regression formulas for a and b.

$a = (n\Sigma t\, y - \Sigma t \Sigma y) / (n\Sigma t^2 - (\Sigma t)^2) = (5\times368 - 10\times142) / (5\times30 - 10^2) = 8.4$

$b = (1/n)(\Sigma y - a\, \Sigma x) = (1/5)(142 - 8.4\times10) = 11.6$

b) In 2012, t = 2012 - 2005 = 7

The estimated sales in 2012 are: $y = 8.4 \times 7 + 11.6 = 70.4$ million dollars.

Exercise 6.6

1. As Earth's population continues to grow, the solid waste generated by the population grows with it. Governments must plan for disposal and recycling of ever growing amounts of solid waste. Planners can use data from the past to predict future waste generation and plan for enough facilities for disposing of and recycling the waste. Given the following data on the waste generated in Florida from 1990-1994, how can we predict the waste that was generated in the years 1995-1999? .

Year	1990	1991	1992	1993	1994
Tons of solid waste generated (in thousands)	19,358	19,484	20,293	21,499	23,561

Predict the average tons of waste in 2000 and 2005.

2. The numbers of insured commercial banks *y* (in thousands) in the United States for the years 1987 to 1996 are shown in the table. (Source: Federal Deposit Insurance Corporation).

Year	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
Y	13.70	13.12	12.71	12.34	11.92	11.46	10.96	10.45	9.94	9.53

Make a scatterplot of the data, letting *x* represent the number of years since 1987.

a) Graph the function of best fit with the scatterplot of the data.

b), Predict the average number of insured commercial banks in 2000 and 2005

2. *U.S. Farms.* As the number of farms has decreased in the United States, the average size of the remaining farms has grown larger, as shown in the table below.

Year	*1910*	*1920*	*1930*	*1940*	*1950*	*1959*	*1969*	*1978*	*1987*	*1997*
Ave acreage per farm	*139*	*149*	*157*	*175*	*216*	*303*	*390*	*449*	*462*	*487*

a) Make a scatterplot of the data, letting x represent the number of years since 1900.

b) Graph the function of best fit with the scatterplot of the data.

c) Predict the average acreage in 2000 and 2010 and determine which function gives the most realistic predictions.

REVISION EXERCISE UNIT 6

1. At a recent chess tournament, all 10 of the participants had to fill out a form that gave their names, address and age. The ages of the participants were recorded as follows:
36, 48, 54, 92, 57, 63, 66, 76, 66, 80.
Construct a frequency distribution table, cumulative frequency table, histogram, ogive for the given data above.

Answer

Ages (class limits)	Frequency	Class marks	Class boundaries	Cumulative frequency
35-44	1	39.5	34.5-44.5	1
45-54	2	49.5	44.5-54.5	3
55-64	2	59.5	54.5-64.5	5
65-74	2	69.5	64.5-74.5	7
75-84	2	79.5	74.5-84.5	9
85-94	1	89.5	84.5-94.5	10

2. Thirty AA batteries were tested to determine how long they would last. The results, to the nearest minute, were recorded as follows:
423, 369, 387, 411, 393, 394, 371, 377, 389, 409, 392, 408, 431, 401, 363, 391, 405, 382, 400, 381, 399, 415, 428, 422, 396, 372, 410, 419, 386, 390

Construct a frequency distribution table, cumulative frequency table, histogram, ogive for the given data above.

Answer

Batter life (limits)	Frequency	Class marks	Class boundaries	Cumulative frequency
360-369	2	346.5	395.5-369.5	2
370-379	3	374.5	376.5-379.5	5
380-389	5	384.5	375.5-389.5	10
390-399	7	394.5	385.5-399.5	17
400-409	5	404.5	395.5-409.5	22
410-419	4	414.5	409.5-419.5	26
420-429	3	424.5	419.5-429.5	29
430-439	1	434.5	429.5-439.5	30

3. A football fan records how many goals his team scored in their last 15 football matches. Here are the results:

2, 3, 0, 0, 1, 3, 4, 2, 5, 0, 1, 1, 2, 1, 1

What is the average mean, median, modal number of goals scored?

Answer

Mean=1.73 median = 1mode=1

4. A golfer records her scores for the last 8 rounds of golf she plays. Here are her results:

72, 75, 80, 81, 76, 77, 77, 74

What is the mean, median, modal score?

Answer

Mean=68 median = 76mode=77

5. The weekly wages of 10 people in a factory are:

R220, R250, R250, R305, R450, R210, R200, R305, R265, R305, R450

What is the mean, median, modal average wage?

Answer

Mean=R321 median = R305 mode=R305

6 The following table shows the number of hours per day of watching TV in a sample of 500 people.

Hours	0-1	2-3	4-5	6-7	8-9	10-11	12-13
Frequency	55	87	145	90	73	35	15

What is the mean number of TV viewing hours in this group?

Answer 5.316

7. The following table shows the annual salaries earned by employees of a company

Annual Salary	Frequency
R35550	3
R42770	5
R51000	5
R99000	1
R150000	1

What is the mean salary?

Answer =R54950

8. Given the set of numbers {5, 9, 4, 2, 6}:

Find the standard deviation?

Answer=2.588435

9. Given the set of numbers {34, 22, 24, 26, 36}:

Find the standard deviation?

Answer=6.22896

10.The following gives frequency distribution of the daily commuting time (in minutes) from home to work for all 25 employees of a company.

Daily commuting time	Number of employees
0-10	4
10-20	9
20-30	6
30-40	4
40-50	2

Calculate the standard deviation of the daily commuting times?

Answer= 11.62

11. Determine an estimate of the standard deviation of the following data for the marks obtained in a test by 88 students.

Marks(x)	Frequency
0-10	6
10-20	16
20-30	24
30-40	25
40-50	17

Calculate the standard deviation

Answer=11.78

12 compute the value of the correlation coefficient between the age(x) and the glucose level (y) from the table below ant interpret it.

subject	age(x)	glucose level (y)
1	43	99
2	21	65
3	25	79
4	42	75
5	57	87
6	59	81

Answer r=0.529809

13 Below are the data for six participants giving their number of years in college(x) and their subsequent income (y)

Years of college (x)	Income (y)
0	15
1	15
3	20
4	25
4	30
6	35

Calculate the coefficient correlation(r) between the years of college and income and interpret it.

Answer r=0.95

Answers to exercises in Unit 1

Exercise 1.5

1.

a)$32 + 80x + 80x^2 + 40x^3 + 10x^4 + x^5$

b) $16x^4 - 96x^3y + 216x^2y^2 - 216xy^3 + 81y^4$

c) $8x^{12} - 12x^8y + 6x^4y^2 - y^3$

d) $1 - x - \frac{x^2}{2} - \frac{x^3}{2} - \cdots$

e) $\frac{1}{9x^2} - \frac{10y}{27x^3} + \frac{25y^2}{27x^4} - \frac{500y^3}{243x^5} + \cdots$

2.

a) $T_3 = 336x^2$ b) $T_{12} = -21840x^{11}$ c) $T_6 = \frac{63x^5}{8}$ d) $T_6 = \frac{189x^5}{4}$

e) $T_4 = 32x^3$ f) $-\frac{28}{27}x^{10}y^5$

3. $1 - 3x + 6x^2 - 10x^3 + \cdots$ 4. 1760

5. $2^{-7}\left(1 + \frac{7x}{4} + \frac{7x^2}{4} + \frac{21x^2}{16} + \cdots\right)$; $-4 < x < 4$

Exercise 1.6

1.a) $\frac{7\pi}{36}$ b) $\frac{3\pi}{4}$ c) $\frac{5\pi}{4}$ d) $-\frac{4\pi}{3}$ e) $\frac{2\pi}{3}$ f) $\frac{5\pi}{3}$

2. a) $120°$ b) $-15°$ c) $72°$

Exercise 1.7

1.a) 425,505 b) 0,18487 c)29,3282 d) 0,06742

e) 8,20896

2. a) 1363,378 b) 737,021 c) 6,4199 d) 16,35

Answers to revision exercise: Unit 1

1.a) $8x^3 - 12x^2y + 6xy^2 - y^3$

b) $81 + 54x + \frac{27x^4}{4} + \frac{3x^3}{2} + \frac{x^4}{16}$

c) $a^5 - 15a^4b + 45a^3b^2 - 270a^2b^3 + 405ab^4 - 243b^5$

2. a) $1 + 2x + 4x^2 + 8x^3 + \cdots$

b) $\frac{1}{9}\left[1 - \frac{2x}{3} + \frac{x^2}{3} - \frac{4x^4}{27} + \cdots\right]$ c) $1 - \frac{x}{3} - \frac{x^2}{18} - \frac{x^3}{54} - \cdots$

3. a) $1 + \frac{x}{4} - \frac{x^2}{32} + \frac{x^3}{128} - \cdots$ 1,0025

b) $1 - \frac{2x}{3} - \frac{x^2}{9} - \frac{4x^3}{81} - \cdots$ 0,825

4. a) $15120x^4y^3$ b) $70x^4$ c) $405x^4$

5. a) 1512 b) 16016

6. a) $1694{,}137mm^2$ b) $4783{,}058m^2$ c) $4043{,}846cm^2$

Answers to Exercises in Unit 2

Exercise 2.3

$12x^3 - 3x^2$ b) $\frac{-1}{\sqrt{x^3}} + \frac{1}{x^2} + 2$ c) $\frac{1}{\sqrt{x}} - \frac{2}{x^3} + \frac{5}{x^2}$

Exercise 2.4

$1 - 8x - 6x^2$ b) $1 + \frac{1}{x^2}$ c) $2x - 6x^2$

Exercise 2.5

$\frac{-6}{(3x-1)^2}$ b) $\frac{2x+5}{(x+1)^2}$ c) $\frac{5}{(x-3)^2}$

Exercise 2.6

$24x(3x^2 - 5)^3$ b) $6x(3 - x^2)^{-4}$ c) $\frac{5(\sqrt{x}+1)^4}{2\sqrt{x}}$

Exercise 2.7

a) $-2x^3 \sin 2x + 3x^2 \cos 2x$ b) $2 \sin x \cos x$ c) $2x \cos x^2 + 2x \sin(x^2 + 2)$

d) $\frac{1}{1-\sin x}$ e) $sec^2 x$

Exercise 2.8

1. $-3e^{-3x} + e^{0,5x}$ 2. $2xe^{x^2} - 2e^{-3x} + 2ae^{2ax}$ 3. $2(3^{2x-1}) \ln 3 + 3a(5^{3ax}) \ln 5$

4. $2x(7^{x^2}) \ln 7 + \frac{1}{2\sqrt{x}}(5^{\sqrt{x}}) \ln 5$

Exercise 2.9

$\frac{2x}{(x^2+5)\ln 3}$ b) $2x\left(\frac{1}{\ln 5} + \log_5 3x^2\right)$ c) $e^x\left[\log(1-x) - \frac{1}{(1-x)\ln 10}\right]$

Exercise 2.10

$a)\ -6x^{-4} - 6x + 5\ :\ 24x^{-5} - 6$ b)$-3 \sin 3x - 3 \cos 3x$:
$-9 \cos 3x + 9 \sin 3x$

1. $v = e^{-t}(1 - t)$ $a = e^{-t}(t - 2)$

Exercise 2.11

a) $\frac{-x^2}{y^2}$ b) $\frac{1+2y-2x}{2y-2x-1}$ c) $\frac{3\cos(3x+4y)}{1-4\cos(3x+4y)}$ d) $\frac{4e^{4x}-ye^{xy}}{xe^{xy}+5e^{5y}}$

2. -5 3. 1

Exercise 2.12

$2x \times 5^{x^2} \ln 5$ b) $\cos x\, e^{\sin x}$ c) $\frac{1}{x}+\frac{4x}{x^2+1}-\frac{3x^2}{2x^{3-1}}$

d) $y\left[3+4\cot x-\frac{2}{x}-\frac{3x^2}{2(x^3-1)}\right]$

Exercise 2.14

$a)\ v = 25-9.8t$ $a = -9.8$ b) 31.888

1. $a)\ \alpha = -\sin\left(2t+\frac{\pi}{6}\right)$ b) $-\frac{\pi}{12}$

Exercise 2.15

1.$y_1 = 2.1,\ y_2 = 2.0946,\ y_3 = 2.0946$

2. a) 0.739 b) 1.404 c) -2.457

Answers to revision exercise Unit 2

1.a) $2x+2$ b). $2x+4$ c). $6x^{-3}$

2.a) $24(4x-5)^5$ b) $2\cos 2x\, e^{\sin 2x}$ c) $-9\sin 3x\, cos^2 3x$ d) $\frac{2}{(2x-1)\ln 10}$

e)$x^2 sec^2 x + 2x\tan x$ f) $e^{5x}(3x+1)(15x+11)$ g) $\frac{(4x+10)\cos 2x - 2\sin 2x}{(2x+5)^2}$

h) $\frac{\frac{x}{\sqrt{x+1}}-2\sqrt{x+1}}{2x^2} = \frac{-(x+2)}{2x^2\sqrt{x+1}}$

3. a) $\frac{-(x+y)}{x+3y}$ b) $\frac{y^2-x^2}{y^2-2xy}$ c) $\frac{-\cos x}{\cos y}$ d) $2+8\cos 4x+\frac{x}{2-x^2}$

e) $y\left[\frac{4}{x-1}-5\tan 5x-\frac{1}{x}-\frac{3sec^2 3x}{2\tan 3x}\right]$ f) $y\left[\cos x\ln(3x^2-5)+\frac{6x\sin x}{3x^2-5}\right]$

5. $\left(\frac{1}{2},4\right)\ min$ $\left(-\frac{1}{2},-4\right)\ max$

6. a) $v = 5-8t$ $a = -8$ b) 1.5625

7. a) $v = e^{-t}(2\cos 2t - \sin 2t)$ b) $v(0) = 2$

8.a) $v = -3\sin 3t + 6\cos 3t$b) $t = o.37s$ c) -18

9. a) $t = 1\ and\ t = 2$ b) $a(1) = -1,\ a(2) = 1$

10. 28×10^{-6} 11. $v = e^{-0.t}(0{,}5t-5)$ b. $t = 10$

12. a) 16 b) $t = \frac{4}{3}$ or 3 c) $t = 3$

13. a) 2.196 b) 1.576 c) 0.739

15. $\frac{-1}{\rho^2}$ 16. $\frac{-E}{C}e^{-t/RC}$ 17. $60(15+t)^{-2}$ 18. $\frac{1}{2\sqrt{t}}e^{-\sqrt{t}}$

Answers to exercises in Unit 3

Exercise 3.1

1. $\frac{x^4}{4}+C$ 2. $2\sqrt{x^3}+C$ 3. $x^{2/3}+C$ 4. $6x-3x^2+C$ 5. $-\frac{1}{x^4}+C$

6. $\frac{2}{5}x^5-x^3+\frac{a}{2}x^2+C$ 7. $\frac{x^3}{6}+\frac{x^4}{12}+C$ 8. $x+\frac{3}{2}x^2+x^3+\frac{x^4}{4}+C$

9. $\frac{21}{3}\sqrt[4]{x^3}+C$ 10. $x+\frac{2}{x^2}+\pi^2x+C$

Exercise 3.2

1. $\frac{1}{18}(3x^2-6x+1)^3+C$ 2. $2\sqrt{t^2-1}+C$ 3. $(7x-2)^4+C$

4. $-\frac{1}{3(1+\sqrt{y})^3}+C$ 5. $\frac{1}{3}(s^3+2s^2-5s+6)^3+C$ 6. $\frac{3}{8}(x^4+3)^{2/3}+C$

Exercise 3.3

1. $4\tan\frac{x}{4}+C$ 2. $\frac{2}{3}\sqrt{tan^3x}+C$ 3. $\frac{1}{6}(1-cos3x)^2+C$ 4. $\frac{sin^4\theta}{4}+C$

5. $-2cos\sqrt{x}+C$ 6. $4\sqrt{5-4cost}+C$ 7. $\frac{1}{5}\sqrt{(3-cos2t)^5}+C$ 8. $\frac{1}{6cos^3 2x}+C$

9. $sec\left(\theta+\frac{\pi}{2}\right)+C$ 10. $cosec\left(\frac{1}{x}\right)+C$

Exercise 3.4

1. $\frac{2^{x^2}}{2ln2}+C$
2. $2e^{\sqrt{x}}+C$
3. $\frac{4a^{x^3-1}}{3lna}+C$
4. $ln(sine^x)+C$
5. $\frac{1}{4}ln(x^4+1)+C$
6. $-\frac{3}{2}ln(1-tan2x)+C$
7. $ln(2x^2-4x+1)+C$

Exercise 3.5.

1. 3
2. $-\frac{1}{2}$
3. π
4. -3
5. 4

Exercise 3.6

1. $\frac{2}{3}$
2. 2
3. 8

Exercise 3.7

1. 4 units^2
2. 28.333 units^2
3. $\frac{9}{2}$ units^2
4. $\frac{64}{3}$ units^2
5. 18 units^2

Exercise 3.8.

1. 0.766
2. 20.667
3. $0.999 \approx 1$
4. 1623394.75

Answers to exercises in Unit 4

Exercise 4.2

1. $|\boldsymbol{a}| = \sqrt{38}$ $|\boldsymbol{m}| = \sqrt{14}$ $|\boldsymbol{n}| = \sqrt{29}$

Exercise 4.3

a) $x = 1.397$ $y = 5.216$ b) $x = -68.481$ $y = -47.951$

c) $x = 176.453$ $y = -649.456$

Exercise 4.4

$\frac{1}{5}(3\boldsymbol{i} - 4\boldsymbol{j})$ 2. $\frac{1}{\sqrt{5}}(2\boldsymbol{i} + \boldsymbol{j})$ 3. $\frac{1}{\sqrt{14}}(\boldsymbol{i} + 3\boldsymbol{j} - 2\boldsymbol{k})$ 4. $\frac{1}{\sqrt{6}}(-2\boldsymbol{i} + \boldsymbol{j} - \boldsymbol{k})$

Exercise 4.5

$\frac{2}{\sqrt{2}}(3\boldsymbol{i} - 4\boldsymbol{j} - 5\boldsymbol{k})$ 2. $4(-8\boldsymbol{i} - 4\boldsymbol{j} + \boldsymbol{k}) = -32\boldsymbol{i} - 16\boldsymbol{j} + 4\boldsymbol{k}$

Exercise 4.6

$4\boldsymbol{b} = \boldsymbol{a}$ and $-4\boldsymbol{d} = \boldsymbol{a}$ $\therefore$ $\boldsymbol{b}$ and $\boldsymbol{d}$ are parallel to $\boldsymbol{a}$

Exercise 4.7

a) $-9\boldsymbol{i} + 7\boldsymbol{j} + 8\boldsymbol{k}$ b) $-17\boldsymbol{i} + 5\boldsymbol{j} + 8\boldsymbol{k}$ c) $\boldsymbol{i} - 7\boldsymbol{j} - 10\boldsymbol{k}$ d) $-8\boldsymbol{i} - 4\boldsymbol{j}$

$e)$ $12\boldsymbol{i} + 9\boldsymbol{j} - 5\text{k}$

Exercise 4.8

a) 5 b) 60 c) 121

Exercise 4.9

a) 29.5° b) 50.7° 2) 84.9° 3) $x = -\frac{1}{2}$ 4) $x = \pm 4$

Exercise 4.10

a) 5J b) 60J

Exercise 4.11

a) $-\boldsymbol{i} + \boldsymbol{j} + 3\boldsymbol{k}$ b) 0 c) $-7\boldsymbol{i} + 5\boldsymbol{j} - 4\boldsymbol{k}$

Exercise 4.12

1. $22\boldsymbol{i} - 2\boldsymbol{j} - 16\boldsymbol{k}$

Answers to revision exercise Unit 4

1. a) $18\boldsymbol{i} + 17\boldsymbol{j} - 31\boldsymbol{k}$ b) $-7\boldsymbol{i} - 14\boldsymbol{j} + 24\boldsymbol{k}$ c) $-\boldsymbol{i} + \boldsymbol{j} - \boldsymbol{k}$
2. $k = -1,\ m = -2$
3. $x = 30$
4. $|F| = \sqrt{405}$
5. $|a| = \sqrt{14}$ $\hat{a} = \frac{1}{\sqrt{14}}(2\boldsymbol{i} - \boldsymbol{j} - 3\boldsymbol{k})$ $|b| = 7\sqrt{3}$ $\hat{b} = \frac{1}{\sqrt{3}}(\boldsymbol{i} - \boldsymbol{j} - \boldsymbol{k})$

   $|c| = \sqrt{118}$ $\hat{c} = \frac{1}{\sqrt{118}}(9\boldsymbol{i} - 6\boldsymbol{j} + \boldsymbol{k})$
6. $x = -339.33$ $y = -160.55$
7. $x = 50.56$ $y = 68.33$
8. $4\boldsymbol{i} - 5\boldsymbol{j} + 11\boldsymbol{k},\ \sqrt{262} = 16.1864$
9. i) 9 $ii) -6$ $iii) -211$
10. 90° 11. $16J$ 12. $32J$

13. $r \times s = 5\boldsymbol{i} + 26\boldsymbol{j} + 17\boldsymbol{k}$ $r \times s = -5\boldsymbol{i} - 26\boldsymbol{j} - 17\boldsymbol{k}$

14. $7\boldsymbol{i} - 7\boldsymbol{k}$ 15. $\frac{-1}{3}$ 16. ± 4

Answers to exercises in Unit 5

Exercise 5.1

a) $\pm 2j$ b) $\frac{-3\pm\sqrt{31}j}{4}$ c) $-1 \pm 6j$

Exercise 5.2

1. $17 + 8j$ 2. $11 - 4j$ 3. $-5 + j$ 4. $5 + 3j$ 5. $-11 - 6j$

Exercise 5.3

1. $26 + 7j$ 2. $16 + 30j$ 3. 25

Exercise 5.4

1. a) $3 - 4j$ b) $3 + 5j$ c) $-7j$
2. a) $\frac{22}{53} - j\frac{29}{53}$ b) $2.8 + 2.4j$

Exercise 5.5

a) $x = 3,\ y = -1.5$ b) $x = -7,\ y = 9$ c) $x = -5,\ y = -12$ d) $x = 4,\ y = -7$

Exercise 5.6

1. $5cis53.3°$ 2. $5cis(-53.3°)$ 3. $\sqrt{8}cis(-135°)$ 4. $2cis60°$

Exercise 5.7

1. $2 - 2j$ 2. $0.02 + 0.69j$ 3. $-41 + 38j$ 4. $52 - 36j$

Exercise 5.8

1. a) $1.10 - 0.46j;\ -1.10 + 0.46j$ b) $0.79 + 0.79j;\ -1.08 + 0.29j;\ 0.29 - 1.08j$
c) $1.63 + 0.52j;\ -1.26 + 1.15j;\ -0.36 - 1.67j$ d) $\sqrt{2}j\ ;\ -\sqrt{2}j$
2. a) $2;\ -1 + \sqrt{3}j\ ;\ -1 - \sqrt{3}j$ b) $1;\ j;\ -1;\ -j$ c) $-2;\ 1.62 + 1.18j;$
$-0.62 + 1.90j;\ -0.62 - 1.90j;\ 1.62 - 1.18j.$

Answers to revision exercise Unit 5

1. a) $\pm 3j$ b) $\frac{-1\pm\sqrt{3}j}{2}$ c) $\frac{3\pm 4j}{5}$
2. a) $-1 - 3j$ b) $-10 - 5j$ c) $24 - 10j$ d) 25 e) 17
3. a) $\frac{-1+18j}{25}$ b) $\frac{-1-7j}{5}$ c) $\frac{4j}{5}$ d) $\frac{2-8j}{17}$
4. a) $x = -1.5,\ y = 4$ b) $x = \frac{3}{25},\ y = \frac{4}{25}$ c) $x = 4,\ y = 5$
5. $3 - 4j$ b) $-5 + 12j$ c) $-1 - 3j$ d) $\sqrt{3} + j$
6. a) $-0.016 + 0.088j$ b) $-4.5 + 6.5j$ c) $-72.027 + 103.981j$

7.1 a) $j, -j$ b) $1.25 + 0.72j,\ -1.25 + 0.72j,\ -1.44j$

7.2 a) $-1.732,\ 1.732$ b) $1.2599, -0.630 + 1.091j,\ -0.630 - 1.091j$